

# Methodologies, level 3 — full formulation

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## Welch's t-test

### Concept

Welch's t-test evaluates whether the means of two independent populations are statistically different, based on two random samples, without assuming the two populations share the same variance. It's a generalization of Student's two-sample t-test that resolves the so-called Behrens-Fisher problem: how to test for a difference in means between two normal populations with possibly different, unknown variances.

The core statistical intuition is this: uncertainty about a sample mean depends on two things — how much variability exists within the group, and how many observations were used to estimate that mean. When the two groups have different variances and different sample sizes, the correct way to combine that uncertainty for testing the difference of means isn't symmetric between the groups — and that's exactly what Welch models, without forcing an equal-variance assumption that's rarely justifiable a priori.

### Mathematical formulation

Let there be two independent samples:

- Group 1:  $n_1$  observations, sample mean  $\bar{x}_1$ , sample variance  $s_1^2$ .
- Group 2:  $n_2$  observations, sample mean  $\bar{x}_2$ , sample variance  $s_2^2$ .

Welch's test statistic is:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

The numerator is the observed difference between the sample means. The denominator is the standard error of that difference, computed by summing the variance of each sample mean ( $s_i^2/n_i$ ) separately — without pooling the two variances into a single combined variance, the way Student's test does.

The degrees of freedom aren't  $n_1 + n_2 - 2$ , as in Student's test. Instead, the Welch-Satterthwaite approximation is used:

$$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

What matters here isn't memorizing the formula but understanding the idea:  $\nu$  is an *effective* number of degrees of freedom, typically non-integer, that weights how much each group contributes to the total uncertainty in the standard error. A small group with high variance “pulls”  $\nu$  down — reducing the test's confidence — in a way that's more faithful to reality than the simple degrees-of-freedom count used by Student's test.

## Assumptions

1. **Independence:** observations within each group, and between the two groups, are independent of one another. When there's dependence structure (several observations from the same sampling unit, or repeated measurements on the same person, say), the assumption is violated and the computed standard error understates the true uncertainty — cluster-robust standard-error methods are more appropriate in that case.
2. **Approximate normality, or a large sample:** the test assumes the sample means are approximately normally distributed. That holds exactly if the underlying data are normal, and approximately for large samples, by the Central Limit Theorem, even when the underlying data aren't normal. With small samples and heavily skewed data, the approximation can break down — it's worth inspecting the data's distribution or considering non-parametric alternatives (like the Mann-Whitney test) in those cases.
3. **Does not require equal variances** — that's precisely the assumption Welch relaxes relative to Student's t-test. There's no need to test for equal variances before choosing Welch: since Welch reduces to Student's test when the variances really are equal (with only a small efficiency loss in that case), it's safe to use Welch by default, even when it isn't known whether the variances are equal.

## Hypotheses

- $H_0$ :  $\mu_1 = \mu_2$  (the population means are equal).
- $H_1$ :  $\mu_1 \neq \mu_2$  (two-tailed test, the most common case), or  $\mu_1 > \mu_2$  /  $\mu_1 < \mu_2$  for one-tailed tests, when there's a specific direction of interest defined before looking at the data.
- Significance level ( $\alpha$ ): typically 0.05, but should be chosen according to context — for higher-stakes decisions or with multiple simultaneous comparisons, a more conservative  $\alpha$  (0.01, say) is more appropriate.
- Test statistic:  $t$ , computed as above, compared against a Student's t distribution with  $\nu$  degrees of freedom.
- P-value: the probability of observing a t statistic as extreme or more extreme than the one observed, assuming  $H_0$  is true.
- Confidence interval: a 95% CI for the difference  $\mu_1 - \mu_2$  is given by  $(\bar{x}_1 - \bar{x}_2) \pm t_{\nu, 0.975} \cdot \text{SE}$ , where SE is the standard error of the difference (the denominator of the t statistic) and  $t_{\nu, 0.975}$  is the appropriate quantile of the reference distribution.

## Interpretation

Rejecting  $H_0$  with a low p-value is statistical evidence that the population means differ — it isn't proof, and it says nothing about why the difference exists. A t-test (Welch or Student) measures the association between belonging to a group and the average value of a variable; it doesn't isolate why that association exists, unless the groups were defined by controlled random assignment (as in an experiment).

It's essential to distinguish statistical significance from practical relevance. With very large samples, differences in means that are practically irrelevant can still be statistically significant. With small samples, large differences may fail to reach significance simply for lack of statistical power. Always report the effect size (the difference in means itself, in its original unit, and ideally a confidence interval too) alongside the p-value.

## Limitations

- The test compares only means — two distributions can have equal means and completely different shapes (variance, skew). If the question is about the entire distribution, not just the mean, methods that look at other points of the distribution (like quantile decompositions) are more informative.
- Multiple testing: running Welch repeatedly across many subgroups or many time periods without correction increases the chance of finding a “significant” difference purely by chance. In analyses with many simultaneous comparisons, it's worth considering a correction (Bonferroni or false-discovery-rate control, for instance).
- Extremely small samples (fewer than about 10 observations per group, say) make the Welch-Satterthwaite approximation less reliable — in that regime, exact or permutation-based methods are preferable.
- The test alone doesn't handle panel data or correlated time series — each application of the test assumes fresh, independent samples.

## Example

Consider a hypothetical scenario: an operations team wants to know whether average handling time (in minutes) differs between two branches of a support network, Branch A and Branch B, which have different call volumes and internal processes.

Data (invented):

- Branch A:  $n_1 = 40$ ,  $\bar{x}_1 = 8.1$  min,  $s_1 = 1.3$  min.
- Branch B:  $n_2 = 95$ ,  $\bar{x}_2 = 9.6$  min,  $s_2 = 3.4$  min.

### Step 1 — standard error of the difference:

$$SE = \sqrt{\frac{1.3^2}{40} + \frac{3.4^2}{95}} = \sqrt{0.0423 + 0.1217} \approx 0.405$$

### Step 2 — t statistic:

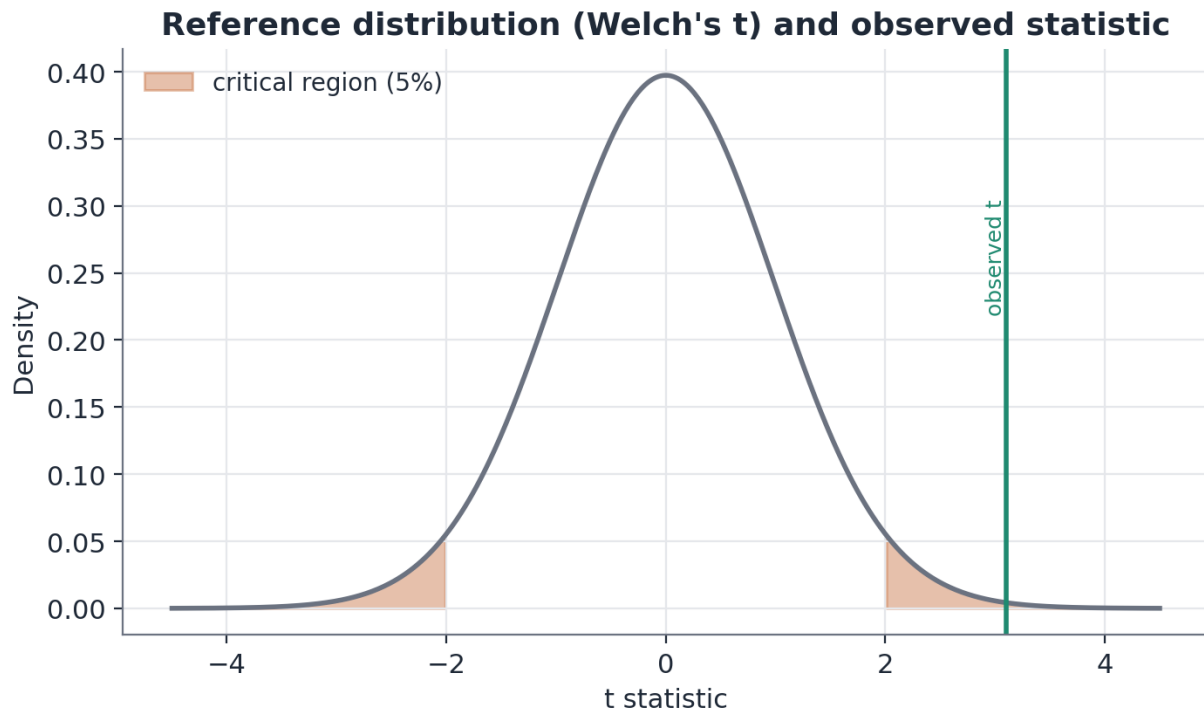
$$t = \frac{9.6 - 8.1}{0.405} \approx \frac{1.5}{0.405} \approx 3.10$$

**Step 3 — effective degrees of freedom** (applying Welch-Satterthwaite to the same numbers): this yields  $\nu \approx 61.4$  — a non-integer number, pulled down relative to  $n_1 + n_2 - 2 = 133$  because Branch B, despite having more observations, also has much higher variability, which reduces the effective confidence in the comparison.

**Step 4 — p-value:** comparing  $t \approx 3.10$  against  $\nu \approx 61.4$  degrees of freedom yields a two-tailed p-value well below 0.01 — strong evidence that the observed difference isn't just sampling noise.

**Step 5 — confidence interval:** the 95% CI for the difference in means comes out to roughly 0.7 to 2.3 minutes — Branch B likely has a longer average handling time, and the interval gives a sense of how large that difference probably is, not just that it exists.

The figure below shows the reference distribution (the t distribution with Welch's degrees of freedom) and where the observed statistic falls relative to the 5% critical region:



Illustrative data, not drawn from any real project.

Figure 1: Welch's t reference distribution with the observed statistic marked and the 5% critical region highlighted

# Oaxaca-Blinder decomposition

## Concept

The Oaxaca-Blinder decomposition is a regression-based accounting technique that splits the observed difference in average outcomes (typically wage or income) between two groups into two parts: one attributable to differences in the groups’ average observable characteristics (the explained component) and one that remains even after adjusting for those characteristics (the unexplained component). It was proposed independently by Ronald Oaxaca and Alan Blinder in 1973, in the context of wage differences between demographic groups.

The central intuition is counterfactual: the method asks “what would Group B’s average outcome be if it had Group A’s distribution of characteristics, but kept being paid according to its own structure of returns?” The answer to that hypothetical question is what allows separating “compositional difference” from “difference in returns to the same characteristics.”

## Mathematical formulation

For each group  $g \in \{A, B\}$ , fit a linear model:

$$Y_g = X_g\beta_g + \varepsilon_g$$

where  $Y_g$  is the outcome variable (wage, say),  $X_g$  is the matrix of observable characteristics (including an intercept),  $\beta_g$  is the vector of coefficients estimated separately for group  $g$ , and  $\varepsilon_g$  is the error term.

The total difference in means is:

$$\bar{Y}_A - \bar{Y}_B = \bar{X}_A\hat{\beta}_A - \bar{X}_B\hat{\beta}_B$$

The classic two-term decomposition (the “twofold” form) rewrites this by adding and subtracting a counterfactual term  $\bar{X}_B\hat{\beta}_A$ :

$$\bar{Y}_A - \bar{Y}_B = \underbrace{(\bar{X}_A - \bar{X}_B)\hat{\beta}_A}_{\text{explained}} + \underbrace{\bar{X}_B(\hat{\beta}_A - \hat{\beta}_B)}_{\text{unexplained}}$$

- The **explained component** is the difference in average characteristics  $(\bar{X}_A - \bar{X}_B)$  weighted by Group A’s returns  $(\hat{\beta}_A)$  — how much of the gap would come purely from compositional differences, if both groups were paid according to Group A’s structure.
- The **unexplained component** is the difference in coefficients  $(\hat{\beta}_A - \hat{\beta}_B)$  weighted by Group B’s average characteristic — how much of the gap comes from different returns to the same characteristics.

This formulation uses  $\hat{\beta}_A$  as the reference structure (the “Group A perspective” variant). An equivalent symmetric formulation exists using  $\hat{\beta}_B$ , and formulations using a weighted combination of both (Reimers’ or Cotton’s proposals, for instance) avoid the arbitrariness of picking one group as the reference — the choice can moderately shift the split between the two components, especially when the two coefficient sets differ a lot.

## Assumptions

1. **Correct specification of the outcome model:** the linear-in- $X_g$  model must reasonably capture the relationship between characteristics and outcome within each group. Important nonlinearities not captured (say, diminishing returns to experience) bias the decomposition.
2. **No relevant omitted-variable bias:** any characteristic that affects the outcome and is correlated with both the group and the other included variables, but wasn't measured, gets absorbed into the unexplained component — artificially inflating it.
3. **Support comparability:** the decomposition presupposes real overlap in the distribution of characteristics between the groups. When one group has almost no observations at certain combinations of characteristics present in the other group, the counterfactual ends up depending on model extrapolation outside the observed data — a form of instability worth checking (by comparing the distributions of  $X$  across groups, say).
4. **No reverse causality between the outcome and the characteristics:** if the characteristic itself is partly determined by the outcome (occupation chosen in response to wage expectations, say), the interpretation of “how much  $X$  explains” gets trickier.

## Hypotheses

The decomposition itself isn't a single hypothesis test — it's a descriptive technique. But each component can (and should) have its own standard error and significance test, usually obtained via the delta method or bootstrap:

- $H_0$  for the explained component: the explained share of the difference equals zero.
- $H_0$  for the unexplained component: the unexplained share equals zero.
- Standard errors: since the decomposition involves products of estimated quantities ( $\bar{X}$  and  $\hat{\beta}$ ), analytically deriving the standard error of each component isn't trivial — bootstrapping (resampling both groups and recomputing the entire decomposition at each resample) is the more robust approach and the one most widely used in practice.

## Interpretation

The explained component has a relatively straightforward reading: it's the part of the gap that can be attributed to observable compositional differences between the groups, given the reference structure of returns chosen.

The unexplained component demands much more caution. It's common — and problematic — to describe it directly as “discrimination.” That interpretation is only valid under the strong assumption that every characteristic relevant to determining the outcome was measured and included in the model. In practice, that's almost never true: unobserved skill, network quality, differences in salary negotiation, bias in hiring and promotion processes, and many other omitted variables can sit inside the unexplained component alongside any genuine discrimination effect. The unexplained component is better read as an approximate upper bound on what can be attributed to factors the model doesn't capture — including, but not limited to, discrimination.

Moreover, the method doesn't establish causality in the characteristics themselves: saying “education explains  $X$  units of the gap” doesn't mean raising one group's education would necessarily close that part of the gap — it assumes a constant general equilibrium and ignores market-composition effects that could shift alongside it.

## Limitations

- **Sensitivity to specification:** adding or removing control variables can substantially shift the split between explained and unexplained — it’s important to report how the result changes across different specifications (a robustness check).
- **Only decomposes the mean:** to understand whether the gap changes across the distribution (larger at the tails, say), a mean decomposition isn’t enough — see RIF decomposition, which extends this logic to other points of the distribution.
- **Ambiguity in the choice of reference group:** different conventions (Group A reference, Group B reference, combined reference) produce numerically different results even from the same data — always report which convention was used.
- **Doesn’t correct for selection:** if who ends up in the sample (who’s employed, say) is itself the result of a selective process related to the group, the decomposition inherits that selection bias.

## Example

Consider a hypothetical scenario: an organization wants to understand the difference in average monthly salary between two departments, Department A and Department B, controlling for years of experience.

Aggregate data (invented):

- Department A: average wage = 5,800, average experience = 8 years.
- Department B: average wage = 4,620, average experience = 5 years.
- Fitted model for Department A: wage = 3,800 + 250 × experience.
- Fitted model for Department B: wage = 3,570 + 210 × experience.

**Step 1 — total gap:**

$$5,800 - 4,620 = 1,180$$

**Step 2 — explained component** (using Department A’s return structure as the reference):

$$(\bar{X}_A - \bar{X}_B)\hat{\beta}_{\text{exp},A} = (8 - 5) \times 250 = 750$$

**Step 3 — unexplained component:**

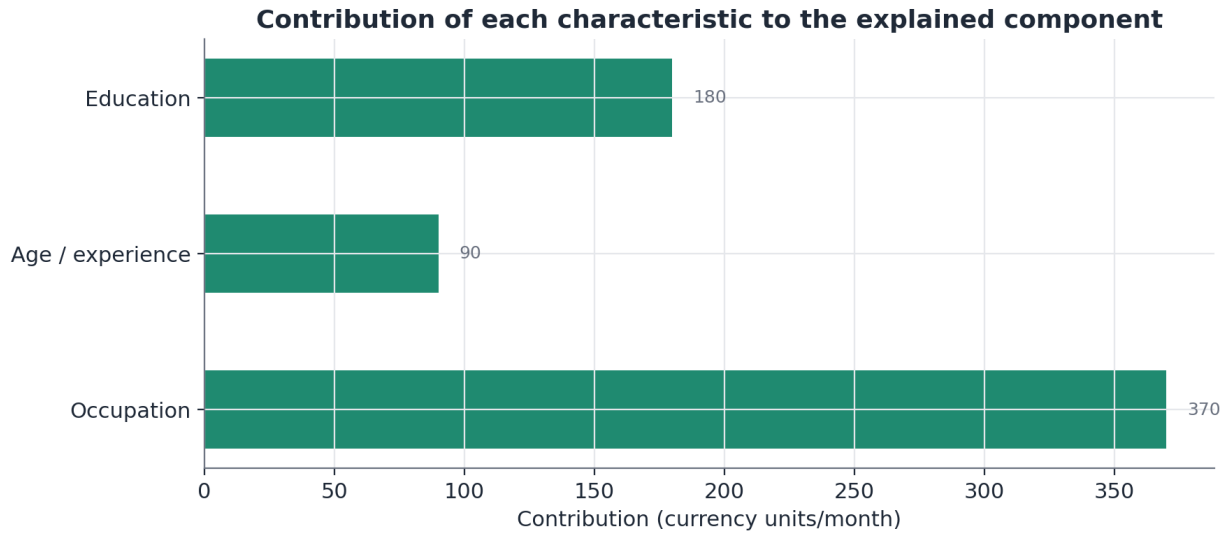
$$1,180 - 750 = 430$$

This can be verified with the coefficient-level formulation: the difference in intercepts (3,800 − 3,570 = 230) plus the difference in slope applied to Department B’s average experience ((250 − 210) × 5 = 200) sums to 430 — consistent with the coefficient-level decomposition.

**Interpretation:** roughly 64% of the wage gap (750 of 1,180) can be attributed to the difference in average experience between the departments, given that both were paid according to Department A’s structure. The remaining 36% reflects both a difference in intercept and a difference in slope (return per year of experience) between the two departments — something that can’t be attributed

to experience itself, but also shouldn't be automatically labeled discrimination without further investigation into other potentially omitted variables (role type within the department, say).

The figure below breaks down the contribution of different hypothetical characteristics to the explained component, in a scenario with more than one control variable:



Illustrative data, not drawn from any real project.

Figure 2: Horizontal bar chart showing the contribution of education, age/experience, and occupation to the explained component

# RIF (Recentered Influence Function) decomposition

## Concept

RIF decomposition, proposed by Firpo, Fortin and Lemieux (2009), generalizes Oaxaca-Blinder decomposition to distributional statistics beyond the mean — quantiles, the Gini index, the variance, or any other functional of the distribution that can be written as the expectation of its recentered influence function.

Before formalizing this, it's worth understanding what an influence function is. In robust statistics, the influence function of an estimator  $\nu(F)$  (a functional applied to the variable's distribution  $F$ ) measures, in a mathematically precise way, how much the value of  $\nu$  would change if the distribution  $F$  were slightly perturbed by an extra unit of probability mass concentrated at a single point  $y$ . Formally:

$$IF(y; \nu, F) = \lim_{\epsilon \rightarrow 0} \frac{\nu((1 - \epsilon)F + \epsilon\delta_y) - \nu(F)}{\epsilon}$$

where  $\delta_y$  is a degenerate distribution putting all its mass on  $y$ . Intuitively, the influence function answers: “if I added one more observation at value  $y$ , with infinitesimal weight, how much would that pull  $\nu$  up or down?” — points with high influence are those whose presence or absence most affects the statistic.

The **recentered** influence function (RIF) is defined as:

$$RIF(y; \nu, F) = \nu(F) + IF(y; \nu, F)$$

The key property — and the reason the method works — is that the expectation of the RIF over the distribution recovers exactly the original functional:

$$\mathbb{E}[RIF(Y; \nu, F)] = \nu(F)$$

This means  $\nu(F)$  (the median, a percentile, the Gini index) can be treated, for regression purposes, as if it were the mean of a transformed variable — the RIF evaluated at each observation. That equivalence is what lets you reuse the entire regression-and-decomposition machinery from Oaxaca-Blinder, now applied to  $RIF(y)$  instead of  $y$  directly.

It's worth stating clearly: RIF decomposition **is not the same method** as Oaxaca-Blinder — it's an extension that uses Oaxaca-Blinder as an internal step, after transforming the data via the RIF. Oaxaca-Blinder decomposes means directly; RIF decomposes any distributional statistic that can be expressed as the expectation of an influence function, using the same counterfactual logic underneath.

## Mathematical formulation

**RIF for a quantile.** For the  $q$ -th order quantile (0.5 for the median, say), with value  $Q_q$  and density  $f_Y(Q_q)$  of the variable  $Y$  evaluated at  $Q_q$ :

$$RIF(y; Q_q) = Q_q + \frac{q - \mathbb{1}\{y \leq Q_q\}}{f_Y(Q_q)}$$

Here,  $\mathbb{1}\{y \leq Q_q\}$  is an indicator equal to 1 if the observation falls below the quantile and 0 otherwise. Observations just below the quantile receive an RIF value slightly above  $Q_q$ ; observations just above receive a value slightly below — the density in the denominator controls the magnitude of that adjustment (the lower the density there, the more sensitive the quantile is to change, and the larger the magnitude of the RIF).

**Regression and decomposition.** Once  $RIF_i = RIF(y_i; \nu)$  is computed for every observation  $i$  in each group, fit:

$$RIF_i = X_i \gamma_g + u_i, \quad g \in \{A, B\}$$

and apply the Oaxaca-Blinder decomposition exactly as in the mean version, but using  $\gamma_g$  in place of  $\beta_g$ :

$$\nu_A - \nu_B \approx \underbrace{(\bar{X}_A - \bar{X}_B) \hat{\gamma}_A}_{\text{explained, at point } \nu} + \underbrace{\bar{X}_B (\hat{\gamma}_A - \hat{\gamma}_B)}_{\text{unexplained, at point } \nu}$$

The approximation sign is there because this is a first-order approximation (via the influence function) — valid locally, not an exact decomposition the way the mean case is.

## Assumptions

1. **Consistent density estimation:** for quantiles, the RIF's denominator depends on the density  $f_Y$  evaluated at the quantile, which needs to be estimated (typically via a kernel). The choice of kernel bandwidth affects the estimate, especially in regions with sparse data.
2. **All Oaxaca-Blinder assumptions apply** to the regression-and- decomposition step on the transformed variable: correct specification, no relevant omitted-variable bias, and real overlap in characteristics between the groups.
3. **Validity of the linear approximation:** RIF decomposition is fundamentally a first-order approximation — exact for small shifts in the distribution, but it can lose precision for comparisons with very large differences between groups. This matters particularly when interpreting changes over long stretches of time, not just between two groups at a single moment.

## Hypotheses

As with Oaxaca-Blinder, each component of the decomposition (explained and unexplained, at each chosen point of the distribution) can be tested individually:

- $H_0$ : the component equals zero at that specific point of the distribution.
- Standard errors: the two-step process (estimating the RIF, then estimating the regression) introduces additional uncertainty that needs to be propagated correctly — bootstrap (resampling the original data and redoing both steps at each resample, including re-estimating the density) is the standard approach, since closed-form analytical formulas for the standard error are complex.

- When reporting several points of the distribution at once (a gap profile by percentile, say), the multiple comparison across points deserves the same caution as any multiple-testing situation — a single “significant” point standing out among several non-significant ones warrants careful interpretation.

## Interpretation

RIF decomposition allows a much richer reading than the mean alone: it shows whether a gap between groups is uniform across the distribution or concentrated in certain regions — a “sticky floor” pattern (larger gap at the bottom) indicates that even among lower earners in both groups, the disparity is large; a “glass ceiling” pattern (larger gap at the top) indicates the disparity grows specifically among those at the top of the income distribution, typical of barriers to reaching higher positions.

The same Oaxaca-Blinder caveats about causality and omitted-variable bias apply, point by point: the unexplained component at any percentile can’t be automatically attributed to discrimination. Moreover, each point has its own margin of error, usually wider at the tails of the distribution — a gap profile that looks dramatically different between two adjacent percentiles may actually fall within their combined margin of error.

## Limitations

- **Less precise estimates at the tails:** the density  $f_Y$  tends to be lower at the extremes of the distribution (fewer observations there), which inflates the RIF formula’s denominator and, consequently, the variance of the estimate.
- **First-order approximation:** for very large differences between groups, or for reweightings far from the observed distribution, the RIF’s linear approximation may not capture nonlinear shifts in the statistic of interest well. Full reweighting methods (like DiNardo-Fortin-Lemieux) are alternatives that avoid this approximation, at the cost of greater computational complexity.
- **Choice of percentile grid:** reporting a limited number of points (P10, P25, P50, P75, P90, say) may miss even more extreme behavior (P1, P99) — the choice of grid should reflect the research question, not convenience.
- **Inherits all of Oaxaca-Blinder’s causal limitations,** point by point: it’s an accounting decomposition, not a causal inference method.

## Example

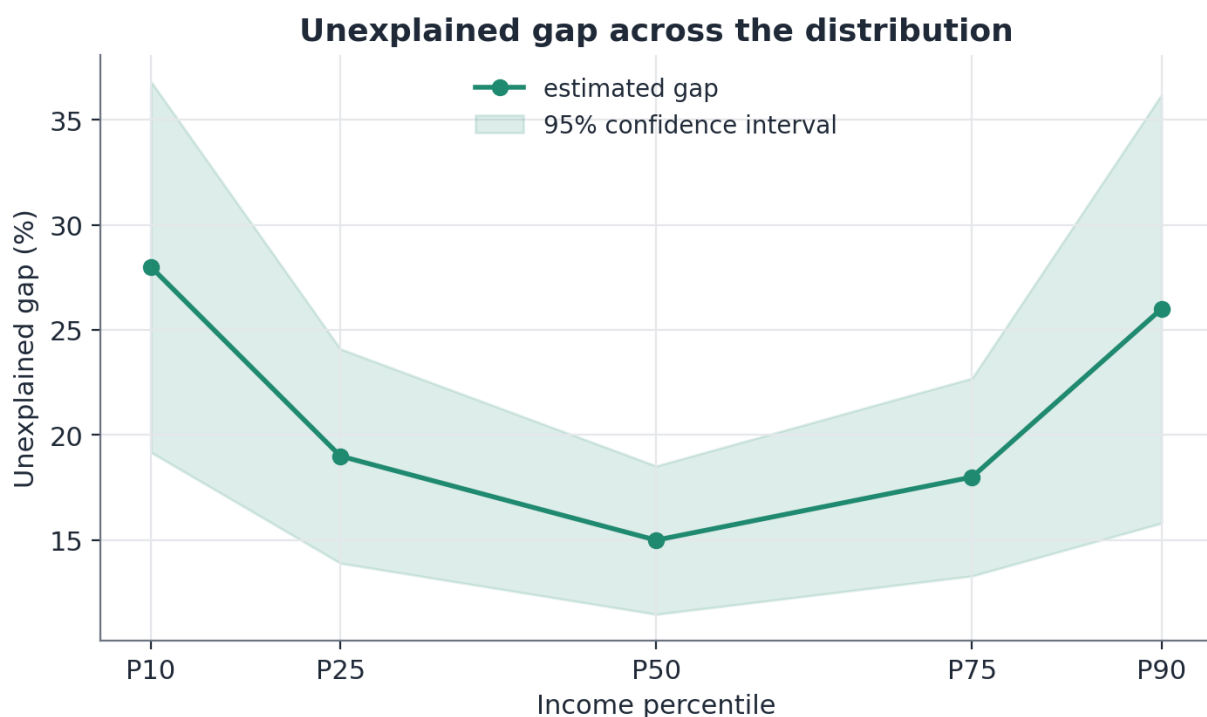
Consider a hypothetical scenario: an annual wage survey collects income data from two regions, North Region and South Region, along with years of education. An Oaxaca-Blinder decomposition at the mean shows an unexplained gap of 15%. RIF decomposition is applied at five percentiles to investigate whether that gap is uniform:

| Percentile | Total gap | Explained gap | Unexplained gap |
|------------|-----------|---------------|-----------------|
| P10        | 34%       | 6 p.p.        | 28%             |
| P25        | 27%       | 8 p.p.        | 19%             |
| P50        | 22%       | 7 p.p.        | 15%             |
| P75        | 25%       | 7 p.p.        | 18%             |
| P90        | 33%       | 7 p.p.        | 26%             |

| Percentile | Total gap | Explained gap | Unexplained gap |
|------------|-----------|---------------|-----------------|
|------------|-----------|---------------|-----------------|

The explained component (attributable to the education difference between the regions) stays fairly stable, between 6 and 8 percentage points, across the distribution. The unexplained component, however, follows a U-shaped pattern: 28% at P10, dropping to 15% at P50, and climbing back to 26% at P90. This suggests the education difference explains a similar slice of the gap at every income level, but what's left over — the unexplained part — is considerably larger both among lower earners and higher earners, and smaller in the middle of the distribution.

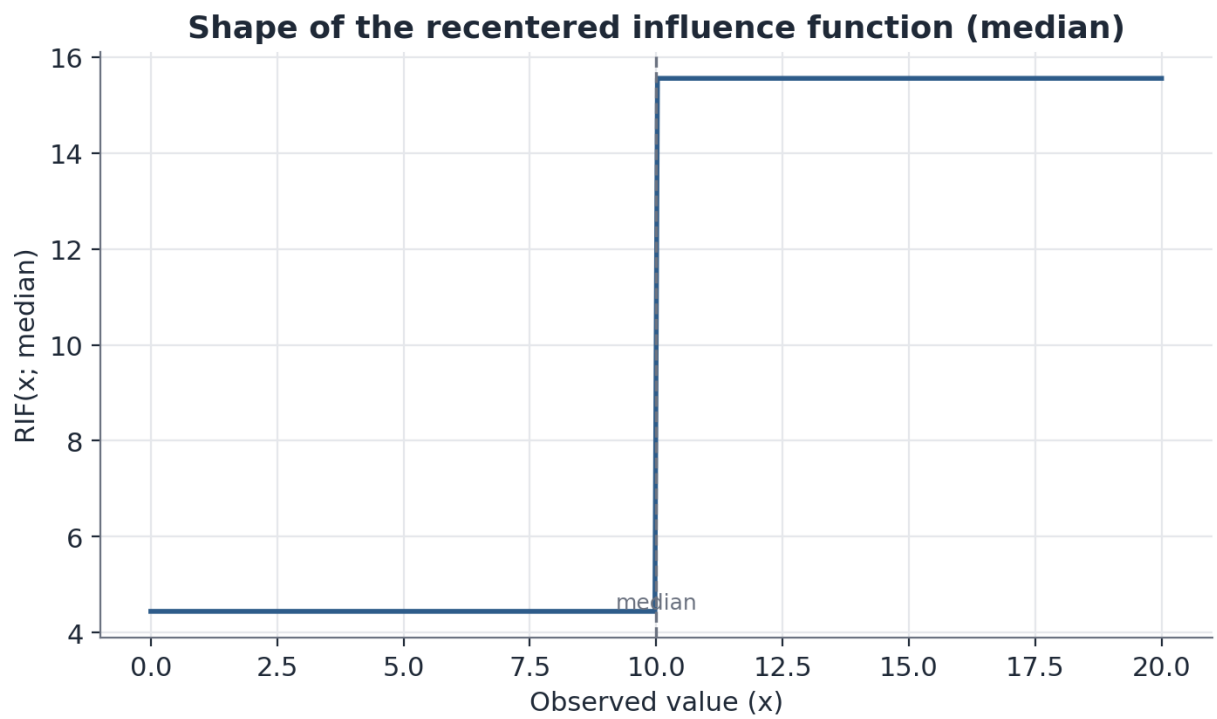
A result like this is editorially meaningful: reporting only the mean figure (15%) would hide that the experience of people at the bottom or top of the income distribution is quite different from the experience of people in the middle.



Illustrative data, not drawn from any real project.

Figure 3: Line chart showing the estimated unexplained gap at each percentile, with a wider 95% confidence interval at the tails

The following figure illustrates the shape of the recentered influence function itself for the median — the characteristic “step” around the median value, which explains why observations just below and just above the median receive opposite weights in the regression:



Illustrative data, not drawn from any real project.

Figure 4: Step-shaped recentered influence function around the median value

# Inequality decomposition (Theil, Gini, weighted quantiles)

## Concept

Measuring income inequality with a single number (like the mean or the median) hides the entire shape of the distribution. This section covers three complementary tools for characterizing inequality more completely: the Theil index (with its property of exact decomposability into “between-group” and “within-group” components), the Gini index (a widely used concentration measure based on the Lorenz curve), and weighted quantiles (estimates of distribution cutoff points — like “what defines the richest 10%” — that account for a survey’s sampling design).

The question these tools answer, together, is: where does observed inequality come from, and how does it compare across different slices of the population?

## Mathematical formulation

**Theil index (T index, entropy-based).** For an income variable  $y_i > 0$  with mean  $\bar{y}$ , computed over  $n$  observations:

$$T = \frac{1}{n} \sum_{i=1}^n \frac{y_i}{\bar{y}} \ln \left( \frac{y_i}{\bar{y}} \right)$$

$T = 0$  corresponds to perfect equality (everyone has the same income); higher values indicate greater inequality. Unlike the Gini index, Theil has no fixed upper bound of 1.

**Exact between/within decomposition.** Splitting the population into  $G$  groups, each with income share  $s_g$  and internal Theil  $T_g$ :

$$T = \underbrace{\sum_{g=1}^G s_g \ln \left( \frac{\bar{y}}{\bar{y}_g} \right)}_{T_{\text{between}}} + \underbrace{\sum_{g=1}^G s_g T_g}_{T_{\text{within}}}$$

where  $\bar{y}_g$  is group  $g$ ’s average income.  $T_{\text{between}}$  is the value Theil would take if every person received their group’s average income (capturing only the difference between group means);  $T_{\text{within}}$  is the weighted average of the Theil indices computed within each group, weighted by each group’s income share. The sum of the two equals exactly the total Theil index — there’s no residual term, which is the central property that makes Theil especially useful for this kind of decomposition (the Gini index, by comparison, only decomposes additively under special conditions, generally leaving an extra residual overlap term between groups).

**Gini index.** For an ordered sample  $y_{(1)} \leq y_{(2)} \leq \dots \leq y_{(n)}$ :

$$G = \frac{2 \sum_{i=1}^n i \cdot y_{(i)} - (n+1) \sum_{i=1}^n y_{(i)}}{n \sum_{i=1}^n y_{(i)}}$$

Geometrically,  $G$  is twice the area between the Lorenz curve (which shows the cumulative income share held by the poorest  $p\%$ ) and the line of perfect equality (the  $45^\circ$  diagonal).  $G = 0$  is perfect equality;  $G \rightarrow 1$  is extreme concentration.

**Weighted quantiles.** Given a set of sampling weights  $w_i$  (reflecting how many units of the population each sample observation represents, given the sampling design), the  $q$ -th order quantile is the value  $Q_q$  such that the sum of weights of observations with  $y_i \leq Q_q$ , divided by the total sum of weights, equals  $q$ . Ignoring the weights and computing a simple sample quantile produces a biased estimate whenever the selection probability isn't constant across observations.

## Assumptions

1. **Strictly positive values** (Theil): the T index formula requires  $y_i > 0$  for every observation — zero or negative incomes require special handling (exclusion with an explicit methodological note, say, or using a Theil variant less sensitive to it, like the L index / mean logarithmic deviation).
2. **Mutually exclusive and exhaustive groups:** the Theil between/within decomposition assumes every observation belongs to exactly one group — overlapping categorizations break the exact decomposition.
3. **Representativeness of the sampling weights:** weighted quantiles and indices computed from survey data depend on the quality of the design weights (and, where applicable, post-stratification) — poorly calibrated weights produce biased estimates regardless of the formula used.
4. **Stability in small subgroups:** when decomposing across many groups at once (crossing region and race, say), some subgroups may have few observations, making that subgroup's  $T_g$  unstable — it's worth reporting each subgroup's sample size.

## Hypotheses

Inequality indices like Theil and Gini don't traditionally have a single, universally accepted hypothesis test the way a difference in means does — but comparisons between indices (is Group A's Gini larger than Group B's?, say) can and should be accompanied by uncertainty measures:

- Standard errors for Gini and Theil are typically obtained via bootstrap (resampling the data with replacement, respecting the sampling design where applicable, and recomputing the index at each resample).
- $H_0$ : the two indices (Group A's Gini and Group B's Gini, say) are equal — tested by comparing the observed difference against the bootstrap distribution of the difference.
- For the Theil decomposition, the “between-groups” component can have its own confidence interval via bootstrap, but by construction it's always non-negative — it's worth reporting both its value in absolute Theil units and its share of the total.

## Interpretation

The Theil between/within decomposition contextualizes — but doesn't replace — a direct comparison of means between groups. It's entirely possible, and common, for a small fraction of total inequality (below 10%, say) to come from the difference between broad groups like race or gender, even when that difference in means is large and highly statistically significant (see Welch's t-test and Oaxaca-Blinder). The two things aren't in competition: the “between-groups” share speaks to the total variance of income in the population; the average gap between groups speaks to the typical positional difference between them. Both readings are valid and should be presented together, not one in place of the other.

On the Gini index: comparing Gini across groups requires similar care. A group can have a lower Gini simply because its income distribution is more compressed near the bottom — that is, almost everyone in that group earns relatively similar, low amounts. That isn’t “more equality” in the positive sense; it’s compression within a low-income range. A lower Gini shouldn’t be read, on its own, as a “better situation” without also looking at that group’s income level (its mean or median).

## Limitations

- The Theil index, despite being exactly decomposable between and within groups, is sensitive to how the groups are defined — carving up the population differently (two broad groups vs. ten specific subgroups) changes the fraction attributed to “between groups.”
- Neither index, on its own, says why inequality within a group is high — that requires further investigation (decomposition by other variables, or RIF decomposition across the distribution, say).
- Comparisons of Gini or Theil over time can be affected by changes in survey coverage, the definition of the income variable, or inflation adjustments — it’s worth ensuring methodological comparability before interpreting a change as a real change in inequality.
- Weighted quantiles in small subgroups (low sample-observation counts, even if the population weight is large) carry more uncertainty — always check the sample size behind a reported quantile.

## Example

Consider a hypothetical scenario: a household survey measures income across two regions of a country, North Region and South Region, with the following aggregate data (invented):

- North Region: average income = 3,200, share of total income = 35%, internal Theil = 0.28.
- South Region: average income = 5,100, share of total income = 65%, internal Theil = 0.33.

**Step 1 — between-groups Theil.** Using the decomposition formula, with the overall average income  $\bar{y}$  (a weighted average by each region’s income share, coming out to about 4,430):

$$T_{\text{between}} = 0.35 \ln \left( \frac{4,430}{3,200} \right) + 0.65 \ln \left( \frac{4,430}{5,100} \right) \approx 0.35 \times 0.325 - 0.65 \times 0.140 \approx 0.022$$

**Step 2 — within-groups Theil:**

$$T_{\text{within}} = 0.35 \times 0.28 + 0.65 \times 0.33 = 0.098 + 0.215 = 0.313$$

**Step 3 — total Theil:**

$$T = T_{\text{between}} + T_{\text{within}} = 0.022 + 0.313 = 0.335$$

**Interpretation:** the “between-regions” share accounts for about  $0.022 / 0.335 \approx 6.6\%$  of total inequality — even with an income difference of nearly 60% between the two regions (3,200 vs. 5,100). The vast majority of inequality (93.4%) sits within each region, reflecting that, within the North Region and the South Region separately, there’s already substantial variation between low and high earners. That doesn’t make the regional difference unimportant — it’s substantial and measurable — but it contextualizes its relative weight within the country’s overall inequality picture.

The figure below illustrates this kind of decomposition, with the “between-groups” share typically small relative to the “within-groups” share:

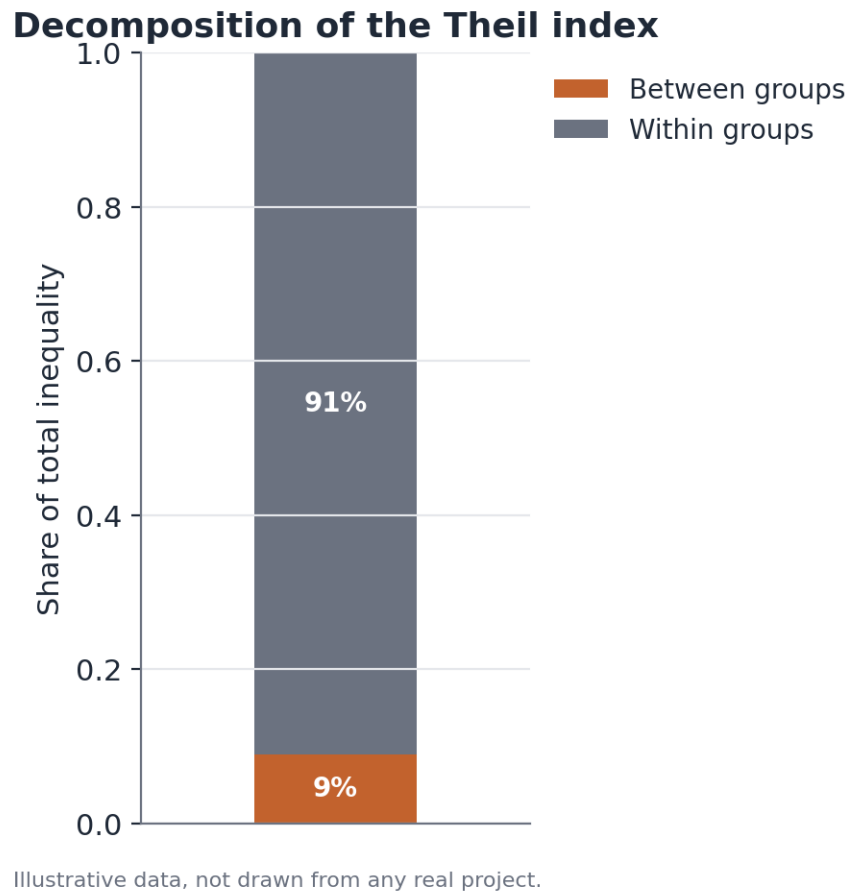
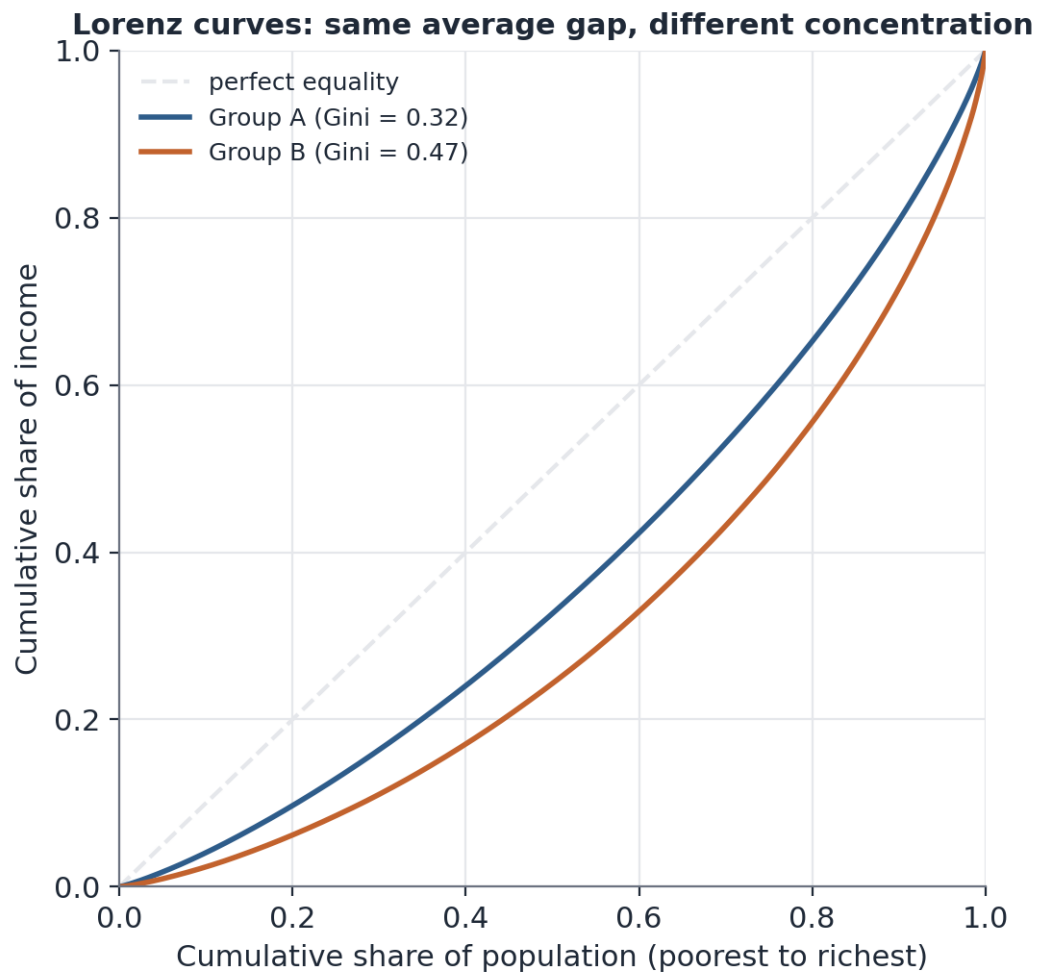


Figure 5: Stacked bar showing the Theil index decomposition between a small between-groups share and a large within-groups share

The second figure shows hypothetical Lorenz curves for two groups with different Ginis — illustrating that a lower Gini corresponds to a curve closer to the perfect-equality diagonal, but that, on its own, says nothing about each group’s income level:



Illustrative data, not drawn from any real project.

Figure 6: Lorenz curves for two hypothetical groups with Ginis of 0.32 and 0.47, showing different degrees of concentration

# Duncan Dissimilarity Index

## Concept

The Duncan dissimilarity index (Duncan & Duncan, 1955) quantifies how much two percentage distributions — for two population groups spread across the same set of categories — differ from each other. It’s the classical segregation measure: occupational, residential, educational, or any other discrete categorization where it makes sense to ask “do the two groups occupy the same categories, in the same proportions?”

The core intuition is geometric: if you draw the two distributions as overlapping bars, the index measures the non-overlapping area, divided by two. Equivalently, it’s the minimum fraction of one group that would need to be reallocated across categories for the two distributions to match exactly.

The index belongs to a broader family of segregation measures (alongside the isolation index, the exposure index, and more recent multigroup segregation indices), but it remains the most widely used because of its simple calculation and direct interpretation.

## Mathematical formulation

Let two populations, A and B, be distributed across a set of  $K$  mutually exclusive categories. For each category  $i = 1, \dots, K$ , define:

- $a_i$  = the share of population A in category  $i$  (sums to 1 across all categories)
- $b_i$  = the share of population B in category  $i$  (sums to 1 across all categories)

The dissimilarity index is:

$$D = \frac{1}{2} \sum_{i=1}^K |a_i - b_i|$$

Where:

- $|a_i - b_i|$  is the absolute share difference in category  $i$ ;
- the sum accumulates this difference across all  $K$  categories;
- the  $\frac{1}{2}$  factor normalizes the index to the  $[0, 1]$  range — without it, the sum of absolute differences would range from 0 to 2, because every point of “excess” for A in one category corresponds to a point of “deficit” for A (excess for B) in another, counted twice across the sum.

$D = 0$  occurs when  $a_i = b_i$  for every category  $i$  — the two distributions are identical.  $D = 1$  occurs when, for every category where  $a_i > 0$ ,  $b_i = 0$  holds and vice versa — the two populations never share a category.

An equivalent reading, useful in practice:  $D$  is also the maximum, over any subset of categories  $S$ , of the difference between the share of A and the share of B contained in  $S$ :

$$D = \max_{S \subseteq \{1, \dots, K\}} \left| \sum_{i \in S} a_i - \sum_{i \in S} b_i \right|$$

This alternative formulation clarifies the “minimum fraction to reallocate” interpretation:  $D$  is exactly the share of one group that is “in excess” in the worst possible subset of categories.

## Assumptions

The Duncan index is not a hypothesis test — it’s a descriptive statistic — but using it correctly depends on a few conditions:

- **Mutually exclusive, collectively exhaustive categories:** every observation belongs to exactly one category. Overlapping categories break the index’s logic.
- **Stable classification across the groups compared:** comparing indices computed with different categorization schemes (say, a 6-category occupational classification versus a 20-category one) is not valid —  $D$  tends to rise mechanically with the number of categories, simply because there are more opportunities for a difference to show up.
- **Sufficient sample within each category, for each group:** with few observations, the shares  $a_i$  and  $b_i$  are estimated with considerable noise, and the index inherits that instability. A bootstrap confidence interval (resampling individuals within each group, recomputing  $D$  on each resample) is the most direct way to quantify this uncertainty, since the exact sampling distribution of  $D$  has no simple closed form.
- **Independence within the sampling design:** if the data come from a complex design (stratified, clustered, with sampling weights), the shares  $a_i$  and  $b_i$  should be computed using the correct weights — ignoring the sampling design produces biased shares (and therefore a biased  $D$ ).

## Hypotheses

The index itself does not come with a formal hypothesis test — it is possible, however, to informally test whether  $D$  is “larger than expected by chance” by comparing it to the distribution of  $D$  obtained by randomly shuffling category assignment between individuals from the two groups (a permutation test). In that case:

- $H_0$ : the category an individual occupies does not depend on which group they belong to (under  $H_0$ , the two populations would be samples from the same categorical distribution).
- $H_1$ : there is an association between group and category.
- Test statistic: the observed  $D$  itself.
- p-value: the fraction of permutations in which the simulated  $D$  is at least as large as the observed  $D$ .

In practice, with large samples (like national household surveys), the permutation test will almost always reject  $H_0$  even for small values of  $D$  — which reinforces that the relevant question isn’t “is segregation statistically detectable?” but “how much segregation is there, and is it large enough to matter in practice?”

## Interpretation

$D$  answers a purely descriptive question: **how unequal is the distribution of two groups across categories.** It does not tell you:

- **Why** that distribution exists. A high  $D$  is consistent with access barriers, real and unobserved differences in qualifications between the groups, distinct voluntary preferences, accumulated

historical discrimination, or any combination of these causes — the index does not separate one explanation from another.

- **Whether segregation is rising or falling in a causal sense.** A drop in  $D$  over time is consistent with genuine integration, but it can also reflect changes in the category classification itself or in the composition of the workforce.
- **Which group is better off.**  $D$  has no sign — it measures distance, not direction. To know which group is concentrated in the better-paid or higher-prestige categories, you need to look directly at the table of category shares, not just the aggregate index.

A recommended practice is to always report  $D$  alongside the table of category shares (or, at minimum, highlight the categories that contribute most to the difference) — the index alone compresses too much information to carry a complete narrative on its own.

## Limitations

- **Sensitivity to category granularity.** As mentioned,  $D$  tends to grow mechanically with the number of categories. Comparisons across studies or countries are only valid if the category classification is equivalent.
- **Does not incorporate ordering or distance between categories.** Two “close” categories (say, two adjacent salary bands) count the same as two “distant” categories (the lowest and the highest band). When ordering matters, alternative measures (such as vertical segregation indices, or a Theil decomposition applied by category) can be more informative.
- **Not simply decomposable across subgroups or over time,** the way the Theil index is for inequality — comparing  $D$  across subpopulations or years requires recomputing the index for each cut, and there is no direct way to attribute the total change to specific factors.
- **Estimation under complex sampling weights requires care.** Standard errors for  $D$  under complex sampling designs have no simple closed form; bootstrap or replicate-weight methods (repeated jackknife, balanced repeated replication) are the standard approaches.

## Example

Consider a hypothetical study comparing the distribution of technology professionals at a mid-sized company, split by gender, across six seniority levels: Intern, Junior, Mid-level, Senior, Staff, and Technical Lead.

Suppose the following observed shares:

| Level          | % Group A | % Group B |
|----------------|-----------|-----------|
| Intern         | 8%        | 14%       |
| Junior         | 22%       | 30%       |
| Mid-level      | 30%       | 28%       |
| Senior         | 24%       | 18%       |
| Staff          | 11%       | 7%        |
| Technical Lead | 5%        | 3%        |

Computing the absolute differences by level: 6, 8, 2, 6, 4, 2. The sum is 28. Dividing by two,  $D = 14$ , or 0.14 on a 0-to-1 scale.

Interpretation: about 14% of one group would need to change seniority level for the two distributions to coincide. Looking at the table, it's clear that most of this difference comes from the entry end (Intern and Junior, where Group B is overrepresented) and the top end (Senior and above, where Group A is overrepresented) — a pattern consistent with a “narrowing pipeline” hypothesis for career progression, but one the index alone does not confirm or explain. Investigating whether this pattern is causal (say, tied to a specific promotion barrier) would require a complementary design — such as a survival analysis of time-to-promotion, controlling for tenure and performance ratings — which is outside the scope of the dissimilarity index.

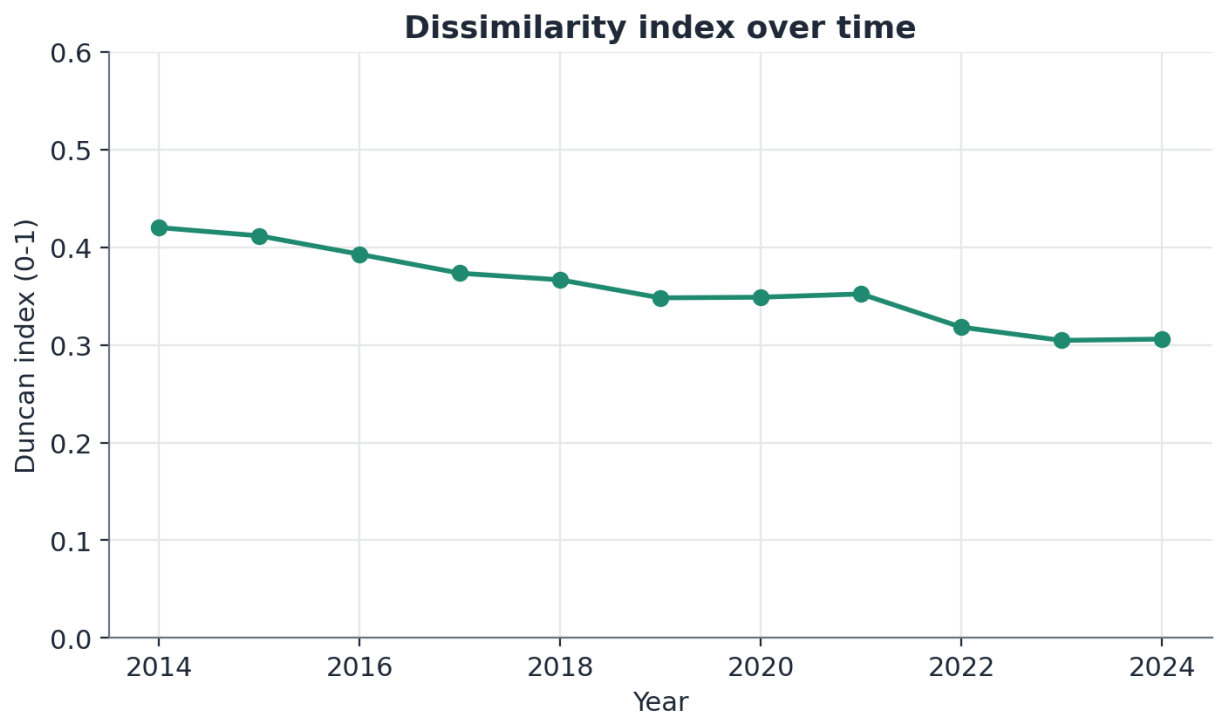
A simple permutation test, randomly shuffling seniority-level assignment 500 times between employees of the two groups (holding group sizes fixed), would in this hypothetical example produce a very low p-value — confirming that this observed distribution would be rare under the hypothesis that gender and level are independent, but, as always, that does not substitute for an investigation of causes.

```
import numpy as np
import pandas as pd

levels = ["intern", "junior", "mid", "senior", "staff", "lead"]
share_a = np.array([0.08, 0.22, 0.30, 0.24, 0.11, 0.05])
share_b = np.array([0.14, 0.30, 0.28, 0.18, 0.07, 0.03])

D = 0.5 * np.sum(np.abs(share_a - share_b))
print(f"Duncan index: {D:.3f}")

# permutation test (sketch): shuffle level assignment
# among simulated individuals from both groups, recompute D,
# compare against the observed D to get an empirical p-value.
```



Illustrative data, not drawn from any real project.

Figure 7: Evolution of the dissimilarity index over time

# Structural Break Test (Chow-Type)

## Concept

The Chow test (Chow, 1960) checks whether a statistical model — typically a linear regression — has stable coefficients across an entire time series, or whether those coefficients change at a specific point, known a priori. It's the canonical tool for answering “did something structurally different happen starting from this known date?”, in contrast with methods that detect multiple breaks at unknown dates (such as Bai-Perron), which solve a broader and more data-demanding problem.

The logic is that of a nested F-test: a “restricted” model, which imposes the same coefficients before and after the cutoff date, is compared to an “unrestricted” model, which allows different coefficients in each of the two periods. If the unrestricted model reduces the fitting error by more than would be expected purely from the mechanical gain of having more free parameters, that's evidence against the stability hypothesis.

## Mathematical formulation

Consider the linear regression model:

$$y_t = \beta_0 + \beta_1 x_t + \varepsilon_t, \quad t = 1, \dots, n$$

Where  $y_t$  is the variable of interest at time  $t$ ,  $x_t$  are the explanatory variables,  $\beta_0$  and  $\beta_1$  are the coefficients to estimate, and  $\varepsilon_t$  is the random error.

Let  $t^*$  be the a priori known break point, splitting the sample into two subperiods of sizes  $n_1$  (before  $t^*$ ) and  $n_2$  (from  $t^*$  onward), with  $n_1 + n_2 = n$ .

Define:

- $RSS_p$  — the residual sum of squares of the model fitted to the **entire** sample (the restricted model, which assumes equal coefficients in both periods).
- $RSS_1$  — the residual sum of squares of the model fitted to the first subperiod only.
- $RSS_2$  — the residual sum of squares of the model fitted to the second subperiod only.
- $k$  — the number of parameters estimated in each regression (here,  $k = 2$ : intercept and slope).

The Chow test statistic is:

$$F = \frac{(RSS_p - (RSS_1 + RSS_2)) / k}{(RSS_1 + RSS_2) / (n - 2k)}$$

Where:

- the numerator measures how much the error dropped by allowing different coefficients in the two periods, adjusted by the additional degrees of freedom used ( $k$ );
- the denominator is the average residual error of the unrestricted model, adjusted by the remaining degrees of freedom ( $n - 2k$ ).

Under the null hypothesis of coefficient stability,  $F$  approximately follows an  $F(k, n - 2k)$  distribution.

## Assumptions

- **The break date is known before looking at the data.** This is the most important assumption, and the one most often violated in practice. If the date is chosen by visually inspecting the series (looking for where it “seems” to have changed), the resulting p-value no longer carries the standard interpretation — the correct procedure in that case is different (unknown-break tests, such as Andrews’ supF, or multiple-break methods).
- **Homoscedastic, non-autocorrelated errors within each subperiod,** the standard condition for ordinary least-squares regression. When this assumption fails — common in economic time series — the standard errors underlying the F-test can be wrong; robust forms of the test (using heteroscedasticity- and autocorrelation-consistent standard errors, such as Newey-West) are preferable in those cases.
- **A minimum sample size in each subperiod.** With  $n_1$  or  $n_2$  small relative to  $k$ , the estimate in each segment becomes unstable and the test loses power — the ability to detect a real break that does exist drops proportionally.
- **Correct functional form of the model in each period.** The test compares the stability of a specific model’s coefficients (say, linear); if the true relationship changes in a nonlinear way the linear model doesn’t capture, the test may fail to detect a real change, or detect a change that’s actually just a specification problem.

## Hypotheses

- $H_0$ : the coefficients  $\beta_0$  and  $\beta_1$  are the same before and after  $t^*$  — there is no structural break at the tested date.
- $H_1$ : at least one coefficient differs between the two periods.
- Test statistic:  $F$ , as defined above.
- Reference distribution under  $H_0$ :  $F(k, n - 2k)$ .
- Conventional significance level:  $\alpha = 0.05$ , though context (sample size, number of tests run) should inform this choice.
- Decision rule: reject  $H_0$  (evidence of a break) when the p-value associated with  $F$  is below  $\alpha$ .

## Interpretation

Rejecting  $H_0$  is evidence that the data-generating process changed pattern at the tested date — not necessarily that the tested date **caused** the change. The Chow test evaluates temporal coincidence with statistical precision, but causal attribution requires more:

- **Isolating concurrent events.** If another relevant event occurred close to the same date (a simultaneous currency crisis alongside a regulatory change, for instance), the test cannot distinguish which of the two — or whether both combined — produced the observed change.
- **Distinguishing a level shift from a slope shift.** A break can show up as a discrete jump in the series’ level (the intercept changes, the slope stays the same), as a change in the growth rate (the slope changes), or as both at once. The test’s specification (which coefficients are left free before/after) determines what kind of change can be detected — it’s always worth reporting not just significance, but the magnitude and direction of the change in each coefficient.
- **Statistical significance is not the same as practical magnitude.** With large samples, small changes with no practical relevance can produce very low p-values. Reporting the estimated size of the change — not just the p-value — is essential for an honest interpretation.

## Limitations

- **This is not a method for detecting unknown breaks.** The Chow test assumes  $t^*$  is given externally. Applying it repeatedly, testing several candidate dates and reporting only the most significant one, seriously inflates the false-positive rate (it’s effectively an uncorrected multiple-comparisons problem) — for that use case, the correct procedure is a method designed for unknown break-point search.
- **Does not control for concurrent events in the same time window.** A significant result is compatible with multiple simultaneous causal explanations; the test alone does not separate one from another.
- **Sensitive to model specification.** A poorly specified functional form can produce a “spurious” break (the linear model fits both periods poorly in different ways, even without a real change in the process) or mask a real one.
- **Limited power with little data in either subperiod.** A break near the beginning or end of the series is particularly hard to detect with confidence.

## Example

Consider a hypothetical study on the participation rate of a specific demographic group in a professional training program, measured monthly over 40 months. A new incentive policy, with a known start date, took effect in month 25.

Before the policy (months 1-24), the participation rate grows at an estimated average pace of 0.15 percentage points per month. After the policy (months 25-40), the estimated pace rises to 0.55 percentage points per month, and the series’ average level also rises noticeably.

Fitting the three models:

- Restricted model (the entire series, a single trend coefficient):  $RSS_p = 210.4$
- First-subperiod model:  $RSS_1 = 62.1$
- Second-subperiod model:  $RSS_2 = 58.7$

With  $k = 2$  and  $n = 40$ :

$$F = \frac{(210.4 - (62.1 + 58.7))/2}{(62.1 + 58.7)/(40 - 4)} = \frac{44.8}{3.36} \approx 4.35$$

Comparing against the  $F(2, 36)$  distribution, this statistic corresponds to a p-value of approximately 0.021 — below 0.05.  $H_0$  is rejected: there is statistical evidence that the series’ pattern changed in month 25.

Careful interpretation: this result is consistent with the incentive policy having changed the series’ pattern, but it does not prove it on its own. If, say, a mass-media awareness campaign also started around month 25, coincidentally or not, the Chow test on its own cannot separate the effect of one from the other — that separation would require an additional design, such as comparing the same series across regions that received the policy but not the campaign, and vice versa.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
```

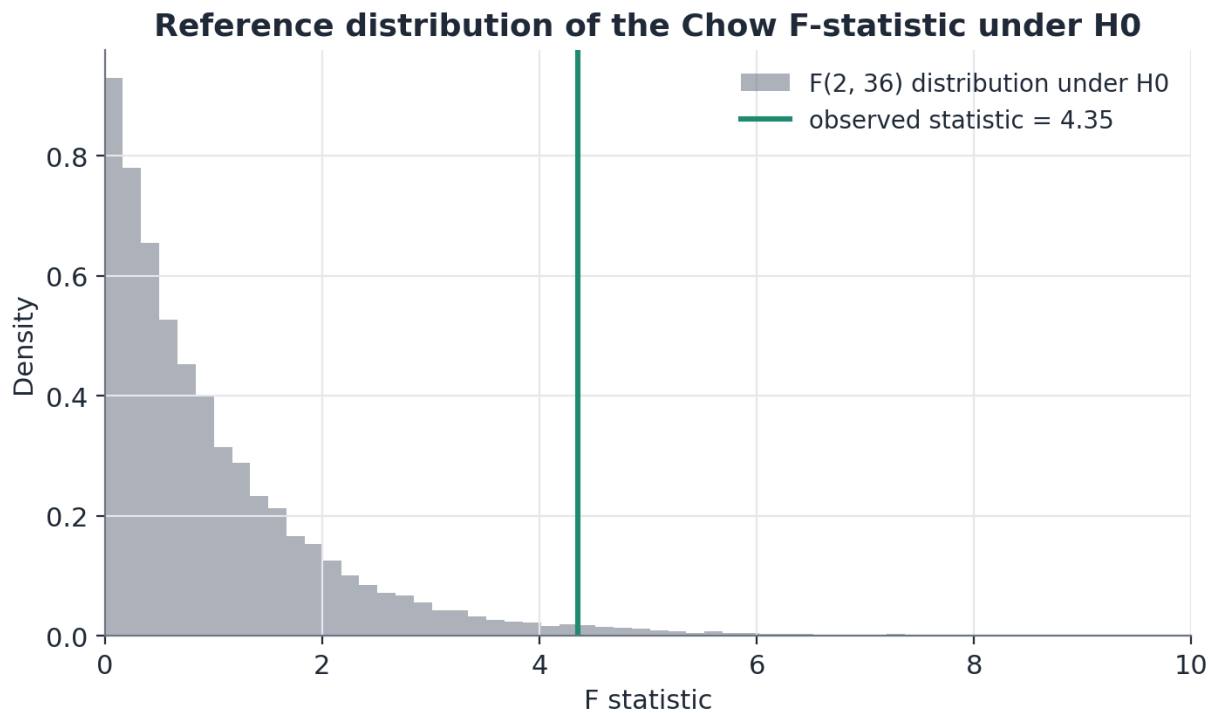
```

def chow_test(y, x, break_index):
    X = sm.add_constant(x)
    rss_p = sm.OLS(y, X).fit().ssr

    X1 = sm.add_constant(x[:break_index])
    X2 = sm.add_constant(x[break_index:])
    rss1 = sm.OLS(y[:break_index], X1).fit().ssr
    rss2 = sm.OLS(y[break_index:], X2).fit().ssr

    k = X.shape[1]
    n = len(y)
    f_stat = ((rss_p - (rss1 + rss2)) / k) / ((rss1 + rss2) / (n - 2 * k))
    p_value = 1 - stats.f.cdf(f_stat, k, n - 2 * k)
    return f_stat, p_value

```



Illustrative data, not drawn from any real project.

Figure 8: Reference distribution of the Chow statistic under H0, with the observed value

# Difference-in-Differences (DiD): Cohort Design, Event-Study, Placebo, and Parallel Trends

## Concept

Difference-in-differences (DiD) is a causal-inference design for observational data, applicable when an intervention (law, policy, event) affects a specific group at a known moment, while another comparable group remains unaffected. The central idea is to use the unaffected group to estimate what would have happened to the affected group **absent** the intervention — the counterfactual — and attribute the gap between the observed outcome and that counterfactual to the intervention's effect.

The design's strength comes from comparing **changes**, not levels. This automatically wipes out any fixed difference between the groups that already existed before the intervention (say, one group already earning more than the other for unrelated historical reasons), and any shock that affects both groups equally at the same time (a nationwide recession, for instance). What remains, under the method's central assumption, is the intervention's specific effect on the affected group.

This document covers four components of the same design, treated together because they rarely appear in isolation in practice: the basic 2x2 DiD design, the extension to **exposure cohorts**, the extension to **event-study**, and the **placebo and parallel-trends** tests that support (or fail to support) the design's validity.

## Mathematical formulation

### Basic 2x2 DiD

With two groups (treated  $T$ , control  $C$ ) and two periods (pre  $t = 0$ , post  $t = 1$ ), the standard regression specification is:

$$y_{it} = \alpha + \beta \cdot \text{Treated}_i + \gamma \cdot \text{Post}_t + \delta \cdot (\text{Treated}_i \times \text{Post}_t) + \varepsilon_{it}$$

Where:

- $y_{it}$  is the outcome of interest for unit  $i$  in period  $t$ ;
- $\text{Treated}_i$  is a (0/1) indicator of belonging to the affected group;
- $\text{Post}_t$  is a (0/1) indicator of the period being after the intervention;
- $\alpha$  is the control group's average level in the pre period;
- $\beta$  captures the fixed difference between groups that already existed before the intervention;
- $\gamma$  captures the change common to both groups between the two periods (the background trend);
- $\delta$ , the interaction coefficient, is the **DiD estimate**: the additional change specific to the treated group, beyond the common trend — the quantity of interest.

Equivalently, without regression:

$$\hat{\delta} = (\bar{y}_{T,\text{post}} - \bar{y}_{T,\text{pre}}) - (\bar{y}_{C,\text{post}} - \bar{y}_{C,\text{pre}})$$

the difference of the two mean differences, which gives the method its name.

## Exposure cohort design

When an intervention affects people differently depending on when they were born, entered some system, or were exposed, the cohort design replaces the binary  $\text{Treated}_i$  indicator with a set of cohort indicators  $c \in \{1, \dots, C\}$ , each defined by a birth or entry window relative to the intervention's cutoff date:

$$y_{ic} = \alpha + \sum_{c \neq c_0} \theta_c \cdot \mathbb{1}[\text{cohort}_i = c] + \mathbf{X}'_i \beta + \varepsilon_i$$

Where  $c_0$  is the reference cohort (typically the one closest to the cutoff date, on the unexposed side),  $\theta_c$  is the estimated effect for cohort  $c$  relative to the reference cohort, and  $\mathbf{X}_i$  are observable controls (age, region, etc.) that should not, in theory, vary systematically across cohorts near the cutoff date — if they do, that's a sign of a design problem (see Limitations).

## Event-study

The event-study generalizes the basic DiD to multiple periods, estimating a separate effect for each period relative to the event date, instead of a single aggregate coefficient:

$$y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \delta_k \cdot \mathbb{1}[t - t_i^* = k] \cdot \text{Treated}_i + \varepsilon_{it}$$

Where:

- $\alpha_i$  are unit fixed effects (control for fixed characteristics of each unit over time);
- $\lambda_t$  are period fixed effects (control for shocks common to all units at each moment);
- $k = t - t_i^*$  is time relative to the event (negative before, positive after, zero at the event period);
- period  $k = -1$  (immediately before the event) is typically omitted as the reference — every  $\delta_k$  is interpreted relative to that period;
- $\delta_k$ , for  $k < 0$ , should be statistically indistinguishable from zero if the parallel-trends assumption is reasonable — these coefficients are themselves the visual/statistical test of parallel trends;
- $\delta_k$ , for  $k \geq 0$ , traces the effect's trajectory over time since the event — revealing whether the effect is immediate, gradual, or grows and then stabilizes.

## Parallel trends test

There is no single, definitive test for parallel trends — standard practice combines:

1. **Visual inspection** of the group-level mean series in the pre-event period.
2. A **formal joint test** that all  $\delta_k$  coefficients for  $k < -1$  in the event-study are simultaneously equal to zero (a joint F- or Wald-test), against the hypothesis that at least one differs from zero.
3. When applicable, a test for a **differential linear trend** in the pre-period: estimating whether the treated group's pre-event trend slope is statistically different from the control group's slope.

## Placebo test

A placebo test repeats the same design (2x2, cohort, or event-study) using a fake “event” date, placed entirely within a period known to have had no real policy change, or applying the treatment to a group known not to have been affected. The statistic of interest is the estimated  $\delta$  (or  $\delta_k$ ) coefficient under this fake assignment — under a valid design, it should be statistically indistinguishable from zero.

## Assumptions

- **Parallel trends:** absent the intervention, the treated group would have followed, on average, the same trajectory as the control group. This is the method’s central and strongest assumption — not directly observable for the post-event period (that’s exactly the missing counterfactual), but partially assessable in the pre-event period.
- **No anticipation:** units do not change behavior before the event’s official date because they already know it’s coming. When anticipation is present (for instance, firms that adjust hiring before a law takes effect, knowing it’s on the way), the “pre-event” period is already contaminated, and the event-study will often show a small nonzero effect even at  $k = -2$  or  $k = -3$ .
- **SUTVA (Stable Unit Treatment Value Assumption):** one unit’s treatment does not affect other units’ outcomes (no spillover/contamination between treatment and control). When the control group is indirectly affected by the intervention (say, a general-equilibrium effect that spreads to sectors not directly regulated), the estimated DiD effect is biased.
- **Stable group composition over time:** in designs using repeated cross-sectional data (not a panel of the same individuals), who belongs to each group in each period should be reasonably stable — if the treated group’s composition changes systematically around the event (say, selective migration into or out of the affected group), the comparison of means no longer reflects the treatment effect alone.
- **No other concurrent shocks specific to the treated group:** no other event besides the intervention of interest should affect the treated group specifically (and not the control group) in the same time window.

## Hypotheses

For the coefficient of interest  $\delta$  (aggregate DiD effect) or each  $\delta_k$  (event-study effect by period):

- $H_0$ :  $\delta = 0$  (or  $\delta_k = 0$ ) — no effect from the intervention.
- $H_1$ :  $\delta \neq 0$  (or  $\delta_k \neq 0$ ) — there is an effect.
- Test statistic:  $t = \hat{\delta} / \widehat{SE}(\hat{\delta})$ , where the standard error should, in practice, be robust to heteroscedasticity and, when there are multiple observations per unit over time, clustered at the unit level — ignoring this clustering structure typically understates the true standard error and artificially inflates significance.
- Confidence interval:  $\hat{\delta} \pm z_{1-\alpha/2} \cdot \widehat{SE}(\hat{\delta})$ .
- For the joint parallel-trends test:  $H_0$ : all  $\delta_k = 0$  for  $k < -1$ ; joint F- or Wald-test statistic.

## Interpretation

### **A well-executed DiD gives a stronger causal reading than before/after — but it’s still conditional**

A simple before-and-after comparison for a single group confounds the intervention’s effect with anything else that changed over time. DiD removes the trend common to both groups, better isolating the effect specific to the treated group. That’s a real improvement in causal credibility — but the result is still conditional on the parallel-trends assumption being valid. DiD is not a substitute for a randomized experiment; it’s an approximation, whose quality depends entirely on how reasonable the central assumption is in the specific context.

### **“Parallel trends holding up” reflects insufficient power, not proof**

This is the most important — and most frequently misunderstood — interpretive point: when the joint pre-period parallel-trends test **fails to reject** the hypothesis that all  $\delta_k$  ( $k < -1$ ) are zero, this is often read as “the parallel-trends assumption was confirmed.” That reading is statistically incorrect. Failing to reject  $H_0$  only means the data doesn’t provide enough evidence **against** the assumption — which can happen both because the assumption is genuinely reasonable, and because the test has low statistical power (few pre-event periods, small sample, noisy metric). A low-power test fails to reject almost anything. The correct way to phrase this is: “the pre-event data does not contradict the parallel-trends assumption” — never “the assumption was confirmed” or “proven.”

## Distinguishing association from causation

The coefficient  $\delta$  is, technically, always an estimate of association conditional on a specific design. It earns a causal reading to the extent that the assumptions above (especially parallel trends and no anticipation) are plausible in context. The robustness of that reading increases when: (a) placebo tests show no spurious effect; (b) the event-study shows a pattern of flat pre-trends followed by a visible shift starting at the event period; and (c) the result is stable across different specifications of the control group and analysis window. None of these points, alone or combined, constitutes definitive proof — they constitute cumulative evidence.

## Limitations

- **Sensitivity to the choice of control group.** Different control groups (all “reasonable” at first glance) can produce substantially different effect estimates. It’s good practice to report robustness of the result to more than one control-group definition.
- **Staggered adoption with heterogeneous effects.** When different units are treated at different times (staggered policy adoption, for instance) and the treatment effect varies over time or across units, the classic two-way fixed-effects specification (unit + period) can produce biased estimates, including the wrong sign, because already-treated units end up serving as an implicit “control” for units treated later. More recent estimators designed specifically for this scenario (Callaway-Sant’Anna, Sun-Abraham, among others) avoid this problem.
- **Placebo tests do not cover every threat to validity.** A “clean” placebo (no spurious effect at a fake date) increases confidence in the design, but does not rule out, for instance, a concurrent shock that specifically affected the treated group at the real event date.
- **Limited statistical power is common and underappreciated.** With few groups (few regions, few sectors) or short series, even a real effect of meaningful magnitude may go

undetected — the absence of significance should not automatically be read as “there is no effect.”

- **Inadequate standard errors are a common mistake.** Ignoring the temporal clustering structure within the same unit (not using cluster-robust standard errors) is one of the most frequent ways significance gets artificially inflated in DiD applications.

## Example

Consider a hypothetical study of a subsidy policy for vocational training courses, implemented in a set of municipalities starting in a specific year, while comparable neighboring municipalities did not receive the subsidy.

### Basic 2x2 design

The completion rate for technical courses (%) in both groups of municipalities, in the year immediately before and immediately after the subsidy’s introduction:

|                               | Pre-subsidy | Post-subsidy | Change   |
|-------------------------------|-------------|--------------|----------|
| Subsidized municipalities     | 40%         | 61%          | +21 p.p. |
| Non-subsidized municipalities | 40%         | 47%          | +7 p.p.  |

DiD estimate:  $\hat{\delta} = 21 - 7 = 14$  percentage points attributable to the subsidy, above the trend already shared by both groups.

### Cohort design

Suppose the subsidy, in practice, only applied to people who entered the technical-education system from a cutoff date onward. Comparing entry cohorts one year before and one year after the cutoff, controlling for age and region, the estimated effect for the cohort immediately after the cutoff is +9 percentage points in the completion rate, relative to the cohort immediately before — somewhat smaller than the aggregate 2x2 estimate, which is expected, since the cohort design better isolates who was actually exposed to the subsidy from the start of their course.

### Event-study

Extending the analysis to five years before and five years after the subsidy’s introduction, the  $\delta_k$  coefficients relative to the immediately preceding year ( $k = -1$ ) show: in years  $k = -5$  through  $k = -2$ , small coefficients (between -1 and +1.5 percentage points) that are statistically indistinguishable from zero — consistent with (not the same as “proof of”) parallel trends in the pre-period. From  $k = 0$  onward, the coefficients rise progressively: +6 p.p. in the event year, +11 p.p. one year later, +14 p.p. two years later, stabilizing near that level — a growing-then-stabilizing effect pattern, more informative than the single aggregate number from the 2x2 design.

### Placebo test

Repeating the same 2x2 design, but using as a “fake” event date a year three years before the real subsidy introduction (a period in which no policy change occurred), the placebo “effect” estimate is +0.8 percentage points, with a p-value of 0.71 — not significant, as expected under a valid design. This result reinforces (without definitively proving) the credibility of the main estimate.

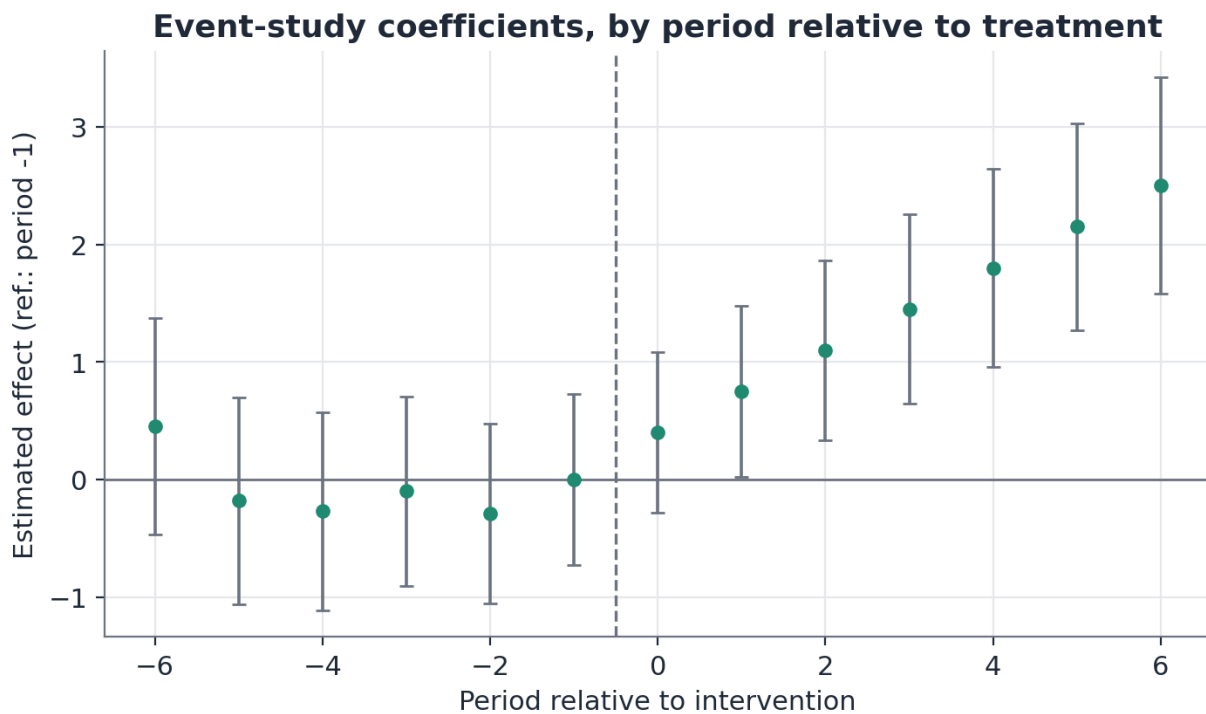
```

import pandas as pd
import statsmodels.formula.api as smf

# basic 2x2 design
model_2x2 = smf.ols(
    "completion_rate ~ subsidy * post", data=data
).fit(cov_type="cluster", cov_kwds={"groups": data["municipality"]})
print(model_2x2.params["subsidy:post"])

# event-study: build time-relative-to-event indicators,
# omitting k = -1 as the reference
data["relative_time"] = data["year"] - data["event_year"]
dummies = pd.get_dummies(data["relative_time"], prefix="k").drop(columns=["k_-1"])
data_es = pd.concat([data, dummies], axis=1)
terms = " + ".join([f"subsidy:{c}" for c in dummies.columns])
model_es = smf.ols(
    f"completion_rate ~ C(municipality) + C(year) + {terms}", data=data_es
).fit(cov_type="cluster", cov_kwds={"groups": data_es["municipality"]})

```



Illustrative data, not drawn from any real project.

Figure 9: Event-study coefficients, by period relative to the intervention

# Rotating Panel Transition Analysis

## Concept

A rotating panel is a sampling design used in several continuous household surveys, in which a fraction of the sample is replaced at each interview round, while another fraction stays on for a limited number of consecutive rounds before also being replaced. This design combines two advantages: it allows representative cross-sectional estimates at each individual round (the full sample remains representative of the population at each moment), while also allowing a fraction of the same individuals to be tracked across a few consecutive rounds — enough to study individual transitions between states.

Transition analysis exploits that second advantage: instead of comparing aggregate estimates from two different moments — which mixes real individual state change with mere turnover of who's in the sample — it matches records for the same person across consecutive interviews and builds a transition matrix, which directly describes the conditional probability of moving (or staying) in a given state from one period to the next.

## Mathematical formulation

Let  $S = \{s_1, s_2, \dots, s_K\}$  be the set of possible states (for example, {formal employment, informal employment, unemployed, out of the labor force}). For an individual  $i$  successfully matched between the interview at period  $t$  and the interview at period  $t+h$  (where  $h$  is the interval between interviews, typically one quarter), let  $S_{i,t}$  and  $S_{i,t+h}$  be the states observed at each moment.

The **transition matrix** for a group  $g$  is defined by its entries:

$$P_{jk}^{(g)} = \Pr(S_{t+h} = s_k \mid S_t = s_j, \text{group} = g) \approx \frac{\sum_{i \in g} \mathbb{1}[S_{i,t} = s_j] \cdot \mathbb{1}[S_{i,t+h} = s_k]}{\sum_{i \in g} \mathbb{1}[S_{i,t} = s_j]}$$

Where:

- $P_{jk}^{(g)}$  is the estimated transition probability from state  $s_j$  to state  $s_k$ , within group  $g$ , between the two interview moments;
- the numerator counts how many individuals in group  $g$  were in state  $s_j$  at  $t$  and state  $s_k$  at  $t+h$ ;
- the denominator counts the total individuals in group  $g$  who were in state  $s_j$  at  $t$  (and could be matched through  $t+h$ ).

Each row  $j$  of the matrix  $P^{(g)}$  sums to 1, since every individual who was in state  $s_j$  ends up in some state  $s_k$  (including  $s_k = s_j$ , i.e., remaining in the same state).

When sampling weights are available (essential in complex household surveys), the correct estimate weights each individual by their longitudinal sampling weight — which, in general, differs from the cross-sectional weight of each individual round, and needs to be computed or provided specifically for panel-analysis use.

Comparisons across groups typically focus on one specific cell of interest, for example  $P_{\text{unemployed,employed}}^{(A)}$  versus  $P_{\text{unemployed,employed}}^{(B)}$ , tested as a difference of two proportions:

$$z = \frac{\hat{P}^{(A)} - \hat{P}^{(B)}}{\sqrt{\hat{P}(1 - \hat{P}) \left( \frac{1}{n_A} + \frac{1}{n_B} \right)}}, \quad \hat{P} = \frac{x_A + x_B}{n_A + n_B}$$

where  $x_g$  is the number of transitions of interest in group  $g$ ,  $n_g$  is the total individuals in group  $g$  who started from the initial state, and  $\hat{P}$  is the pooled proportion under the null hypothesis that the two rates are equal.

## Assumptions

- **Correct matching between interviews.** The validity of the whole analysis hinges on correctly identifying that the same individual — not a different person who moved into the same household — is being compared across the two moments. Most surveys provide a household identifier plus a within-household resident position, and correct matching also requires confirming stable demographic characteristics (sex, and age within a tolerance consistent with the interview interval).
- **Sample attrition not systematically different across groups.** If the matching (or sample-retention) success rate differs across the groups being compared — for example, if one group has higher residential mobility and is therefore systematically harder to track — the matched sample stops being representative of its source population differently for each group, introducing selection bias into the comparison.
- **Well-defined, mutually exclusive destination states.** Ambiguities in state classification (say, precisely defining what counts as “out of the labor force” versus “unemployed” under active job-search criteria) need to be handled consistently at both moments of the comparison.
- **Comparable interval between interviews across individuals.** If the interval  $h$  varies (some people matched a quarter apart, others two quarters apart), transition rates are not directly comparable without adjustment, since transitions have more time to occur over longer intervals.

## Hypotheses

To compare a specific transition rate between two groups  $A$  and  $B$ :

- $H_0$ :  $P^{(A)} = P^{(B)}$  — the transition probability is equal in both groups.
- $H_1$ :  $P^{(A)} \neq P^{(B)}$  (or one-sided, if there’s an expected direction a priori).
- Test statistic:  $z$ , as defined above (two-proportion test).
- Conventional significance level:  $\alpha = 0.05$ .
- When multiple destination states are being compared simultaneously (the entire transition matrix, not just one cell), a chi-squared homogeneity test between the two matrices is more appropriate than separate cell-by-cell tests, which don’t correct for multiple comparisons.

## Interpretation

The transition matrix describes mobility **observed among the matched individuals**, not necessarily the entire population. Three interpretive points deserve attention:

- **Composition versus real mobility.** A stable aggregate rate over time can hide either high individual turnover (many people flowing in and out of a state in an offsetting pattern) or low turnover (the same people staying put). Only transition analysis distinguishes these two

scenarios — a crucial difference for policy decisions, since the appropriate interventions differ between them.

- **Representativeness of the matched subsample.** If matching succeeds in, say, 85% of eligible cases, the results directly describe that 85% — extrapolating to the unmatched 15% requires the additional assumption that matching fails randomly with respect to the very state being studied, which is rarely guaranteed (people who move households, for instance, may be systematically more likely to also be changing employment status).
- **Transitions do not imply causation.** A higher unemployment-exit rate in one group is an observed mobility difference, not an explanation of why that mobility is higher — compositional factors (age, education, region, prior occupation type) may explain part or all of the difference, and isolating them would require additional analysis (say, a proportional-hazards model adjusting for covariates).

## Limitations

- **A rotating panel is not a full longitudinal panel.** The distinction is fundamental: in a longitudinal panel, the same (or a fixed) sample is followed for years. In a rotating panel, each individual is observed only during the window they're in the sample (typically a few consecutive rounds) before being replaced — transition analyses are only possible within that limited window, and most of a rotating-panel survey's data, outside that overlap window, is effectively repeated cross-section: each round is representative of the population, but individuals interviewed in non-overlapping rounds are not the same people.
- **Matching can fail non-randomly.** Beyond household moves, matching can fail due to recording errors, changes in household composition (birth, death, separation), or refusal in subsequent interviews — each of these mechanisms can correlate with the very state being studied, creating differential attrition bias across groups.
- **Few overlapping rounds limit the analysis horizon.** If the panel design allows tracking each individual for, say, at most five interviews, transitions can only be measured within that window — longer-term trends require chaining several panel cohorts together, a more complex approach subject to more sources of noise.
- **State-classification errors accumulate across two measurements.** Because the analysis depends on correctly classifying the state at two moments (not one), any measurement error or classification ambiguity at either moment contaminates the estimated transition.

## Example

Consider a hypothetical study of labor-market mobility for two groups of workers, A and B, using a household survey with a rotating panel design in which each household is interviewed for up to four consecutive quarters.

Out of a total of 500 unemployed people in Group A at the first interview, 420 (84%) could be matched to an interview one quarter later. Out of Group B's 480 unemployed people, 390 (81%) were matched. The matching rates are similar, which reduces (but does not eliminate) concern about differential attrition.

Within the matched sample, the one-quarter transition matrix is:

**Group A** (starting from unemployed,  $n=420$ ):

| Destination            | %     |
|------------------------|-------|
| Formal employment      | 14.0% |
| Informal employment    | 7.9%  |
| Unemployed             | 55.7% |
| Out of the labor force | 22.4% |

**Group B** (starting from unemployed, n=390):

| Destination            | %     |
|------------------------|-------|
| Formal employment      | 9.2%  |
| Informal employment    | 8.5%  |
| Unemployed             | 58.7% |
| Out of the labor force | 23.6% |

The formal-employment transition rate is 14.0% for Group A versus 9.2% for Group B — a 4.8-percentage-point difference. The two-proportion test for this specific difference gives  $z \approx 2.17$ , corresponding to a p-value of approximately 0.030 — statistically significant at the conventional 5% level.

Careful interpretation: this difference describes the experience of the individuals successfully matched in each group — 84% and 81% of the original samples, respectively. If, for instance, people in Group B who moved households (and therefore went unmatched) have systematically different employment outcomes than those who stayed in the same household, the estimated 9.2% rate may not correctly represent Group B as a whole. The 4.8-percentage-point difference also does not, on its own, isolate why the transition is faster in Group A — observable characteristics like average education and prior occupation type would need to be controlled for separately to move that question forward.

```
import pandas as pd
import numpy as np
from statsmodels.stats.proportion import proportions_ztest

# matched_data: one row per successfully matched individual
# columns: group, state_t0, state_t1

def transition_matrix(df, group):
    sub = df[df["group"] == group]
    return pd.crosstab(sub["state_t0"], sub["state_t1"], normalize="index") * 100

matrix_a = transition_matrix(matched_data, "A")
matrix_b = transition_matrix(matched_data, "B")

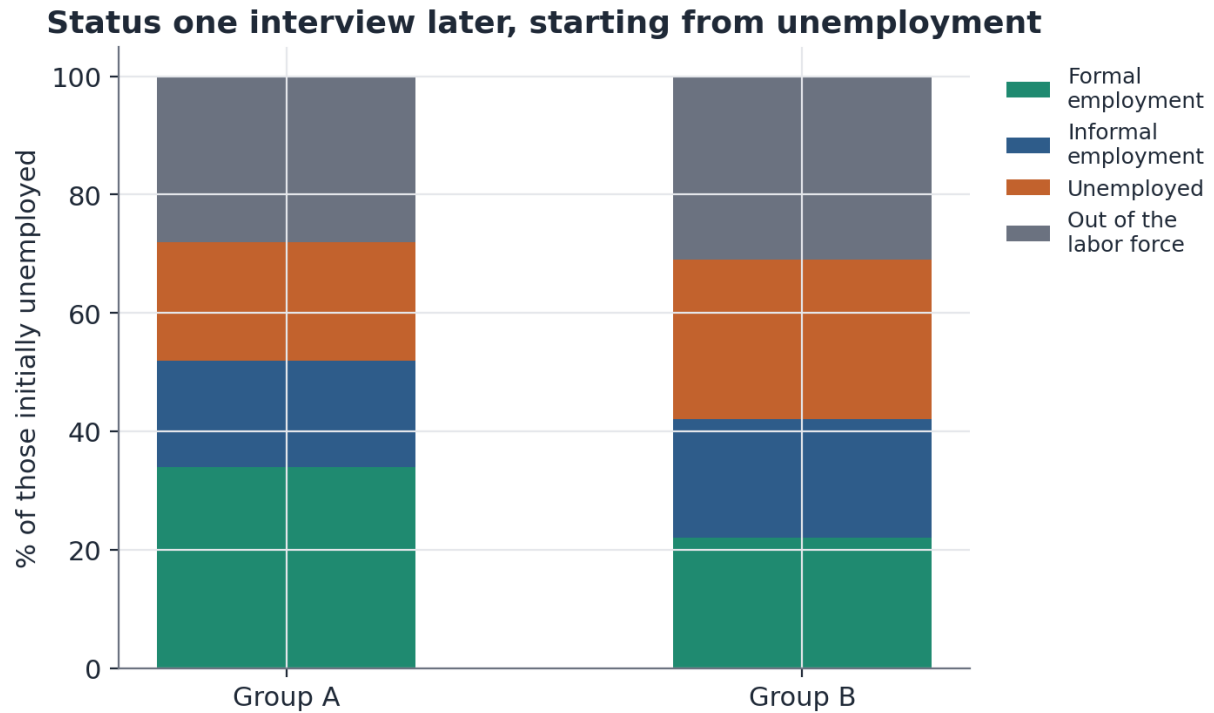
unemp_a = matched_data[(matched_data["group"] == "A") & (matched_data["state_t0"] == "unemployed")]
unemp_b = matched_data[(matched_data["group"] == "B") & (matched_data["state_t0"] == "unemployed")]

x = [(unemp_a["state_t1"] == "formal_employment").sum(),
```

```

(unemp_b["state_t1"] == "formal_employment").sum()]
n = [len(unemp_a), len(unemp_b)]
stat, p_value = proportions_ztest(x, n)
print(stat, p_value)

```



Illustrative data, not drawn from any real project.

Figure 10: Status one interview later, starting from unemployment

## A/B testing fundamentals

This document covers the statistical fundamentals of an A/B test — experimental design, randomization, effect size, standard error, confidence interval, and hypothesis testing for means and proportions. The three documents that follow (Sample Ratio Mismatch, covariate balance, and non-inferiority testing) treat this content as a prerequisite and go straight to what each of them adds.

### Concept

An A/B test is a controlled, randomized experiment applied to a product or business setting: units (usually users) are randomly assigned to a control or treatment condition, and a pre-defined primary metric is compared between the two groups at the end of a fixed period.

The core idea is isolating the causal effect of the change under test. In an observational comparison — say, “before” versus “after” a launch, or users who opted into a new feature versus those who didn’t — any difference found could come from dozens of factors that also shift over time or that systematically differ between who opts in and who doesn’t. Randomization breaks that link: in expectation, random assignment makes the two groups comparable on everything, observed or not, except the condition they were assigned. That is what lets you attribute a difference in outcome to the change being tested, with a quantifiable margin of uncertainty.

### Mathematical formulation

#### Difference in means (continuous metric)

Let control ( $C$ ) and treatment ( $T$ ) have sample means  $\bar{x}_C$  and  $\bar{x}_T$ , sample variances  $s_C^2$  and  $s_T^2$ , and sizes  $n_C$  and  $n_T$ . The estimated effect is:

$$\hat{\Delta} = \bar{x}_T - \bar{x}_C$$

The standard error of that difference, without assuming equal variances across groups (the same logic behind Welch’s t-test), is:

$$SE(\hat{\Delta}) = \sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}}$$

The test statistic and the 95% confidence interval follow as:

$$t = \frac{\hat{\Delta}}{SE(\hat{\Delta})}, \quad 95\% \text{ CI} = \hat{\Delta} \pm z_{0.975} \cdot SE(\hat{\Delta})$$

where  $z_{0.975} \approx 1.96$  under a normal approximation (adequate with large samples; with small samples, use the corresponding quantile of a t distribution with Welch-Satterthwaite degrees of freedom).

### Difference in proportions (binary metric, e.g. conversion)

Let  $\hat{p}_C = x_C/n_C$  and  $\hat{p}_T = x_T/n_T$  be the observed rates in each group, with  $x_C$  and  $x_T$  conversions and  $n_C$ ,  $n_T$  sample sizes. The estimated effect is  $\hat{\Delta} = \hat{p}_T - \hat{p}_C$ . For the hypothesis test, the pooled proportion under the null of equality is used:

$$\hat{p}_{pool} = \frac{x_C + x_T}{n_C + n_T}, \quad SE_0 = \sqrt{\hat{p}_{pool}(1 - \hat{p}_{pool}) \left( \frac{1}{n_C} + \frac{1}{n_T} \right)}$$
$$z = \frac{\hat{p}_T - \hat{p}_C}{SE_0}$$

Note that the standard error used in the test ( $SE_0$ , under  $H_0$ ) differs from the one used to build the confidence interval around the observed effect, which does not assume equal proportions:

$$SE(\hat{\Delta}) = \sqrt{\frac{\hat{p}_C(1 - \hat{p}_C)}{n_C} + \frac{\hat{p}_T(1 - \hat{p}_T)}{n_T}}$$

### Sample size / power

The sample size needed to detect a minimum effect  $\delta$  with power  $1 - \beta$  and significance  $\alpha$  (two-sided test) is approximately, for a proportion metric:

$$n \approx \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 \cdot [p_C(1 - p_C) + p_T(1 - p_T)]}{\delta^2}$$

where  $z_{1-\alpha/2}$  and  $z_{1-\beta}$  are the normal quantiles corresponding to the desired confidence and power, and  $\delta = p_T - p_C$  is the smallest effect you want to be able to detect.

### Assumptions

- **Effective randomization:** every eligible unit has a known (typically equal) assignment probability, independent of every other unit. Without this, the entire causal argument collapses — hence the importance of checking the observed allocation share (see Sample Ratio Mismatch).
- **Independence between units:** one unit’s outcome should not depend on which condition another unit received (no interference, or the “SUTVA” — Stable Unit Treatment Value Assumption). In settings with network effects or a shared marketplace, this can fail.
- **A well-defined, stable metric fixed before the experiment:** the primary metric, its aggregation window, and the eligibility criteria should be locked down before looking at the data, to avoid retroactive selection bias.
- **No uncorrected repeated peeking:** stopping the experiment as soon as the p-value crosses 0.05, without an appropriate sequential method, inflates the false-positive rate well above the nominal .

## Hypotheses

For a continuous or proportion metric, the standard two-sided test is:

$$H_0 : \Delta = 0 \quad H_1 : \Delta \neq 0$$

where  $\Delta$  is the true (population) difference between treatment and control. The significance level  $\alpha$  is typically set at 0.05 before the experiment.  $H_0$  is rejected when the p-value is below  $\alpha$ , or equivalently, when the  $(1 - \alpha)$  confidence interval excludes zero.

## Interpretation

Rejecting  $H_0$  is evidence that the observed effect is not plausibly explained by sampling variation alone — it is not proof of causality on its own; the causal reading comes from randomization, not from the statistical test in isolation. The width of the confidence interval communicates the precision of the estimate: two experiments can share the same point estimate with very different degrees of confidence, depending on sample size and the metric’s variability.

Distinguishing statistical significance from practical relevance is essential. A 0.05-percentage-point effect on conversion can be statistically significant with millions of users, yet not justify the engineering cost of the change. Likewise, “not significant” is not the same as “zero effect” — an underpowered experiment simply lacks the sensitivity to detect effects of the size that actually matter; the correct thing to report there is “insufficient evidence,” not “no effect.”

## Limitations

- An A/B test measures the average population effect under the specific conditions of the experiment (period, user mix, product context at that moment) — generalizing the result to a different time or population takes caution.
- Testing many secondary metrics at once without a multiple-comparisons correction raises the chance of finding spurious “effects.” Interesting secondary findings deserve to be treated as a hypothesis for a dedicated follow-up experiment, not as a final conclusion.
- The aggregate result can hide heterogeneity — the effect could be positive for one subgroup and negative for another, canceling out overall (see heterogeneity/interaction testing).
- Everything hinges on randomization having worked as planned; the hypothesis test itself does not verify that.

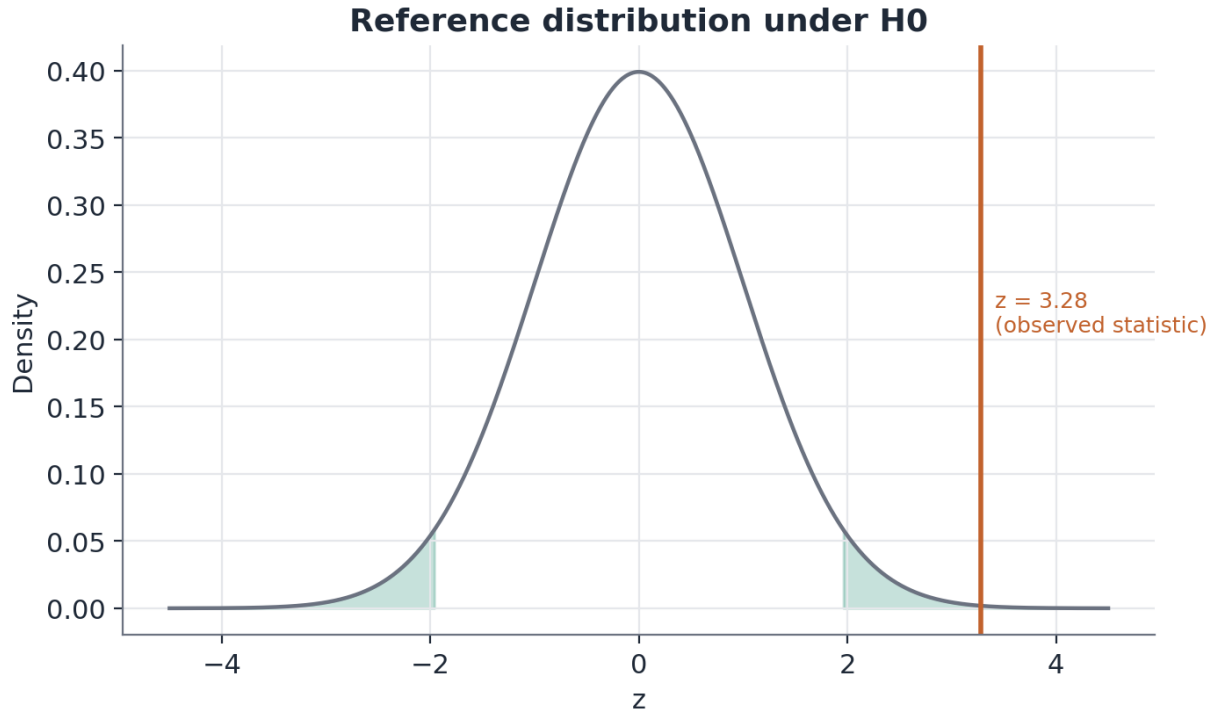
## Example

Suppose an e-commerce site tests a new checkout page layout against the current one. The team randomizes 12,000 visitors into each arm and runs the experiment for two full weeks without stopping early.

Observed result:

|             | Control | Treatment |
|-------------|---------|-----------|
| Visitors    | 12,000  | 12,000    |
| Conversions | 984     | 1,128     |
| Rate        | 8.20%   | 9.40%     |

The estimated effect is  $\hat{\Delta} = 0.094 - 0.082 = 0.012$  (1.2 percentage points). The pooled proportion under  $H_0$  is  $\hat{p}_{pool} = (984 + 1128)/24000 \approx 0.088$ , giving  $SE_0 \approx 0.00366$  and  $z \approx 3.28$ , corresponding to a p-value below 0.01. The 95% confidence interval for the difference, computed without assuming equal proportions, runs roughly from 0.4 to 2.0 percentage points — entirely above zero.



Illustrative data, not drawn from any real project.

Figure 11: Reference distribution under the null hypothesis with the observed statistic marked

The statistical conclusion is clear: the new layout increased conversion, and the size of that increase (somewhere between roughly 5% and 24% relative lift, depending on where in the interval the true effect sits) is large enough to warrant a cost-benefit case for shipping it. Before declaring victory, though, the team would check: (1) whether the observed allocation (12,000/12,000, or close to it) matches what was planned — a Sample Ratio Mismatch test; and (2) whether any guardrail metric, such as page load time or error rate, moved beyond an acceptable margin — a non-inferiority test.

## Sample Ratio Mismatch (SRM) test

*Assumes the A/B testing fundamentals (randomization, groups, primary metric, hypothesis testing) — see [A/B testing fundamentals](#). This document covers specifically the allocation-integrity check that must precede any effect reading.*

### Concept

A Sample Ratio Mismatch (SRM) test checks whether the observed proportion of units in each experiment arm is compatible with the proportion planned at assignment. At its core, it's a goodness-of-fit test: observed counts are compared against expected counts under the design's nominal split (50/50, 90/10, or any other), using a chi-square test — or, equivalently for two groups, an exact binomial test.

The reason it exists as a separate check, rather than being folded into the effect test, is that it answers a logically prior question: “is the comparison I’m about to make even valid?” A successful effect test means nothing if the groups being compared weren’t formed the way the analysis assumes.

### Mathematical formulation

Let there be  $k$  experiment arms, with planned proportions  $\pi_1, \pi_2, \dots, \pi_k$  (summing to 1) and observed counts  $o_1, o_2, \dots, o_k$ , with total  $N = \sum_i o_i$ . The expected count in each arm under the planned proportion is:

$$e_i = \pi_i \cdot N$$

The chi-square goodness-of-fit statistic is:

$$\chi^2 = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}$$

Under  $H_0$  (the real proportion matches the plan),  $\chi^2$  approximately follows a chi-square distribution with  $k - 1$  degrees of freedom. For two arms ( $k = 2$ ), this reduces to 1 degree of freedom, and the test is equivalent to a two-sided binomial test comparing  $o_1/N$  against  $\pi_1$ :

$$z = \frac{\hat{p}_1 - \pi_1}{\sqrt{\pi_1(1 - \pi_1)/N}}, \quad z^2 = \chi^2 \text{ (1 df)}$$

where  $\hat{p}_1 = o_1/N$  is the observed proportion in the first arm.

### Assumptions

- **The planned proportion is known and fixed** — it must be set explicitly before the experiment, not inferred after the fact.
- **The counts are independent of each other** — each unit contributes to exactly one arm, with no double counting or units shared across arms.

- **The chi-square test assumes reasonably large expected counts** (a common rule of thumb: a minimum expected count of 5 per cell) for the asymptotic approximation to be reliable; with very small counts, an exact binomial test is preferable.
- **The check should cover the full eligible population**, not a sample filtered after assignment — filtering before checking can mask exactly the kind of differential loss the test is meant to detect.

## Hypotheses

$$H_0 : \pi_i = \pi_i^{planned} \text{ for all } i \quad H_1 : \pi_i \neq \pi_i^{planned} \text{ for at least one } i$$

The significance level used in practice is typically stricter than the standard 0.05 used for effect tests — values like  $\alpha = 0.001$  are common, reflecting an asymmetry of costs: a false positive here (declaring an SRM when there isn’t one) costs an investigation; a false negative (missing a real SRM) contaminates every subsequent reading of the experiment.

## Interpretation

A significant SRM is evidence that the assignment mechanism, or the counting process, isn’t producing the intended proportion — it is not, on its own, a diagnosis of the cause. The most common causes include:

- latency differences between the tested versions, causing one to drop more events to timeouts before logging;
- bot, fraud, or traffic-quality filters applied asymmetrically across arms;
- bugs in the user-to-group hashing/assignment code itself;
- instrumentation issues that log one arm incompletely.

The practical implication of a detected SRM is direct: any effect estimated in the experiment is not interpretable until the cause is identified and fixed (or until the experiment is rerun). This holds even if the effect looks “reasonable” or matches the team’s prior expectation — an SRM undermines the fundamental premise of comparability between groups, which is exactly what gives the effect test its causal reading.

## Limitations

- The test detects deviation in the aggregate proportion; it does not guarantee the absence of SRM within specific subgroups (by platform, region, acquisition channel) — an SRM can be “hidden” in the overall aggregate when deviations in opposite directions cancel out across segments.
- It doesn’t locate the cause — it only flags that a problem exists. Follow-up investigation (comparing assignment logs, per-segment drop rates, load timing) is a separate manual step.
- An SRM can be intermittent, present only during part of the experiment’s run; looking at the observed proportion over time, rather than only the cumulative total at the end, helps identify that.
- With few arms and a very small sample, the chi-square approximation can be unreliable — in those cases, an exact binomial test (or Fisher’s exact test, for more than two arms) is more appropriate.

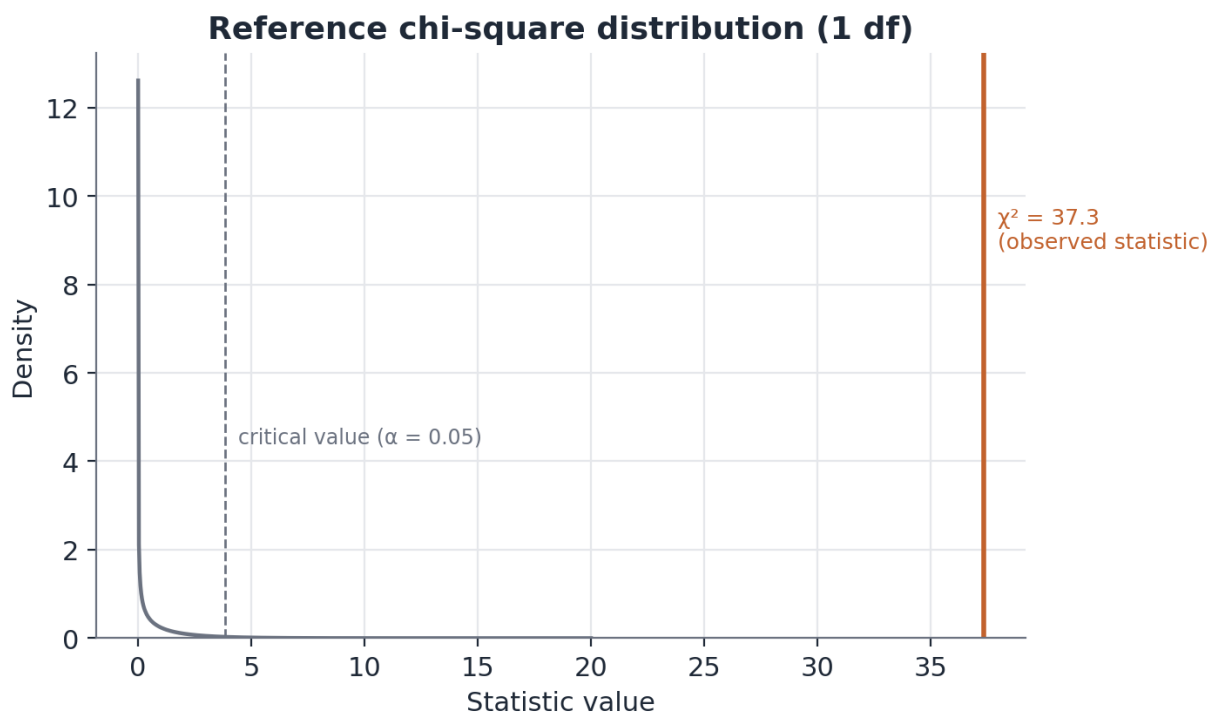
## Example

A subscription app plans to test a new onboarding screen for 50% of new users, keeping the other 50% on the current screen. After two weeks, the team logs the following counts:

| Arm                      | Planned | Observed |
|--------------------------|---------|----------|
| Control (current screen) | 50%     | 10,432   |
| Treatment (new screen)   | 50%     | 9,568    |
| Total                    | 100%    | 20,000   |

Under  $H_0$ , the expected count in each arm is  $e = 0.5 \times 20,000 = 10,000$ . The chi-square statistic:

$$\chi^2 = \frac{(10,432 - 10,000)^2}{10,000} + \frac{(9,568 - 10,000)^2}{10,000} = \frac{186,624}{10,000} + \frac{186,624}{10,000} \approx 37.3$$



Illustrative data, not drawn from any real project.

Figure 12: Reference chi-square distribution with 1 degree of freedom, showing the critical value and the observed statistic

With 1 degree of freedom and  $\chi^2 \approx 37.3$ , the p-value is extremely small — many orders of magnitude below the commonly used 0.001 threshold. The team pauses any effect reading and investigates. They find that the new screen loads a heavier visual component, and users on slow connections are being marked “not eligible” more often in the treatment arm due to a loading timeout in the experiment’s instrumentation — a differential sample loss that produces exactly the observed pattern. After fixing the timeout, the experiment is restarted from scratch, and only then is the effect reading treated as reliable.

# Covariate balance and covariate adjustment

*Assumes the A/B testing fundamentals — see [A/B testing fundamentals](#).*

## Concept

This topic covers two related but conceptually distinct tools, both built on covariates measured before an experiment starts.

**Covariate balance** is a randomization-validity check: it compares the distribution of pre-experiment characteristics between the treatment and control groups. Since assignment is random, those characteristics — which have no causal relationship with the treatment, having been measured before it even existed — should be distributed similarly across groups, up to sampling variation. A systematic imbalance is evidence against the premise that randomization worked as planned.

**Covariate adjustment** (which includes post-stratification and estimators like CUPED) is a variance-reduction technique: it uses pre-experiment covariates correlated with the metric of interest to explain part of the between-unit variation that has nothing to do with the treatment, making the effect estimate more precise — without changing what it estimates. It's important not to conflate the two: balance is a diagnostic check done (ideally) before looking at results; adjustment is an estimation technique applied when computing the effect. Successful randomization doesn't need adjustment to be valid; adjustment is about precision, not about fixing a problem.

## Mathematical formulation

### Standardized mean difference (balance)

For a continuous covariate  $X$ , with means  $\bar{x}_C$  and  $\bar{x}_T$  and standard deviations  $s_C$  and  $s_T$  in the control and treatment groups:

$$SMD = \frac{\bar{x}_T - \bar{x}_C}{\sqrt{\frac{s_C^2 + s_T^2}{2}}}$$

The denominator is the pooled standard deviation, which makes the SMD independent of the covariate's original scale and comparable across different covariates. A commonly cited reference threshold in the causal-inference literature is  $|SMD| < 0.1$  for acceptable balance, though that value is a practical convention, not a statistically derived bound.

For a binary covariate with proportions  $\hat{p}_C$  and  $\hat{p}_T$ :

$$SMD = \frac{\hat{p}_T - \hat{p}_C}{\sqrt{\frac{\hat{p}_C(1 - \hat{p}_C) + \hat{p}_T(1 - \hat{p}_T)}{2}}}$$

### Covariate-adjusted estimator (regression)

The adjusted effect is obtained by including the pre-experiment covariate  $Z$  (mean-centered, for interpretability) in a linear model:

$$Y_i = \alpha + \tau \cdot D_i + \beta \cdot (Z_i - \bar{Z}) + \varepsilon_i$$

where  $D_i$  is the treatment indicator (1 if treatment, 0 if control),  $\tau$  is the adjusted treatment effect, and  $\beta$  captures the relationship between the covariate and the outcome. Since  $D_i$  is (approximately) independent of  $Z_i$  by construction of randomization,  $\hat{\tau}$  estimates the same average effect as the simple difference in means — but with a residual variance  $\varepsilon_i$  smaller than the total variance of  $Y_i$ , which shrinks  $\hat{\tau}$ 's standard error in proportion to how much  $Z$  explains of  $Y$ .

### CUPED (residual variance)

An equivalent form, popular in product experimentation, adjusts the metric directly:

$$Y_i^{CUPED} = Y_i - \theta \cdot (Z_i - \bar{Z}), \quad \theta = \frac{\text{Cov}(Y, Z)}{\text{Var}(Z)}$$

The relative variance reduction achieved is approximately:

$$\frac{\text{Var}(Y^{CUPED})}{\text{Var}(Y)} \approx 1 - \rho^2$$

where  $\rho$  is the correlation between the covariate  $Z$  and the metric  $Y$ . The more correlated the pre-experiment covariate is with the metric, the bigger the precision gain.

### Assumptions

- **Covariates used must be strictly pre-treatment** — measured before the experiment started. Using a post-treatment covariate (for example, number of sessions *during* the experiment) introduces bias, because it may already have been affected by the treatment itself.
- **The choice of adjustment covariates should be specified before looking at results**, to avoid the bias of picking, after the fact, whichever covariate “improves” the desired outcome.
- **Balance is necessary, but not sufficient, for causal validity** — it checks observed covariates; nothing guarantees balance on unmeasured factors, although randomization balances those too in expectation.
- **The adjustment model should be reasonably well specified** — a highly non-linear relationship between the covariate and the metric, forced into a linear adjustment, may not capture all the available precision gain (though it rarely introduces bias, given that  $D$  is random).

### Hypotheses

For the per-covariate balance test, the usual hypothesis test (a t-test or proportions test on the covariate, analogous to the effect test) is:

$$H_0 : \mu_{Z,C} = \mu_{Z,T} \quad H_1 : \mu_{Z,C} \neq \mu_{Z,T}$$

where  $\mu_{Z,C}$  and  $\mu_{Z,T}$  are the population means of covariate  $Z$  in each group. In practice, many teams prefer to report the SMD directly rather than a p-value — when the number of tested covariates is large, a p-value alone has a high false-positive rate (by chance, some covariates will “come up significant”), while the SMD gives a magnitude measure that’s comparable across covariates.

Covariate adjustment doesn’t have a hypothesis of its own to test — it modifies the estimate and standard error of the treatment-effect test, whose hypothesis remains the one from the A/B testing fundamentals document ( $H_0 : \tau = 0$ ).

## Interpretation

A small SMD across all observed covariates is evidence that randomization produced comparable groups along those dimensions — it reinforces, but does not alone prove, that the subsequent effect test will have a valid causal reading. A large SMD on a single covariate, among many tested, can just be expected sampling variation (with  $k$  covariates tested at 5% significance, about  $0.05k$  “imbalances” are expected by chance alone). The pattern worth noticing is: several covariates imbalanced simultaneously, or a business-critical covariate strongly imbalanced, warrant investigation — possibly with a complementary Sample Ratio Mismatch test.

On adjustment: the point estimate of the effect, with and without adjustment, should be very close when randomization worked (the covariate is, in expectation, uncorrelated with the treatment). A large shift in the point estimate when adding the adjustment covariate is itself a warning sign about imbalance — not the expected behavior of a well-behaved variance adjustment.

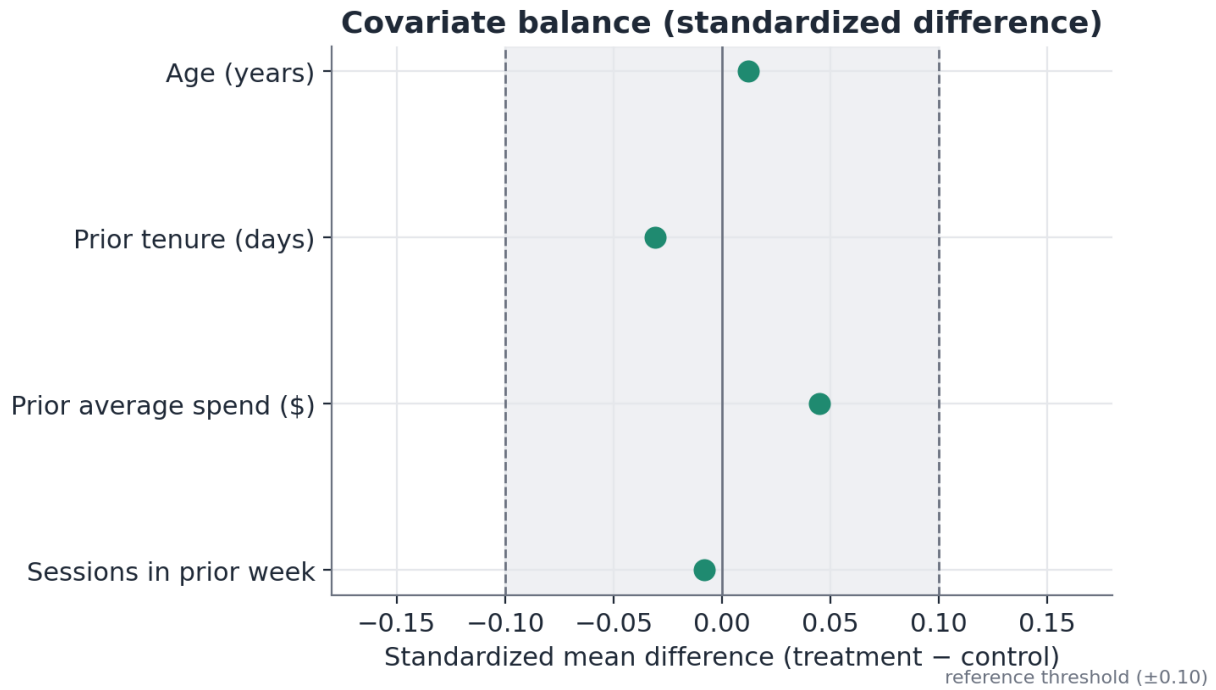
## Limitations

- Balance on observed covariates does not imply balance on unobserved covariates — it’s evidence in favor of randomization, never complete proof.
- The precision gain from covariate adjustment depends entirely on how predictive the covariate is of the metric; weakly correlated covariates bring negligible gain, even when statistically valid to use.
- Choosing the adjustment covariate after seeing that it “improves” the result (indirect p-hacking) invalidates the interpretation of the resulting p-value, even though the point estimate remains unbiased.
- In designs with few units (for instance, few treated geographic regions), both SMD and regression adjustment can be unstable — inference methods more robust to small samples (randomization inference, wild-cluster bootstrap) become more relevant in that regime.

## Example

A subscription app tests a new pricing policy, randomizing 8,000 users to control and 8,000 to treatment. Before looking at any results, the team checks the balance of four pre-experiment covariates:

| Covariate                | Control | Treatment | SMD    |
|--------------------------|---------|-----------|--------|
| Age (years)              | 34.2    | 34.3      | 0.012  |
| Prior tenure (days)      | 210     | 204       | -0.031 |
| Prior average spend (\$) | 42.10   | 42.80     | 0.045  |
| Sessions in prior week   | 5.4     | 5.3       | -0.008 |



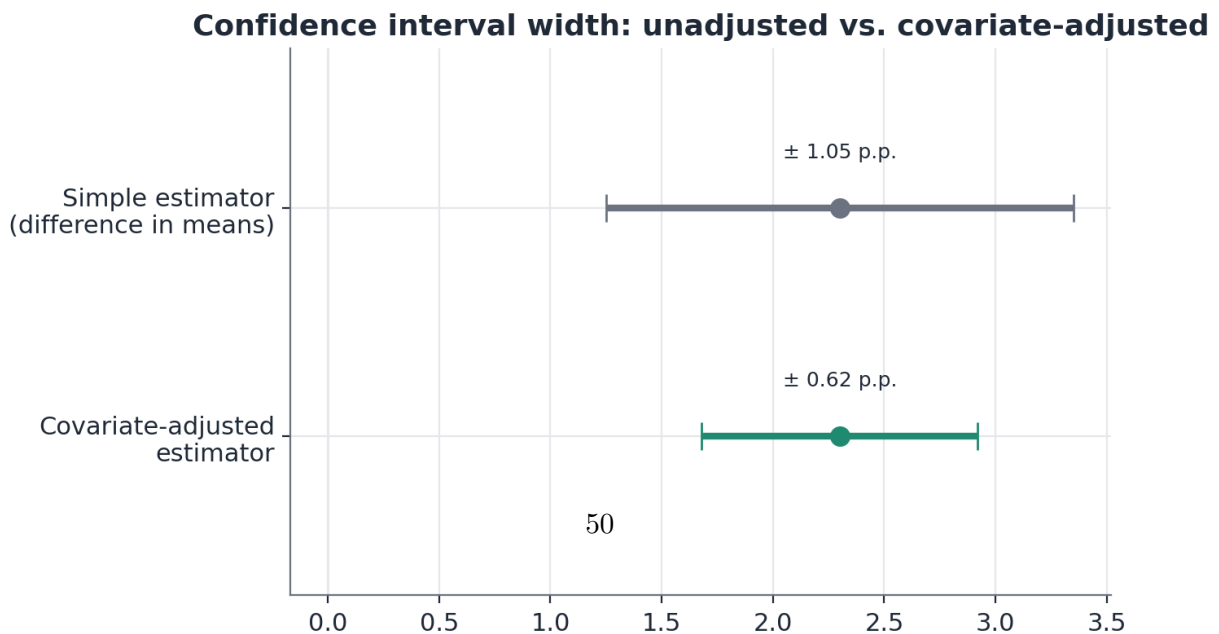
Illustrative data, not drawn from any real project.

Figure 13: Love plot showing each covariate’s standardized difference, all within the  $\pm 0.10$  reference band

All four covariates have  $|SMD| < 0.1$  — acceptable balance, reinforcing that randomization worked.

The team then estimates the pricing policy’s effect on revenue for the experiment month, comparing two estimators: the simple difference in means and an adjusted estimator using prior average spend as a covariate (which has a correlation of roughly 0.55 with the experiment-month revenue).

| Estimator                  | Estimated effect | 95% CI        |
|----------------------------|------------------|---------------|
| Simple difference in means | +2.3 pp          | $\pm 1.05$ pp |
| Adjusted for prior spend   | +2.3 pp          | $\pm 0.62$ pp |



# Non-inferiority testing

*Assumes the A/B testing fundamentals — see [A/B testing fundamentals](#).*

## Concept

A non-inferiority test evaluates whether a treatment metric is not worse than the control metric by more than a pre-defined margin  $\delta$  — rather than testing whether the two are equal. It’s the appropriate test for guardrail metrics: indicators that must not deteriorate beyond an acceptable limit, even when the experiment’s main goal is something else (for instance, gaining speed, cutting cost, or improving a different business metric).

The key difference from a conventional two-sided test isn’t statistical mechanics — it’s how the hypotheses are logically framed. A standard test splits the space of possibilities into “no difference” (null) and “some difference, in either direction” (alternative). A non-inferiority test splits the space asymmetrically: the null hypothesis becomes precisely “deterioration exceeded the margin,” and the alternative is “deterioration did not exceed the margin.” That flip in which statement carries the burden of proof is what makes non-inferiority testing structurally different, not just a rewording of the same test.

## Mathematical formulation

Let  $\Delta = \mu_T - \mu_C$  be the true (population) difference between treatment and control on a metric where higher values are worse (e.g., a fraud rate). Let  $\delta > 0$  be the non-inferiority margin, defined so that a deterioration of up to  $\delta$  is considered acceptable.

The hypotheses are:

$$H_0 : \Delta \geq \delta \quad H_1 : \Delta < \delta$$

Note the asymmetry:  $H_0$  states the metric got worse by at least the margin (the “bad” scenario is the null);  $H_1$  states any deterioration, if present, is smaller than the margin. This is the structural opposite of a superiority test, where the null hypothesis is “no effect” and the alternative is “there’s an effect.”

The test statistic follows the same logic as a standard effect test, but compared against  $\delta$  instead of zero:

$$z = \frac{\hat{\Delta} - \delta}{SE(\hat{\Delta})}$$

$H_0$  is rejected (concluding non-inferiority) when  $z < -z_{1-\alpha}$ , which is, more intuitively, equivalent to checking whether the upper bound of a one-sided  $(1 - \alpha)$  confidence interval for  $\Delta$  falls below  $\delta$ :

$$\hat{\Delta} + z_{1-\alpha} \cdot SE(\hat{\Delta}) < \delta$$

In practice, this check is commonly reported using the upper bound of a two-sided  $(1-2\alpha)$  confidence interval, which is numerically equivalent to the one-sided  $(1 - \alpha)$  upper bound — for example, a two-sided 90% CI yields the same upper bound as a one-sided 95% CI.

## Assumptions

- **The margin  $\delta$  is set before looking at the experiment’s data**, as a business decision — it is not estimated or optimized from the experiment’s own data. Setting the margin after seeing the result invalidates the test’s interpretation.
- **The direction of “worse” is clearly defined** for the metric at hand. For metrics where higher values are bad (fraud, errors, latency), the margin caps how much the metric can rise; for metrics where lower values are bad (satisfaction, retention), the framing flips and the margin caps how much it can fall.
- **The same assumptions as a standard effect test apply** — valid randomization, independence between units, a sample large enough for the normal approximation of the standard error to be reasonable (or use of exact/bootstrap methods when it isn’t).
- **The design has enough statistical power for the test to be informative** — with a small sample and a tight margin, the test can be structurally incapable of concluding non-inferiority even when the true effect is favorable, simply for lack of precision.

## Hypotheses

As detailed above:

$$H_0 : \Delta \geq \delta \quad H_1 : \Delta < \delta$$

The significance level  $\alpha$  (typically 0.05, one-sided) represents the tolerated error rate of concluding non-inferiority when the metric actually got worse than the margin. Unlike a two-sided test, there is only one relevant direction of error here — there’s no symmetric concern about a “false positive for improvement,” because improvement isn’t what’s being tested.

## Interpretation

Concluding non-inferiority is a specific statistical statement: there is evidence, at the chosen confidence level, that the true deterioration does not exceed  $\delta$ . It is not a claim of equivalence (that would require an equivalence test, with margins on both sides) nor a claim of zero cost — the point estimate can indicate a small deterioration, just one that’s within what’s tolerable.

An inconclusive result — when the confidence interval’s upper bound crosses  $\delta$  — should not be read as “the metric got much worse.” It only means the data don’t provide enough evidence to rule that possibility out. The most common cause of inconclusiveness is insufficient statistical power: a small sample, a highly variable metric, or a margin set too tight relative to the precision the design can deliver. Before reporting an inconclusive result as final, it’s worth checking whether the design had adequate power for the chosen margin — a calculation that mirrors the sample-size calculation for standard effect tests, but using  $\delta$  as the reference “minimum effect.”

## Limitations

- The margin is a business choice, and different choices of margin can lead to different conclusions on exactly the same data — the test doesn’t validate the chosen margin, it only tests against it.

- Non-inferiority concluded in a single experiment, with a specific sample and context, doesn't guarantee the metric will stay within the margin indefinitely — shifts in user composition, scale, or context can change the metric's real behavior over time.
- Testing multiple guardrail metrics at once, each with its own non-inferiority test, runs into the same multiple-comparisons issue as any battery of tests — worth accounting for when interpreting “all guardrails passed.”
- A non-inferiority test does not replace an effect test on the primary metric; the two answer different questions and are usually used together, one for the intended gain, the other for the risk being monitored.

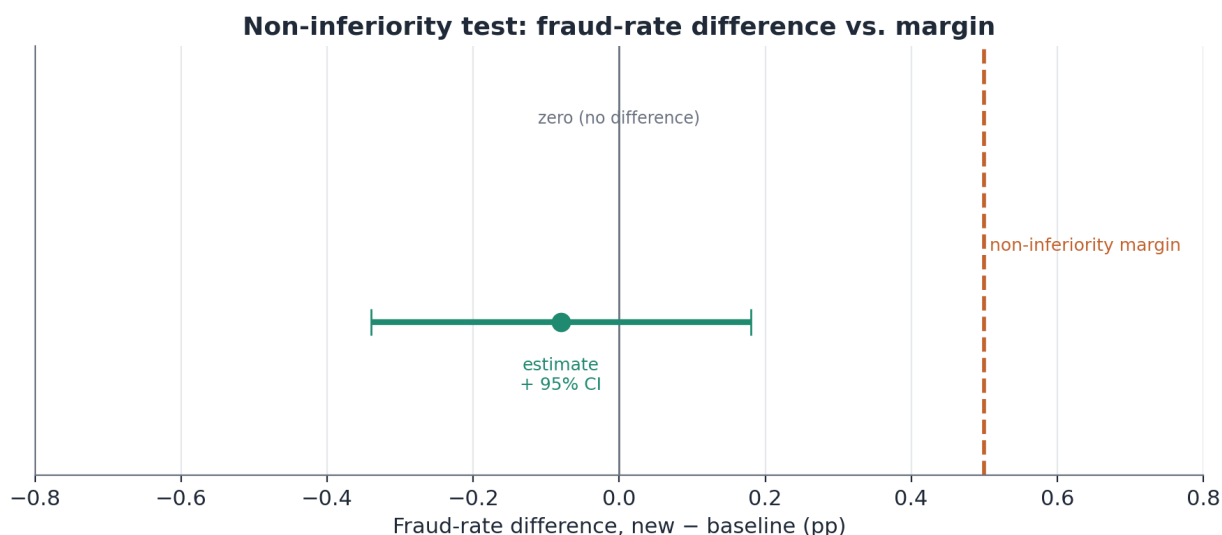
## Example

A payments company tests a new fraud-detection model, significantly faster than the current one but potentially slightly less precise. Before the experiment, the risk team sets the non-inferiority margin at a 0.5 percentage-point increase in fraud rate — a threshold above which the speed gain wouldn't offset the risk taken on.

The experiment runs with 12,000 transactions in each arm:

|              | Control (current model) | Treatment (new model) |
|--------------|-------------------------|-----------------------|
| Transactions | 12,000                  | 12,000                |
| Fraud cases  | 252                     | 242                   |
| Rate         | 2.10%                   | 2.02%                 |

The observed difference is  $\hat{\Delta} = 2.02\% - 2.10\% = -0.08$  percentage points (the new model had slightly lower fraud). The standard error of the difference, computed the same way as in a standard effect test, yields a 95% confidence interval of roughly  $[-0.34, +0.18]$  percentage points.



Illustrative data, not drawn from any real project.

Figure 15: Point estimate and 95% CI for the fraud-rate difference compared against the 0.5-pp non-inferiority margin

Since the interval's upper bound (+0.18 pp) sits below the 0.5 pp margin, non-inferiority is concluded with 95% confidence: there's statistical evidence the new model doesn't push fraud beyond what's tolerable.

To illustrate how the same logic would apply to other possible outcomes of the same experiment, consider three hypothetical scenarios for the CI of the difference, all against the same 0.5 pp margin:

- **Scenario A (superior)**: CI of  $[-0.70, -0.14]$  — entirely below zero. The new model isn't just non-inferior, it shows evidence of a real improvement.
- **Scenario B (non-inferior, not superior)**: CI of  $[-0.20, +0.30]$  — crosses zero, but the upper bound still sits below the 0.5 margin. No evidence of improvement, but evidence of non-inferiority.
- **Scenario C (inconclusive)**: CI of  $[+0.05, +0.65]$  — the upper bound exceeds the margin. Non-inferiority cannot be confidently concluded; the observed deterioration may or may not exceed what's tolerable.

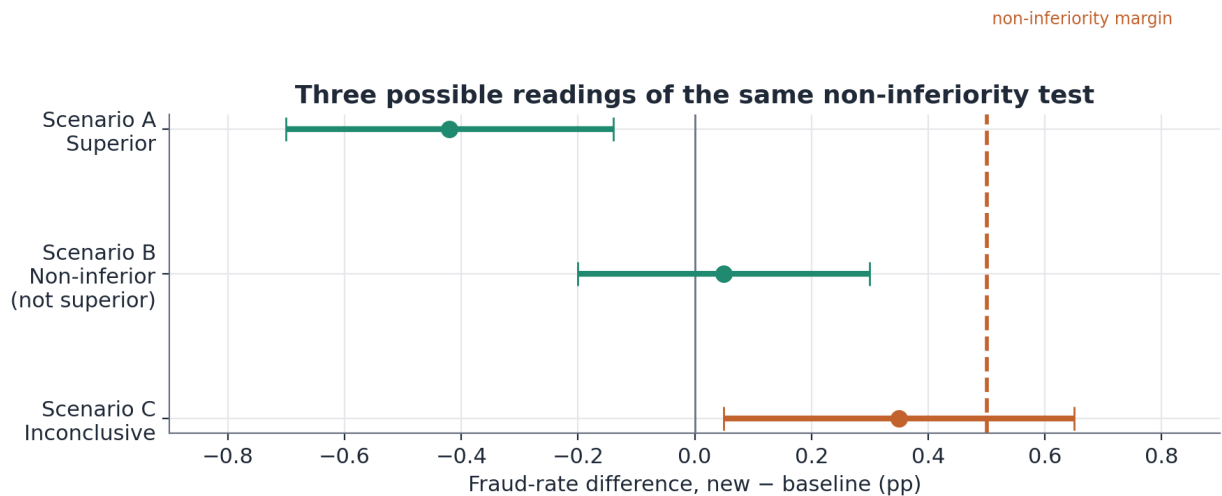


Figure 16: Three hypothetical CI scenarios compared against the same non-inferiority margin

Note that only Scenario C leads to a decision to hold off without further investigation — Scenarios A and B, despite having quite different point estimates and interval shapes, lead to the same practical decision to approve the launch from the guardrail metric's point of view.

# Heterogeneity testing and novelty-effect decay

## Concept

An experiment summarized in a single average-effect estimate answers only “is the effect, on average, different from zero?” Two additional questions, often relevant in practice, fall outside that single number:

- **Effect heterogeneity (interaction):** is the treatment effect constant across subgroups of the population, or does it vary systematically with some observable characteristic (loyalty tier, acquisition channel, device type, region)? This is a question about variation **across units**, measured at a given point in time.
- **Novelty-effect decay:** does the effect of a change stay stable over time since launch, or does part of it reflect a transient curiosity reaction (a “novelty effect,” in the broad sense of the Hawthorne literature) that fades once the change stops being new? This is a question about variation **over time**, measured within the same group.

The two questions are conceptually distinct and should be tested separately, even though they often show up together in practice: an effect can be homogeneous across subgroups and still decay; it can be heterogeneous and stable; it can be both at once, or neither.

## Mathematical formulation

### Interaction test (heterogeneity)

Let  $Y_i$  be the outcome of interest for unit  $i$ ,  $T_i \in \{0, 1\}$  the treatment indicator, and  $S_i$  a subgroup indicator (for simplicity, binary:  $S_i = 1$  for the subgroup of interest,  $S_i = 0$  for the reference subgroup). The interaction model is:

$$Y_i = \beta_0 + \beta_1 T_i + \beta_2 S_i + \beta_3 (T_i \times S_i) + \varepsilon_i$$

where:

- $\beta_1$  is the treatment effect in the reference subgroup ( $S_i = 0$ );
- $\beta_2$  captures the baseline-level difference between subgroups, independent of treatment;
- $\beta_3$  is the **interaction coefficient** — the difference in treatment effect between the subgroup of interest and the reference subgroup. It is the central parameter of the heterogeneity test;
- $\varepsilon_i$  is the error term, subject to the usual care around heteroskedasticity and, when relevant, within-cluster correlation (see the document on cluster-robust standard errors).

The heterogeneity test is the standard hypothesis test on  $\beta_3$ :  $H_0 : \beta_3 = 0$  against  $H_1 : \beta_3 \neq 0$ , using the statistic  $t = \hat{\beta}_3 / \widehat{SE}(\hat{\beta}_3)$ .

With more than two subgroups, this generalizes to a categorical variable  $S_i$  with  $K$  categories, producing  $K - 1$  interaction terms (one per category, against the reference category), and the joint heterogeneity test becomes an F-test on all  $K - 1$  interaction coefficients simultaneously:

$$H_0 : \beta_3^{(1)} = \beta_3^{(2)} = \dots = \beta_3^{(K-1)} = 0$$

## Novelty-effect decay

Let  $w = 1, \dots, W$  index the time window (for example, week) since launch. For each window, estimate the treatment effect  $\hat{\tau}_w$  separately, with its standard error, using only the observations from that window:

$$Y_{i,w} = \alpha_w + \tau_w T_i + \varepsilon_{i,w}, \quad w = 1, \dots, W$$

The object of interest becomes the sequence  $\{\hat{\tau}_1, \hat{\tau}_2, \dots, \hat{\tau}_W\}$ . A direct way to test decay is to regress that sequence of estimates on  $w$ :

$$\hat{\tau}_w = \gamma_0 + \gamma_1 w + u_w$$

with  $\gamma_1 < 0$  and statistically significant as evidence of decay. A more robust alternative, when the number of windows allows it, is to estimate a single model directly with a treatment  $\times$  continuous-time interaction, analogous to the interaction term above, treating time since launch as the subgroup variable.

## Assumptions

- **Subgroups defined before looking at the effect data.** The interaction test presumes the compared subgroups were defined by a reasonable criterion and, ideally, specified before running the experiment — not chosen after trawling through which cuts “look like” they have a larger effect.
- **Correctly specified errors.** As in any regression, valid inference on  $\beta_3$  depends on standard errors suited to the data’s dependence structure (for example, robust to heteroskedasticity, or clustered when observations are not independent within a group).
- **Stable composition over time (decay).** Comparing  $\hat{\tau}_w$  across windows assumes the population composition in each window is comparable — if very different users enter the experiment in different weeks (for example, because of an acquisition campaign), part of the observed variation in  $\hat{\tau}_w$  may reflect a composition shift, not genuine decay of the effect on the same type of user.
- **Sufficient sample per subgroup and per window.** Splitting the data into subgroups or time windows shrinks the sample size in each cell, inflating standard errors — a heterogeneity test without enough statistical power simply fails to detect real differences.

## Hypotheses

### Interaction test:

- $H_0$ :  $\beta_3 = 0$  — the treatment effect is equal across the compared subgroups.
- $H_1$ :  $\beta_3 \neq 0$  — the effect differs across subgroups.
- Test statistic:  $t = \hat{\beta}_3 / \widehat{SE}(\hat{\beta}_3)$ , compared to the t distribution (or normal, in large samples) under  $H_0$ .
- Conventional significance level: 5%, but see the limitations section on multiple comparisons when several subgroups are tested.

### Decay test:

- $H_0$ :  $\gamma_1 = 0$  — there is no downward trend in the effect across time windows.
- $H_1$ :  $\gamma_1 < 0$  (one-sided test, if the question is specifically about decay) or  $\gamma_1 \neq 0$  (two-sided test, if any pattern change matters).
- A 95% confidence interval for  $\hat{\tau}_w$  in each window lets you see directly whether the trajectory of estimates is consistent with a constant effect (all bands overlap) or a systematic decline.

## Interpretation

A significant interaction coefficient says the effect **differs** across the tested subgroups — it does not automatically say which subgroup is “better” in absolute terms, nor why the difference exists. A larger effect in one subgroup could reflect genuinely greater sensitivity to the change, or simply a lower baseline with more room to improve.

Detecting novelty decay does not mean the product “doesn’t work” — it means the long-run effect magnitude is smaller than the initial measurement suggested. Ideally, the launch decision should rely on the stabilized effect (later windows), not the peak effect (earliest windows), when the goal is to understand the sustainable gain.

Neither analysis, on its own, establishes any additional causal claim beyond what the original experiment’s randomization already guarantees — they merely decompose an already-identified average causal effect into how it is distributed across groups and over time.

## Limitations

- **Multiple comparisons.** Testing heterogeneity across several subgroups simultaneously (by tier, channel, device, region, and so on) inflates the false-positive rate. Testing 10 independent subgroups at 5% significance, the probability of at least one “significant” result by pure chance, even with no real heterogeneity, exceeds 40%. Corrections such as Bonferroni, or requiring replication in a follow-up experiment, mitigate this risk. A subgroup defined *a priori*, with a clear theoretical rationale, is always more trustworthy than one discovered by exploring the data after the fact.
- **Reduced statistical power in small subgroups.** Splitting the sample increases the standard error of each subgroup estimate — a lack of significance in a small subgroup may simply reflect insufficient sample, not the absence of an effect.
- **Confounding composition with genuine decay.** Changes in the population composition across weeks (seasonality, acquisition campaigns, calendar effects) can produce a trajectory that looks like decay without the effect, for the same user, actually having fallen.
- **Time windows too short to conclude stabilization.** Observing decay in the first few weeks does not guarantee the effect goes to zero — it may stabilize at a smaller positive level. Extrapolating the trend beyond the observed period is speculative.

## Example

Consider a brokerage app testing a new portfolio-summary screen, aiming to increase weekly logins. The average estimated effect, across the full base, is +6.0% relative to control, with a standard error of 1.2 (statistically significant).

**Heterogeneity by investor profile.** Splitting by account tenure:

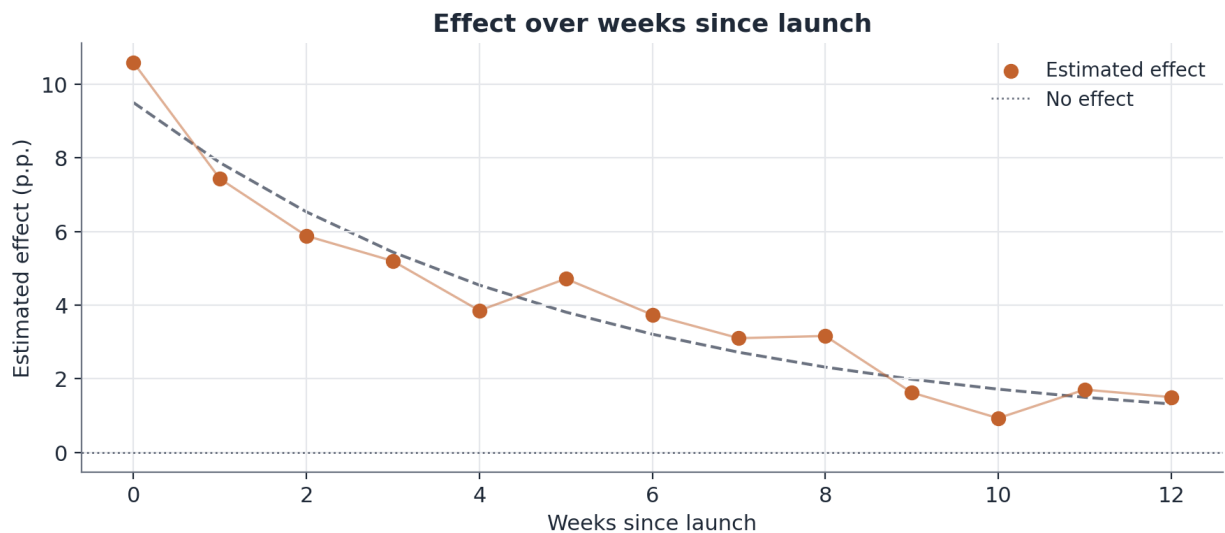
| Subgroup                   | Estimated effect | Standard error |
|----------------------------|------------------|----------------|
| Accounts under 6 months    | +11.4%           | 2.0            |
| Accounts 6 months or older | +2.1%            | 1.4            |

The difference between the two effects is 9.3 percentage points. The standard error of the difference, combining the two standard errors under independence,  $\sqrt{2.0^2 + 1.4^2} \approx 2.44$ , gives a statistic  $t \approx 3.8$  — well above the usual significance threshold. There is solid evidence of heterogeneity: new accounts respond much more strongly to the change than established ones, a plausible pattern, since new users are still forming usage habits.

**Novelty decay.** Looking at the aggregate effect week by week since launch:

| Week | Estimated effect |
|------|------------------|
| 1    | +12.8%           |
| 2    | +9.4%            |
| 4    | +6.7%            |
| 8    | +3.9%            |
| 12   | +3.5%            |

The trajectory declines consistently through week 8 and then stabilizes near 3.5%. The correct reading is not “the change doesn’t work” — it is that the initial peak effect (nearly 13%) included a temporary curiosity component, and the real sustainable gain, relevant for long-run projections, is closer to 3.5%.



Illustrative data, not drawn from any real project.

Figure 17: Effect over weeks since launch, illustrating novelty-effect decay

Combining both analyses: the better-informed product decision is not “roll out to everyone with the +6% average effect,” but something like “prioritize new accounts in the rollout, and project the long-run gain using the stabilized level of roughly +3.5% to +4%, not the initial peak.”

# Statistical power and minimum detectable effect (MDE)

## Concept

Statistical power and minimum detectable effect (MDE) are two faces of the same mathematical relationship between four quantities in a hypothesis test: sample size, the metric's variance, the significance level, and power. Given any three, the fourth is determined.

A test's **power** is the probability of correctly rejecting the null hypothesis when it is in fact false — that is, the probability of detecting a true effect of a given size, given the experiment's design. It is the complement of the Type II error rate ( $\beta$ ):  $\text{power} = 1 - \beta$ .

The **MDE** flips the question: instead of fixing a hypothetical effect and asking what the power is, it fixes the desired power (conventionally 80%) and asks what the smallest true effect is that the design can detect with that probability. In practice, computing the MDE before running an experiment is the most direct way to answer “does this design have any chance of working?” before spending time and resources on it.

The core usefulness of the concept is avoiding two symmetric mistakes: (1) running an underpowered experiment that never had the strength to detect the effect that actually matters for the business decision, producing a “non-significant” result that means “no power to see it,” not “no effect”; and (2) over-powering an experiment beyond what's needed, consuming traffic and time that could go toward other tests.

## Mathematical formulation

For a comparison of two proportions (the most common case in product experiments), with equal sample sizes in each arm, the relationship between the four quantities can be written, under a normal approximation, as:

$$\text{MDE} = (z_{(1-\alpha/2)} + z_{(1-\beta)}) \cdot \sqrt{2 \cdot p \cdot (1-p) / n}$$

where:

- MDE is the minimum detectable effect, in percentage points (the relative version is  $\text{MDE} / p$ ).
- $z_{(1-\alpha/2)}$  is the standard normal quantile corresponding to the chosen two-sided significance level — for example, 1.96 for  $\alpha = 0.05$ .
- $z_{(1-\beta)}$  is the standard normal quantile corresponding to the desired power — for example, 0.84 for 80% power.
- $p$  is the expected baseline conversion rate (or proportion) in the control group.
- $n$  is the sample size per group.

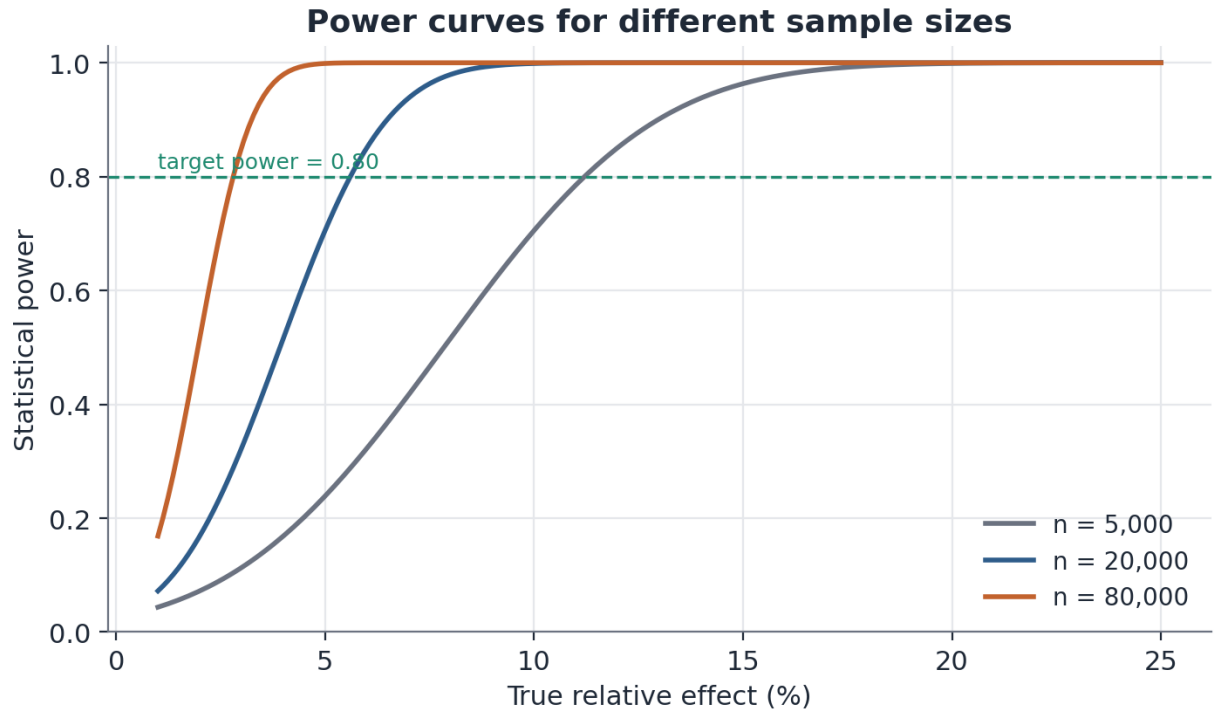
For a continuous metric (say, usage time, purchase value), the equivalent form uses the metric's standard deviation ( $\sigma$ ) instead of the  $p(1-p)$  term:

$$\text{MDE} = (z_{(1-\alpha/2)} + z_{(1-\beta)}) \cdot \sigma \cdot \sqrt{2 / n}$$

Solving for  $n$  gives the sample size required for a target MDE:

$$n = 2 \cdot \sigma^2 \cdot (z_{(1-\alpha/2)} + z_{(1-\beta)})^2 / \text{MDE}^2$$

The structurally important point in these equations is that  $n$  depends on  $1/\text{MDE}^2$ : to cut the MDE in half, the sample needs to quadruple — hence the diminishing returns of throwing more traffic at an experiment.



Illustrative data, not drawn from any real project.

Figure 18: Power curves for different sample sizes

## Assumptions

- **Normal approximation to the distribution of the mean/proportion difference:** valid for reasonably large samples via the central limit theorem; for very small samples or highly skewed metrics, the normal approximation degrades and exact forms (for example, based on the binomial distribution) are preferable.
- **Correctly estimated variance:** the MDE calculation depends on a prior estimate of the metric's variance (or baseline conversion rate). If that estimate comes from an atypical period or a small sample, the computed MDE inherits that error.
- **Independent observations:** the standard calculation assumes each experimental unit contributes independently. When there's cluster structure (say, the same store generating multiple observations over time), the effective variance is larger than the simple formula suggests, and the true MDE is larger than the one computed without that adjustment.
- **Allocation and significance level fixed in advance:** the calculation assumes a single significance check at the end of the experiment, with a fixed  $\alpha$ . If the experiment is monitored repeatedly and stopped as soon as a "significant" result appears, the effective false-positive rate rises above the nominal  $\alpha$  — a separate but related problem, handled by sequential testing methods.

## Hypotheses

Power and MDE aren't, by themselves, a hypothesis test — they're properties of a planned hypothesis test. The underlying test is typically:

- **H0**: there is no difference between the treatment and control group on the metric of interest (the difference in means or proportions equals zero).
- **H1**: there is a nonzero difference.
- **Significance level** ( $\alpha$ ): the accepted probability of incorrectly rejecting H0 (Type I error), typically 5%.
- **Target power** ( $1 - \beta$ ): the desired probability of correctly rejecting H0, given that an effect of size equal to the MDE truly exists, typically 80%.

The MDE is the smallest true effect for which this probability of correct rejection reaches the target power, given the assumed  $n$ ,  $\alpha$ , and variance.

## Interpretation

The MDE computed before an experiment is a property of the **design**, not a forecast of the true effect. Common misinterpretations include treating a 3% MDE as “we expect a 3% effect” (wrong — it’s the detection floor, not a forecast) or concluding, at the end of a non-significant experiment, that “there’s no effect” without checking whether the design ever had the power to detect an effect of the size that would matter for the decision.

Recomputing power mid-experiment is useful and legitimate when the available sample size changes for reasons external to the test’s own result — for example, a business deadline shortens the experiment window, or observed traffic came in below plan. In that case, the MDE (or the power for a specific effect) is recalculated using the  $n$  actually reached, and the change is reported transparently: “the original design was set to detect effects of X%; with the reduced available sample, the detection floor rose to Y%.” This is different from — and legitimate, unlike — repeatedly checking the experiment’s own p-value and deciding to stop as soon as it crosses 0.05, which inflates the false-positive rate.

## Limitations

- The calculation depends on a variance estimate that’s only known precisely after data is collected — beforehand, it relies on an approximation (from similar historical data, a pilot, or a conservative range), which carries its own uncertainty.
- MDE and power assume a single two-sided test at the end of the experiment; tests with multiple secondary metrics, multiple subgroup cuts, or continuous p-value monitoring aren’t covered by the simple calculation without additional adjustment (correction for multiple comparisons, or sequential methods).
- A design with adequate power for the primary metric may have essentially no power for secondary metrics with much higher variance or a much lower baseline rate — it’s worth computing the MDE separately for every metric that matters to the decision.
- High power doesn’t imply that a detected effect will be relevant to the business — a very small MDE can detect effects that are statistically significant but practically irrelevant. The MDE should be compared against the smallest effect worth acting on, not just computed and set aside.

## Example

A hypothetical streaming service wants to test whether a new content recommendation algorithm increases average daily usage time, currently at 42 minutes, with a historical standard deviation of 35 minutes (the metric is fairly noisy, since it mixes occasional users with very engaged ones).

The product team estimates, based on prior recommendation-change tests, that a realistic gain would fall between 2 and 4 minutes per day. Before committing the experiment to four weeks of traffic, the MDE is calculated for different durations:

| Duration | n per group (approx.) | MDE (minutes) |
|----------|-----------------------|---------------|
| 1 week   | 8,000                 | 2.2           |
| 2 weeks  | 16,000                | 1.5           |
| 4 weeks  | 32,000                | 1.1           |

With just one week, the 2.2-minute MDE would already cover the lower end of the expected effect (2 minutes) — but with a thin margin, making the result sensitive to noise. The team opts for two weeks, with an MDE of 1.5 minutes, comfortably below the smallest effect they consider relevant.

Midway through the experiment, a technical issue cuts eligible traffic by 40% over three days. By the end of the planned two weeks, the effective *n* per group came in at 11,200, not 16,000. Recalculating the MDE with the actual *n*:

```
effective_MDE = (1.96 + 0.84) · 35 · sqrt(2 / 11,200) 1.85 minutes
```

The effective MDE (1.85 min) is still below the minimum relevant effect (2 minutes), so the experiment, even reduced, retains enough power for the decision that matters — but the team logs this change in the experiment’s documentation instead of silently assuming the original design (with *n*=16,000) still held.

## Code example

```
import numpy as np
from statsmodels.stats.power import TTestIndPower

analysis = TTestIndPower()

sigma = 35.0          # historical standard deviation of the metric, minutes
alpha = 0.05
target_power = 0.80

# MDE (in minutes) for different sample sizes per group
for n in [8_000, 16_000, 32_000, 11_200]:
    effect_size = analysis.solve_power(nobs1=n, alpha=alpha, power=target_power)
    mde_minutes = effect_size * sigma
    print(f"n={n:>6}: MDE {mde_minutes:.2f} min")
```

The `effect_size` returned by `statsmodels` is standardized (units of standard deviation); multiplying by `sigma` converts it back to the metric’s original unit — minutes, in this example.

# Spatial spillover and cannibalization analysis

## Concept

In experiments with geographically located units — stores, delivery zones, service areas — one of the most frequently violated, and least checked, assumptions is **unit isolation** (SUTVA, the stable unit treatment value assumption): the idea that a control unit’s outcome depends only on it not having been treated, not on what’s happening at treated units nearby.

When treated and control units compete for the same audience — the same delivery radius, the same commercial neighborhood — that assumption fails through a specific mechanism called **spatial spillover**, of which **cannibalization** is the most common case in business contexts: part of the activity that shows up as “gain” at the treated unit is actually displaced activity from a neighboring control unit, not new activity. When this happens and goes undetected, the experiment’s estimated effect systematically overstates the true net gain, because the control unit used as a reference was also affected by the treatment — just in the opposite direction.

Spatial spillover analysis uses the experimental design’s own geography as a diagnostic instrument: by comparing the effect estimated against controls at different distances from the nearest treated unit, it becomes possible to separate, at least approximately, how much of the apparent effect is displacement versus net creation.

## Mathematical formulation

Let  $Y_i$  be the metric of interest (sales, orders, revenue) for unit  $i$ , measured before ( $Y_i^{\text{pre}}$ ) and after ( $Y_i^{\text{post}}$ ) treatment begins, and let  $d_i$  be the distance between control unit  $i$  and the nearest treated unit.

Define two control subgroups using a distance threshold :

Nearby control:  $C_{\text{near}} = \{ i : \text{treated}_i = 0, d_i \leq \tau \}$   
Distant control:  $C_{\text{far}} = \{ i : \text{treated}_i = 0, d_i > \tau \}$

The estimated effect against each group, in a simple difference-in-differences form, is:

$$\begin{aligned} \_near &= ( \bar{y}_T^{\text{post}} - \bar{y}_T^{\text{pre}} ) - ( \bar{y}_{\text{near}}^{\text{post}} - \bar{y}_{\text{near}}^{\text{pre}} ) \\ \_far &= ( \bar{y}_T^{\text{post}} - \bar{y}_T^{\text{pre}} ) - ( \bar{y}_{\text{far}}^{\text{post}} - \bar{y}_{\text{far}}^{\text{pre}} ) \end{aligned}$$

where  $\bar{y}_T$ ,  $\bar{y}_{\text{near}}$ , and  $\bar{y}_{\text{far}}$  are the metric’s averages in the treated group, the nearby control, and the distant control, respectively.

The gap between the two estimated effects is a direct indicator of spillover contamination:

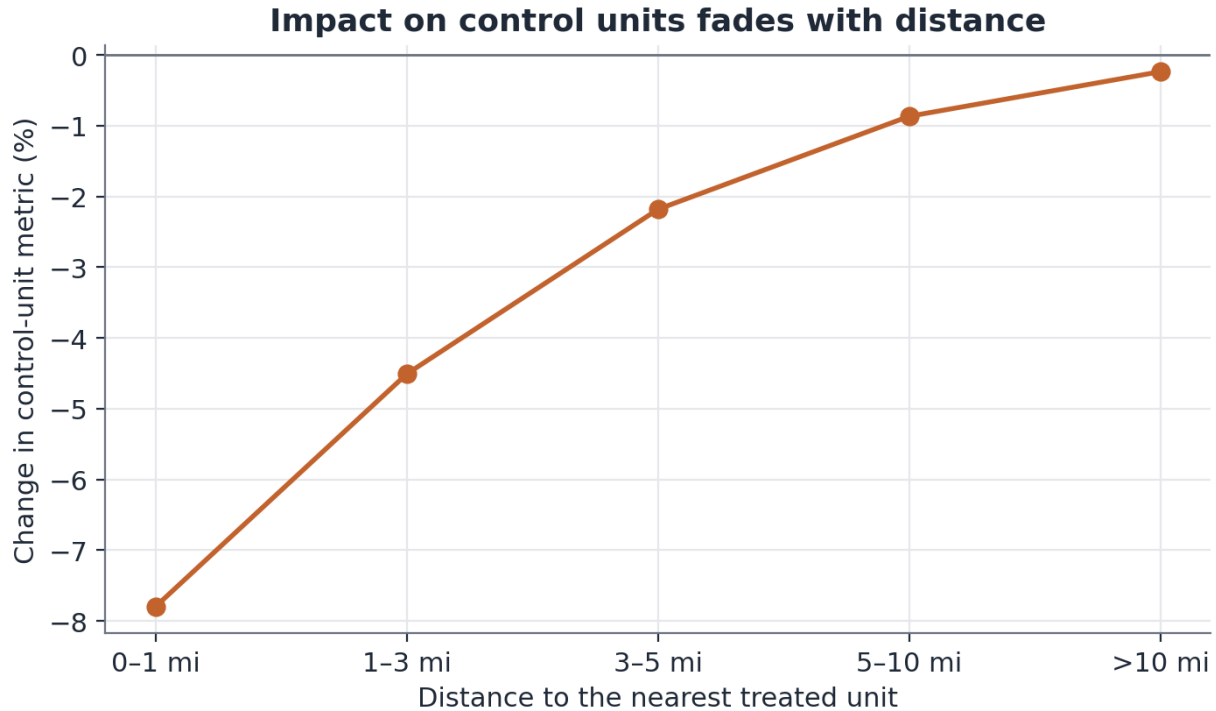
$$\text{Cannibalization indicator} = \_near - \_far$$

A positive, meaningful value indicates that the nearby control is, on average, performing worse than the distant control during the treatment period — consistent with the hypothesis that it’s losing activity to the treated units. Formally, under the assumption that the distant control is free of contamination,  $\_far$  is the estimator of the treatment’s **true net effect**, and  $\_near - \_far$  estimates the size of the displacement effect absorbed by the nearby control.

A continuous extension of the same logic models the effect on control units as a function of distance  $d_i$  to the nearest treated unit — typically with smooth decay, for example via local (kernel) regression or a simple parametric specification such as:

$$\Delta y_i = \alpha + \beta \cdot \exp(-d_i / \gamma) + \delta_i$$

where  $\Delta y_i$  is the percentage change in the metric at control unit  $i$ ,  $\gamma$  is a scale parameter capturing how fast the effect decays with distance, and  $\delta_i$  measures the magnitude of the spillover effect in the limit  $d_i \rightarrow 0$ .



Illustrative data, not drawn from any real project.

Figure 19: Impact on control units as a function of distance to the nearest treated unit

## Assumptions

- **The distant control is genuinely free of contamination:** this is the central identifying assumption. If the true competition radius is larger than the assumed “distant” threshold, even that group carries some bias, and the cannibalization indicator understates true contamination.
- **Which units were treated wasn’t correlated with local characteristics that also affect the metric’s trend** (for example, preferentially treating stores in already-growing neighborhoods) — otherwise, the comparison against any control, nearby or distant, is confounded by that local characteristic, not just by spillover.
- **Stable market boundaries during the experiment:** the analysis assumes the spatial competition pattern (who competes with whom) doesn’t shift substantially during the experiment window.
- **Distance as a reasonable proxy for competition:** straight-line geographic distance (or travel time) needs to be a reasonable proxy for actual customer overlap — in digital contexts, “distance” may need redefining (for example, coverage-area overlap, or user-base similarity) rather than strictly geographic.

## Hypotheses

When formalized as a hypothesis test on the cannibalization indicator:

- **H0:** `_near` = `_far` (no systematic difference between the effect measured against nearby control and against distant control — no evidence of spillover).
- **H1:** `_near`  $\neq$  `_far` (a systematic difference exists, consistent with spatial contamination).

The test can be run via a difference-in-differences model with an interaction term between treatment, period, and a “nearby control” indicator, or via permutation inference when the number of treated and control units is small — common in this type of design, since geographic experiments rarely have dozens of treated units.

## Interpretation

The analysis’s central result isn’t “is there or isn’t there cannibalization” in a binary sense, but **how much** of the apparent effect is displacement. A cannibalization indicator of, say, 4 percentage points against a total apparent effect of 17% means roughly a quarter of the “gain” measured against the nearby control is actually that control’s loss — the true net effect sits closer to 13%.

It’s important not to interpret a smaller net effect as “the change didn’t work.” Demand shifting between units of the same business can be a desirable outcome (for example, migrating customers to a more profitable channel) or an undesirable one (an expansion that in practice just reallocates existing revenue), depending entirely on the strategic goal behind the tested change. Spillover analysis doesn’t decide that — it just separates the two components so the decision can be made with that information.

## Limitations

- The choice of distance threshold is arbitrary to some degree, and results can be sensitive to it — it’s worth reporting the cannibalization indicator for more than one threshold, or using the continuous version (decay with distance) when the data allows.
- With few treated and control units, the estimates of `_near` and `_far` carry considerable uncertainty, and the gap between them inherits that uncertainty — a “positive” cannibalization indicator over small samples may not be distinguishable from zero.
- The method assumes the only relevant form of contamination is geographic. Other forms of spillover (for example, the same customer buying through different channels of the same company, or social-network effects between customers) aren’t captured by physical distance and require different designs.
- Even when the distant control is well chosen, it estimates the net effect **aggregated over the observed region** — it says nothing about whether growth is sustainable outside that region, which is a limitation of extrapolating any local pilot result to a national rollout.

## Example

A hypothetical gym chain tests a new membership promotion in 10 locations across a metro area, measuring monthly new sign-ups.

Control group definitions: 18 locations within 4 km of some treated location (nearby control) and 22 locations more than 15 km from any treated location (distant control), all in the same city.

Aggregated data (average monthly sign-ups per location):

| Group           | Before | After | Change |
|-----------------|--------|-------|--------|
| Treated         | 42     | 58    | +38.1% |
| Nearby control  | 40     | 36    | −10.0% |
| Distant control | 41     | 43    | +4.9%  |

Effect estimated against nearby control:  $38.1\% - (-10.0\%) = \mathbf{48.1 \text{ percentage points}}$ .

Effect estimated against distant control:  $38.1\% - 4.9\% = \mathbf{33.2 \text{ percentage points}}$ .

Cannibalization indicator:  $48.1 - 33.2 = \mathbf{14.9 \text{ percentage points}}$ .

Interpretation: of the roughly 48 percentage points of apparent “advantage” the promotion shows when compared to neighboring locations, about 15 points reflect members who switched from nearby gyms, not net market growth. The true net effect, closer to 33 percentage points, is the relevant number for deciding whether the promotion, rolled out chain-wide (where there would no longer be any “untreated neighboring locations” to cannibalize), would generate comparable growth — a hypothesis that would need separate testing, since cannibalization disappears once every location is treated simultaneously, but true market growth may or may not hold at the same magnitude.

## Code example

```
import numpy as np
import pandas as pd
from scipy import stats

rng = np.random.default_rng(11)

# illustrative data: 10 treated, 18 nearby control, 22 distant control locations
n_treated, n_near, n_far = 10, 18, 22

signups = pd.concat([
    pd.DataFrame({
        "group": "treated",
        "before": rng.normal(42, 4, n_treated),
        "after": rng.normal(58, 5, n_treated),
    }),
    pd.DataFrame({
        "group": "nearby_control",
        "before": rng.normal(40, 4, n_near),
        "after": rng.normal(36, 4, n_near),
    }),
    pd.DataFrame({
        "group": "distant_control",
        "before": rng.normal(41, 4, n_far),
        "after": rng.normal(43, 4, n_far),
    }),
])
```

```

def pct_change(df):
    return (df["after"].mean() / df["before"].mean() - 1) * 100

change_treated = pct_change(signups[signups.group == "treated"])
change_near = pct_change(signups[signups.group == "nearby_control"])
change_far = pct_change(signups[signups.group == "distant_control"])

effect_near = change_treated - change_near
effect_far = change_treated - change_far
cannibalization_indicator = effect_near - effect_far

print(f"Effect vs. nearby control:    {effect_near:.1f} pp")
print(f"Effect vs. distant control:    {effect_far:.1f} pp")
print(f"Cannibalization indicator:    {cannibalization_indicator:.1f} pp")

```

# Cluster-robust standard errors

## Concept

Classical statistical inference (ordinary least squares standard errors, for example) assumes model errors are independent and identically distributed across observations. That assumption is violated whenever the data has a **grouping** structure: observations within the same cluster (store, region, household, classroom, sampling unit of a survey) share unobserved shocks in common, which generates within-cluster correlation in the errors.

Ignoring that structure does not bias the point estimate of the coefficient — the least-squares estimator remains unbiased under the usual assumptions — but it biases the computed standard error **downward**, because the standard formula counts every observation as a fully independent unit of information, when in fact part of the within-cluster variation is repetition, not new information.

The cluster-robust standard error (also called the “cluster-robust variance estimator,” CRVE) corrects for this by allowing arbitrary correlation within each cluster and requiring independence only between clusters.

## Mathematical formulation

For a linear model  $Y_i = X_i'\beta + \varepsilon_i$ , the least-squares estimator is  $\hat{\beta} = (X'X)^{-1}X'Y$ . The classical variance of  $\hat{\beta}$  assumes  $\varepsilon_i$  i.i.d. with variance  $\sigma^2$ :

$$\widehat{\text{Var}}(\hat{\beta})_{\text{classical}} = \hat{\sigma}^2(X'X)^{-1}$$

The cluster-robust estimator, with  $G$  clusters indexed by  $g = 1, \dots, G$ , each with covariate matrix  $X_g$  and residual vector  $\hat{u}_g = Y_g - X_g\hat{\beta}$ , takes the “sandwich” form:

$$\widehat{\text{Var}}(\hat{\beta})_{\text{cluster}} = (X'X)^{-1} \left( \sum_{g=1}^G X_g'\hat{u}_g\hat{u}_g'X_g \right) (X'X)^{-1}$$

where:

- $X_g'\hat{u}_g\hat{u}_g'X_g$  is cluster  $g$ ’s contribution to the central term — it lets residuals from observations in the same cluster be correlated with each other in an arbitrary way, without imposing a specific correlation structure;
- the sum runs cluster by cluster, not observation by observation — that is what preserves the internal correlation instead of treating it as independent noise;
- $(X'X)^{-1}$  at each end (the “bread” of the sandwich) is the same matrix as in the classical estimator.

It is common to apply a finite-sample degrees-of-freedom correction, multiplying the result by a factor such as:

$$c = \frac{G}{G-1} \cdot \frac{n-1}{n-k}$$

with  $n$  the total number of observations,  $k$  the number of estimated parameters, and  $G$  the number of clusters — this correction helps, but does not fully resolve, the problematic behavior with few clusters (see limitations).

## Assumptions

- **Independence across clusters.** The correction is only valid if different clusters are genuinely independent of each other — if there is also dependence between clusters (for example, a macroeconomic shock hitting every region at the same time), the cluster-robust standard error still understates the real uncertainty.
- **Sufficiently large number of clusters.** The estimator’s asymptotic properties rely on  $G \rightarrow \infty$ . In practice, values of  $G$  below 30-40 make the normal approximation to the distribution of  $\hat{\beta}$  unreliable, even with a large  $n$  within each cluster.
- **The clustering variable is correctly specified.** Grouping at too fine a level (understating the true correlation) invalidates the correction just as much as ignoring the grouping entirely. When in doubt, the cluster level should match the level at which the treatment (or the source of dependence) actually varies.

## Hypotheses

The CRVE does not introduce a new hypothesis test — it is a method for computing standard errors, used within the usual tests on regression coefficients:

- $H_0: \beta_j = 0$  (the coefficient of interest is zero in the population)
- $H_1: \beta_j \neq 0$
- Test statistic:  $t = \hat{\beta}_j / \widehat{SE}_{\text{cluster}}(\hat{\beta}_j)$ , compared to the t distribution with  $G - 1$  degrees of freedom (a common and conservative choice, since the effective amount of information is closer to the number of clusters than to the number of observations).
- 95% confidence interval:  $\hat{\beta}_j \pm t_{0.975, G-1} \cdot \widehat{SE}_{\text{cluster}}(\hat{\beta}_j)$ .

## Interpretation

The point estimate of the effect does not change when swapping the classical standard error for the cluster-robust one — only the uncertainty around it changes. An effect that stops being significant under the cluster-robust standard error is not “less real” — the estimate is still the best one available; only the confidence claimed around it was inflated before the correction.

Worth stressing: the cluster-robust standard error fixes inference, not causal identification. If the study design has other problems (omitted variables, selection, lack of randomization), the cluster-robust standard error does nothing about those — it only guarantees that, given the point estimate, the stated margin of error is more faithful to the data’s dependence structure.

## Limitations

- **Few clusters.** This is the most-cited limitation in the applied literature. With few clusters (a common rule of thumb: below 30-40), the CRVE tends to understate the true variance, even with the degrees-of-freedom correction, and the test statistic’s distribution drifts from the assumed t approximation. In that setting, two alternative paths, each covered in its own document in this repository, tend to work better: **randomization inference**, when the number of possible treatment reassignments is small enough to enumerate (or sample) directly; and

**wild-cluster bootstrap**, which resamples residuals multiplied by random signs per cluster and tends to have better finite-sample behavior with few clusters than the asymptotic CRVE. As a rule of thumb: CRVE is the standard, simplest choice when there are enough clusters (dozens or more); with few clusters, prefer wild-cluster bootstrap as a more computationally accessible alternative, or randomization inference when the design’s structure lets you enumerate the possible reassignments naturally.

- **Very unequal cluster sizes.** A cluster much larger than the others can dominate the sum in the sandwich formula, making the variance estimate unstable.
- **Choice of clustering level.** Grouping at too fine a level (understating the true correlation) invalidates the correction just as much as ignoring the grouping entirely. When in doubt, the cluster level should match the level at which the treatment (or the source of dependence) actually varies.

## Example

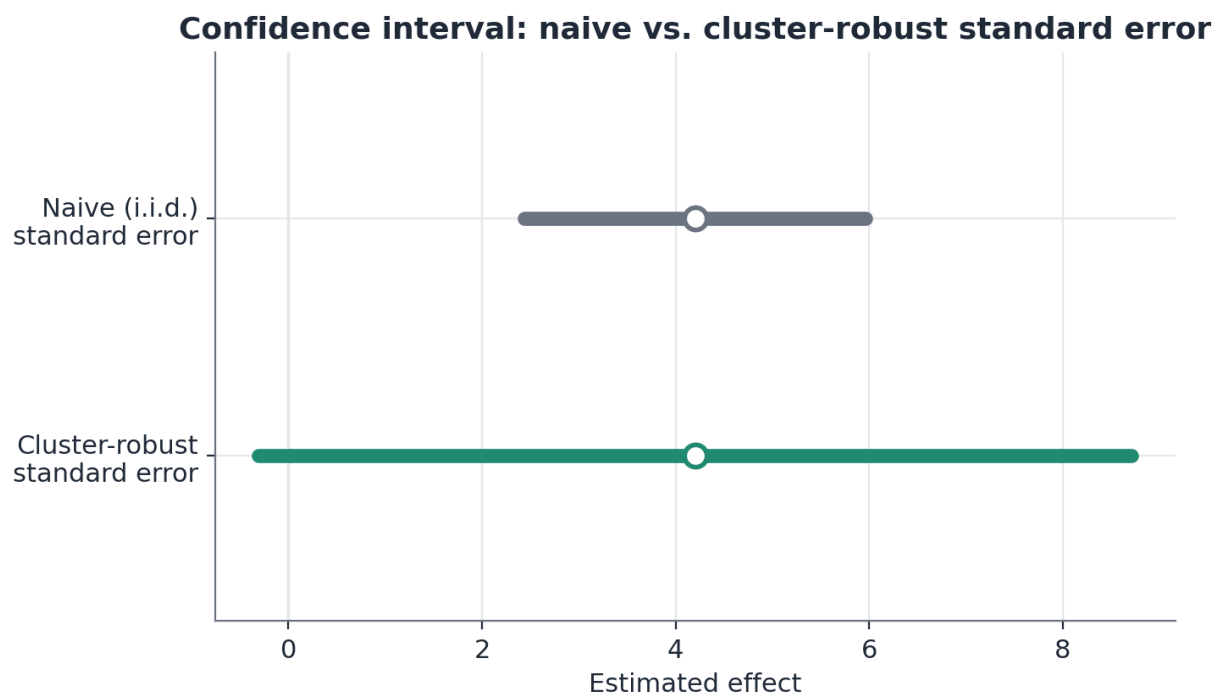
A coffee shop chain tests a new counter layout across 24 locations (12 treatment, 12 control), measuring average customer queue time, with about 150 customer observations per location — 3,600 observations in total.

A simple mean-comparison model, ignoring the cluster structure, estimates an effect of  $-1.8$  minutes on queue time, with a standard error of 0.25 (treating the 3,600 observations as independent), giving  $t = -7.2$  — strongly significant.

Recomputing the standard error as cluster-robust, grouping by location (24 clusters), the standard error rises to 0.68 — almost three times larger — giving  $t = -2.65$ . Still significant at the 5% level, but with a substantially more realistic confidence margin: the 95% interval moves from roughly  $[-2.3, -1.3]$  to  $[-3.2, -0.4]$ .

The reason for the difference: customers at the same location share the same barista, the same peak-hour flow, and the same physical queue layout — factors that make queue times within the same location resemble each other for reasons that have nothing to do with the new layout. The real information available is much closer to coming from 24 locations than from 3,600 independent customers.

With only 24 clusters, still within a reasonable range for the CRVE to work well, but close enough to the edge that it would be worth confirming the result with a wild-cluster bootstrap as an additional robustness check.



Illustrative data, not drawn from any real project.

Figure 20: Confidence interval computed with the naive standard error compared to the interval with the cluster-robust standard error, same point estimate

# Randomization inference (permutation test)

## Concept

Randomization inference (also called a permutation test, or the “Fisher randomization test,” after Ronald Fisher) builds a hypothesis test’s reference distribution directly from the actual randomness of the experimental design, instead of relying on a theoretical approximation (such as the normal or t distribution) that is only valid asymptotically, in large samples.

The central idea is the “sharp” null hypothesis: under  $H_0$ , each unit’s outcome would be exactly the same regardless of whether it was assigned to treatment or control. If that is true, the only source of variation in the observed statistic is the randomness of who was drawn into each group itself — and that source of variation can be enumerated exactly, because the researcher knows the assignment mechanism that was used.

## Mathematical formulation

Let there be  $N$  experimental units, of which  $N_1$  are assigned to treatment and  $N_0 = N - N_1$  to control, according to a known assignment mechanism. Let  $Y_i$  be unit  $i$ ’s observed outcome and  $W_i \in \{0, 1\}$  the treatment indicator that was actually drawn.

The observed test statistic is typically the mean difference:

$$\hat{\tau}_{\text{obs}} = \frac{1}{N_1} \sum_{i:W_i=1} Y_i - \frac{1}{N_0} \sum_{i:W_i=0} Y_i$$

Under the sharp null hypothesis  $H_0 : Y_i(1) = Y_i(0)$  for every unit  $i$  (each unit’s outcome would be identical under either group), the observed values  $Y_i$  do not depend on  $W_i$ . That makes it possible to compute the same statistic under **any** other possible assignment  $W'$ , holding the  $Y_i$  values fixed:

$$\hat{\tau}(W') = \frac{1}{N_1} \sum_{i:W'_i=1} Y_i - \frac{1}{N_0} \sum_{i:W'_i=0} Y_i$$

The set of all possible assignments, given the assignment mechanism (for example, every way of choosing  $N_1$  out of  $N$  units without replacement), has  $\binom{N}{N_1}$  elements when the design is a simple complete randomization. The two-sided p-value is:

$$p = \frac{1}{\binom{N}{N_1}} \sum_{W'} \mathbb{1}[|\hat{\tau}(W')| \geq |\hat{\tau}_{\text{obs}}|]$$

where  $\mathbb{1}[\cdot]$  is the indicator function (1 if the condition holds, 0 otherwise). When  $\binom{N}{N_1}$  is too large to enumerate exhaustively, the sum is approximated by Monte Carlo: a large number  $M$  (typically 10,000 or more) of random reassignments  $W'$  is drawn, and the same proportion is computed over that sample, converging to the exact p-value as  $M$  grows.

## Assumptions

- **Genuine randomness of the treatment assignment.** This is the method’s only fundamental assumption — without it, the reference distribution built does not correspond to any real process that generated the data.
- **Sharp null hypothesis.** The classical (Fisher) form of the test checks the hypothesis that treatment had absolutely no effect on any unit at all — stronger than the usual “average effect equal to zero” hypothesis (Neyman’s). Rejecting  $H_0$  under this design is evidence that treatment affected at least some unit, not necessarily all of them equally.
- **No interference between units (SUTVA).** One unit’s outcome cannot depend on which treatment other units received (no spillovers or general-equilibrium effects) — otherwise, the hypothetical reassignment  $W'$  no longer corresponds to a coherent counterfactual scenario.

## Hypotheses

- $H_0$ : sharp null effect — each unit’s outcome would be identical regardless of which group it was assigned to.
- $H_1$ : treatment affects the outcome of at least some units.
- Test statistic: usually the mean difference between groups, but the method accepts any statistic of interest (median difference, a model coefficient, a ratio of proportions) — the reassignment logic is the same regardless of which statistic is chosen.
- P-value: the proportion of possible (or sampled) reassignments that produce a statistic as extreme as, or more extreme than, the observed one.
- Significance level: conventionally 5%, but subject to the discrete granularity of the number of possible reassignments (see limitations).

## Interpretation

A low p-value obtained by randomization is evidence that the actual assignment produced an unusual result among every possible way the draw could have gone — a causal reading directly tied to the experiment’s own randomization, without relying on assumptions about the data’s population distribution. That is an important advantage over parametric tests: validity does not depend on normality, a large sample, or a specific functional form for the errors.

What the method does not do is automatically generalize the estimated effect beyond the observed units — the inference is about the observed assignment versus the other possible assignments within the experiment, not about a larger population the units were sampled from (that would be a question of sampling inference, separate from inference about the assignment).

## Limitations

- **P-value granularity in small samples.** With few units, the number of possible reassignments is small, limiting how low a p-value can go — with  $\binom{4}{2} = 6$  reassignments, the smallest possible p-value is  $1/6 \approx 0.167$ , even in the face of a large effect. This is a structural limitation, not a computational flaw.
- **Depends on genuine randomization.** If treatment assignment was not actually random (for example, “volunteer” units self-selected into treatment), the built reference distribution does not represent any real mechanism — the method loses validity.
- **Computational cost in large designs.** With many units, exhaustive enumeration becomes infeasible (the number of combinations grows quickly), requiring Monte Carlo approximation

— which, in turn, introduces a small sampling margin of error into the p-value calculation itself, controllable by increasing the number of drawn reassignments.

### Comparison with other methods for inference with few clusters

Randomization inference, cluster-robust standard errors, and wild-cluster bootstrap all answer the same general problem: reliable inference when observations are not independent, often because of few clusters or few randomization units.

- **Cluster-robust standard errors** are the standard starting point, but become unreliable precisely in the few-cluster setting, because they depend on an asymptotic approximation.
- **Randomization inference** is the most direct alternative when the experiment’s own assignment mechanism is known and the number of possible reassignments is enumerable (or sample-able) — it depends on no distributional assumption at all, only on the genuine randomness of the assignment. It is especially attractive when the experimental design is simple (for example, a complete random draw among a small number of units).
- **Wild-cluster bootstrap** is preferable when the effect of interest comes from a more complex regression model (with controls, for example), or when there is no simple, known assignment mechanism to reproduce exactly through reassignment — wild-cluster bootstrap resamples the fitted model’s residuals instead of directly reassigning treatment.

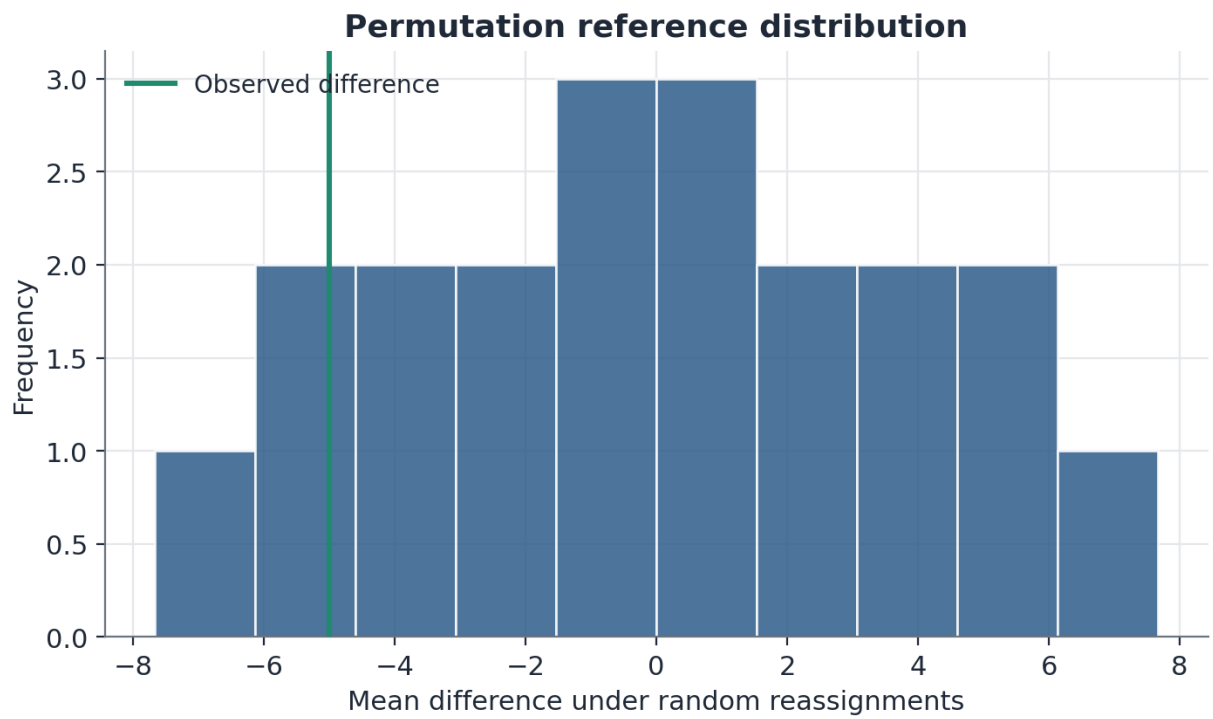
In practice, when both are applicable, it is common to run both as a cross-check: conclusions that agree between randomization inference and wild-cluster bootstrap give more confidence in the result’s robustness than either one alone.

### Example

A restaurant chain tests a new seasonal menu across 6 locations (simple draw: 3 treatment, 3 control), measuring average spend per table during the test week, in dollars: treated locations record 12, 15, and 9; control locations record 22, 18, and 11.

The observed mean difference is  $12.0 - 17.0 = -5.0$ . There are  $\binom{6}{3} = 20$  possible ways to choose which 3 locations were treated. Computing the mean difference for each of those 20 combinations gives the full reference distribution — with no assumption at all about the shape of the distribution of spend per table.

Of the 20 combinations, 6 produce a difference as extreme as (or more extreme than)  $-5.0$  in absolute value, giving a two-sided p-value of  $6/20 = 0.30$ . With only 6 units in the experiment, this result is not unusual enough to be evidence that the new menu actually changed average spend per table — even though the observed difference, taken alone, looks large in percentage terms. The example illustrates the granularity limitation well: with few units, you need a very large effect for the observed assignment to genuinely stand out among the few possible reassignments.



Illustrative data, not drawn from any real project.

Figure 21: Reference distribution formed by all 20 possible reassignments, with the observed difference of  $-5.0$  highlighted

# Wild-cluster bootstrap

## Concept

The wild-cluster bootstrap is a resampling method developed specifically for inference in regression models with grouped data when the number of clusters is small — the setting in which the cluster-robust variance estimator (CRVE) loses its asymptotic guarantees and systematically understates the real uncertainty.

The core idea rests on two design choices:

1. **Resample residuals, not observations or entire clusters.** A conventional cluster bootstrap draws entire clusters with replacement — which, with few clusters, produces replicates with coarse repetition (for example, one cluster appearing three times and another not at all) and an effectively too-small sample space to form a good reference distribution.
2. **Perturb by a random sign, preserving the within-cluster dependence structure.** Instead of resampling, the method keeps every original observation and multiplies each cluster's residuals by a single random value drawn for that cluster (typically  $+1$  or  $-1$ , the “Rademacher distribution”) — preserving exactly the correlation pattern within each cluster, because every observation in that cluster is perturbed as a block, in the same direction.

## Mathematical formulation

Consider the linear model with  $G$  clusters:

$$Y_{ig} = X'_{ig}\beta + \varepsilon_{ig}, \quad g = 1, \dots, G$$

The interest is testing  $H_0 : \beta_j = \beta_j^0$  (often  $\beta_j^0 = 0$ ) for a specific coefficient  $\beta_j$ .

**Step 1 — restricted fit.** Estimate the model imposing  $H_0$  (the “restricted” model), obtaining residuals  $\tilde{u}_{ig}$  and fitted values  $\tilde{Y}_{ig}$  under the restriction.

**Step 2 — sign perturbation.** For each replicate  $b = 1, \dots, B$ , draw a sign  $v_g^{(b)} \in \{-1, +1\}$  **per cluster** (not per observation), usually with probability  $1/2$  for each value:

$$v_g^{(b)} \sim \text{Rademacher}(0.5)$$

**Step 3 — synthetic data.** Build a synthetic dependent variable:

$$Y_{ig}^{*(b)} = \tilde{Y}_{ig} + v_g^{(b)} \tilde{u}_{ig}$$

Note that  $v_g^{(b)}$  does not vary within cluster  $g$  — every observation in that cluster gets the same sign in that replicate, preserving the within-cluster correlation.

**Step 4 — refit and statistic.** Refit the (unrestricted) model to the synthetic data  $Y^{*(b)}$ , obtaining  $\hat{\beta}_j^{*(b)}$  and its cluster-robust standard error  $\widehat{SE}^{*(b)}(\hat{\beta}_j)$ , forming the bootstrap statistic:

$$t^{*(b)} = \frac{\hat{\beta}_j^{*(b)} - \beta_j^0}{\widehat{SE}^{*(b)}(\hat{\beta}_j)}$$

**Step 5 — p-value.** Repeating steps 2 through 4 for  $B$  replicates (typically  $B = 999$ ), the bootstrap p-value is:

$$p^* = \frac{1}{B} \sum_{b=1}^B \mathbb{1} [ |t^{*(b)}| \geq |t_{\text{obs}}| ]$$

with  $t_{\text{obs}}$  the t-statistic computed on the real data (under the unrestricted model), using the conventional cluster-robust standard error.

With  $G$  clusters, there are at most  $2^G$  distinct sign combinations — with  $G = 6$ , for example, 64 combinations, which still leaves a reasonably fine-grained p-value even when enumerating exhaustively instead of sampling.

## Assumptions

- **Model correctly specified under  $H_0$ .** The restricted fit, used to generate the perturbed residuals, needs to reasonably reflect the real relationship between the variables — a misspecified model produces residuals that do not represent the true error structure well.
- **Independence across clusters.** As with the CRVE, the method’s validity depends on clusters being independent of one another — residual dependence between clusters is not fixed by the wild-cluster bootstrap.
- **Symmetry of the sign distribution.** The standard choice of Rademacher signs (+1/−1 with equal probability) works well in most cases; alternative sign distributions exist in the literature, but Rademacher is the most widely used for its robustness and simplicity.
- **Minimum number of clusters.** With extremely few clusters (2 or 3), even the space of  $2^G$  possible signs becomes too small to form a useful reference distribution — the method remains superior to the asymptotic CRVE at that extreme, but with weaker guarantees.

## Hypotheses

- $H_0$ :  $\beta_j = \beta_j^0$  (commonly  $\beta_j^0 = 0$ , “the coefficient is zero”).
- $H_1$ :  $\beta_j \neq \beta_j^0$ .
- Test statistic:  $t = (\hat{\beta}_j - \beta_j^0) / \widehat{SE}_{\text{cluster}}(\hat{\beta}_j)$ , computed on the real data and compared to the empirical distribution of  $t^{*(b)}$  generated by the bootstrap, rather than to a theoretical t distribution.
- Confidence interval: can be obtained by inverting the test (searching for the range of  $\beta_j^0$  values not rejected by the bootstrap test), a more computationally demanding approach, but more accurate than approximating the interval from the raw percentiles of the  $\hat{\beta}_j^{*(b)}$  distribution.

## Interpretation

A low wild-cluster bootstrap p-value is evidence that the observed statistic would be unusual even under simulated variation consistent with the null hypothesis and the data’s actual dependence

structure — a more trustworthy reading than the asymptotic CRVE precisely in the few-cluster range where the latter tends to fail.

As with any inference-correction method, the wild-cluster bootstrap does not solve the original design’s causal-identification problems (omitted variables, non-random selection) — it only produces a more faithful reference distribution for testing hypotheses about the estimated coefficient, given that the model and design are already valid for other reasons.

## Limitations

- **Still depends on a minimum number of useful clusters.** With very small  $G$  (2 or 3), the space of  $2^G$  possible signs is too small to form a good distribution — at that extreme, randomization inference, when applicable to the design, tends to be preferable.
- **Sensitivity to the restricted model’s specification.** Residuals from a poorly fitted model under  $H_0$  propagate that specification error throughout the entire bootstrap distribution.
- **Computational cost.** Each replicate requires refitting the model — with  $B$  in the thousands and complex models, the total cost can be significant, though it is typically manageable at usual applied-work scale.

## Comparison with other methods for inference with few clusters

Wild-cluster bootstrap, cluster-robust standard errors, and randomization inference all answer the same problem: reliable inference when observations within a cluster are not independent and few clusters are available.

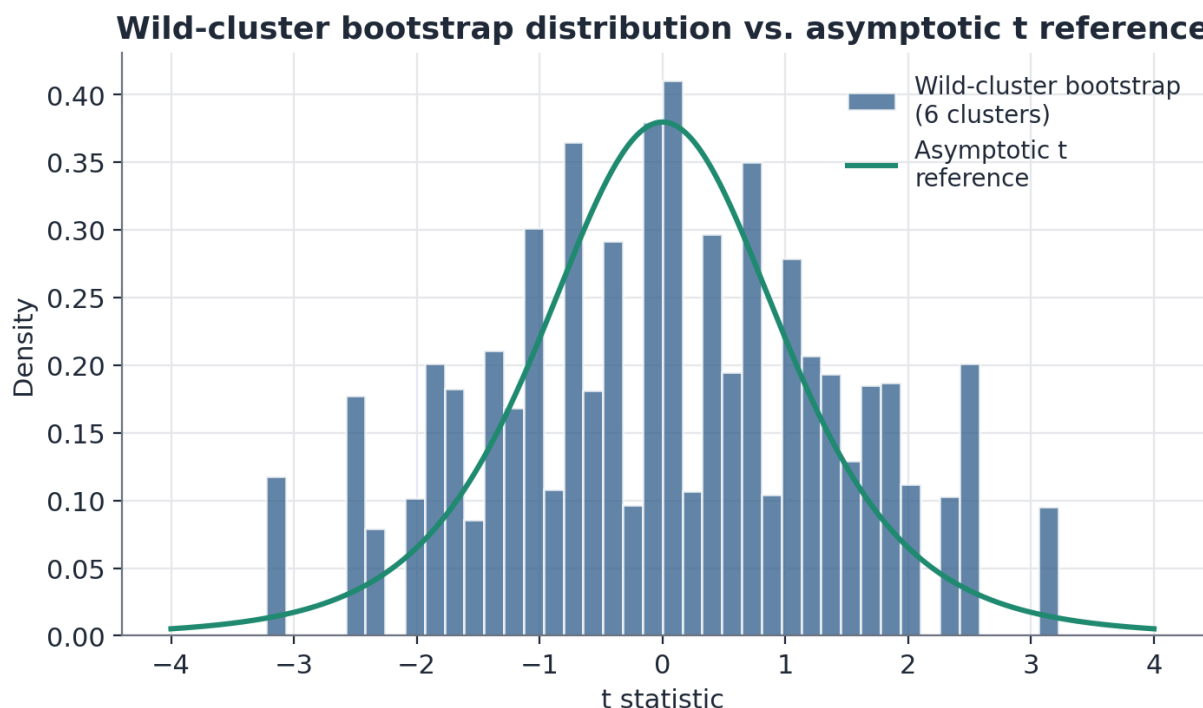
- **Cluster-robust standard errors (CRVE)** are the standard, simple, fast choice, but lose reliability precisely when there are few clusters — the setting where wild-cluster bootstrap and randomization inference become relevant.
- **Randomization inference** is preferable when the experiment’s own assignment mechanism is known and simple enough to enumerate (or sample) the possible reassignments directly — it requires fitting no regression model at all, only recomputing a statistic under treatment reassignments. It works best when the effect of interest comes from a direct comparison between groups, with no additional controls.
- **Wild-cluster bootstrap** is the more natural choice when the effect of interest comes from a regression model with controls or a richer structure (fixed effects, covariates), where there is no simple assignment mechanism to reassign directly — the bootstrap perturbs the already-fitted model’s residuals, instead of reassigning treatment to units.

When both methods are applicable to the same problem, running both and comparing conclusions is a valuable robustness check: agreement between randomization inference and wild-cluster bootstrap reinforces confidence in the result; disagreement is a signal that it is worth investigating further before reporting a conclusion.

## Example

A clinic chain tests a new scheduling protocol across 8 locations (4 treatment, 4 control), measuring average patient wait time, with additional controls in the model (day of week, time slot). The estimated treatment coefficient is  $-6.5$  minutes, with a conventional cluster-robust standard error of 2.1, giving  $t = -3.1$  and an asymptotic p-value of roughly 0.003 (apparently highly significant).

Running the wild-cluster bootstrap with  $B = 999$  replicates over the 8 clusters (drawing Rademacher signs per clinic in each replicate, refitting the restricted model, and recomputing the t-statistic each time), the observed statistic of  $-3.1$  falls at a percentile corresponding to a bootstrap p-value of roughly 0.04 — still significant at 5%, but with a substantially tighter margin than the asymptotic p-value of 0.003 suggested.



Illustrative data, not drawn from any real project.

Figure 22: Wild-cluster bootstrap distribution from 999 replicates compared to the asymptotic t reference, for a design with few clusters

The gap between the two p-values (0.003 vs. 0.04) illustrates why the asymptotic CRVE tends to be excessively optimistic with few clusters — the qualitative conclusion (a significant effect) still holds in this example, but the real confidence margin is much narrower than the first reading suggested, and a slightly weaker result could easily stop being significant under the wild-cluster correction.

# Cohort and retention decomposition

## Concept

An aggregate retention metric — computed over the whole customer base in a given month — blends together two axes of variation that need to be separated for a correct reading: **lifecycle dynamics** (how a group of customers' retention evolves as time passes since they joined) and **composition dynamics** (how the share of customers at different lifecycle stages shifts over the calendar, as the company grows or slows down acquisition).

Cohort decomposition resolves that confusion by reorganizing the data around time since joining (*tenure*) instead of calendar time. A **cohort** is the set of customers who entered the system during the same reference period (a month, a week). A **cohort's retention curve** is the sequence of shares of the original cohort still active in each subsequent period. Comparing curves from different cohorts, aligned by tenure, makes it possible to attribute a change in the aggregate metric to three possible, non-mutually exclusive sources: deterioration specific to one or more recent cohorts, broad-based deterioration (a calendar event that hits every cohort simultaneously, regardless of age), or a shift in the base's composition (more relative weight on young cohorts, which naturally retain worse in their first months than mature cohorts).

## Mathematical formulation

Let  $N_c$  be cohort  $c$ 's initial size (the number of customers who joined during period  $c$ ), and let  $A_{\{c,t\}}$  be the number of customers from that cohort still active  $t$  periods after joining. The cohort's retention curve is:

$$R_c(t) = A_{\{c,t\}} / N_c$$

with  $R_c(0) = 1$  by definition.

The aggregate retention observed at calendar period  $T$ , considering every cohort that existed by  $T$ , is a weighted average of the individual curves, each evaluated at a different tenure point:

$$R_{\text{aggregate}}(T) = \sum_c [ N_c \cdot R_c(T - c) ] / \sum_c N_c$$

where the sum runs over every cohort  $c \leq T$ , and  $T - c$  is cohort  $c$ 's tenure at moment  $T$  (how much time has passed since it joined).

This formula makes the confusion mechanism explicit:  $R_{\text{aggregate}}(T)$  changes from one calendar period to the next both because the individual curves  $R_c(\cdot)$  change (a behavior effect) and because the relative weights  $N_c$  of recent versus older cohorts change (a composition effect), without any individual curve needing to move at all.

A simple decomposition of the change in aggregate retention between two moments ( $T$  and  $T'$ ) can be approximated by separating the two effects:

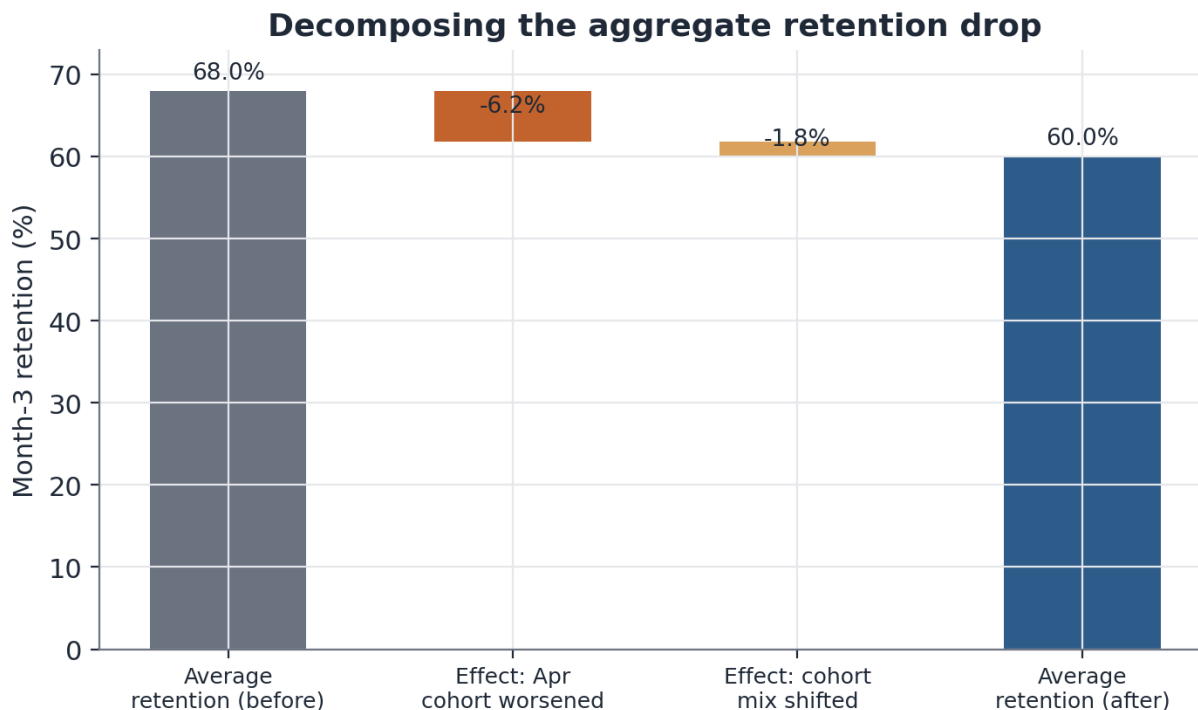
$$\Delta R_{\text{aggregate}} = \text{Behavior effect} + \text{Composition effect}$$

$$\text{Behavior effect} = \sum_c w_c(T') \cdot [ R_c(T' - c) - R_c(T - c) ]$$

$$\text{Composition effect} = \sum_c [ w_c(T') - w_c(T) ] \cdot R_c(T - c)$$

where  $w_c(T)$  is cohort  $c$ 's weight in the total base at  $T$  ( $N_c$  divided by the total active customers considered at  $T$ ). The first term isolates how much of the change comes from retention curves

genuinely moving; the second isolates how much comes purely from the shift in which cohorts make up the base.



Illustrative data, not drawn from any real project.

Figure 23: Waterfall decomposition of the aggregate retention drop into a cohort-specific effect and a mix effect

## Assumptions

- **Comparable tenure across cohorts:** the decomposition assumes “month 3 since joining” means the same thing for every cohort — that is, the product or service and the onboarding process haven’t changed in ways that invalidate comparing tenure between cohorts far apart in time.
- **Consistent definition of “active”:** the activity metric defining  $A_{\{c,t\}}$  (active subscription, login in the period, purchase in the period) needs to be applied identically across every cohort; a change in the operational definition of “active” mid-analysis contaminates the comparison.
- **Cohort size sufficient for statistical stability:** small cohorts produce retention curves with high sampling variance, especially at advanced tenures, where few members of the original cohort remain.
- **No differential censoring:** more recent cohorts have, by definition, fewer observed periods (there’s no way to know the month-6 retention of a cohort that joined 3 months ago) — comparisons that ignore this censoring, comparing tenure points not yet observed for recent cohorts, produce invalid conclusions.

## Hypotheses

Cohort decomposition is primarily descriptive/exploratory, but it can be formalized as a hypothesis test comparing a specific cohort's curve against the historical average curve:

- **H0:** cohort  $c$ 's retention curve at a specific tenure  $t$  doesn't differ from the historical average retention of comparable cohorts at the same tenure ( $R_c(t) = R(t)$ ).
- **H1:**  $R_c(t) \neq R(t)$  — the cohort in question retains significantly differently from the historical pattern.

The test can be run as a comparison of two proportions (the cohort in question versus the weighted historical average, or versus the immediately preceding cohort), with a z-test for proportions or an exact test, depending on cohort size.

## Interpretation

The decomposition's result directly steers the kind of follow-up investigation. A behavior effect concentrated in one specific cohort points the investigation toward what was different for whoever joined during that period — a product change, an alteration to the billing process, an acquisition campaign with a different audience. A behavior effect distributed across every cohort at once, starting from a specific calendar marker, points toward an event that hit the entire base simultaneously, regardless of tenure — a broad price change, a widespread technical issue, a regulatory shift. A dominant composition effect, with no meaningful change in the individual curves, shifts attention to acquisition strategy — why the base is younger on average — rather than to a retention problem per se.

It's important not to read a timing coincidence between a worse cohort and a known change as proof of causation — the decomposition identifies where to look, it doesn't confirm the cause. Confirming the cause typically requires additional information (for example, a change tested through a controlled experiment, or a comparison with a group unaffected by the suspected change).

## Limitations

- The decomposition doesn't control for confounders within a cohort — if a specific cohort also differs in demographic makeup or acquisition channel from the others, part of the “cohort effect” may actually be an effect of those other variables, not of the joining period itself.
- Recent cohorts inherently have fewer observed tenure points — any comparison needs to restrict itself to tenures actually observed across every cohort being compared, or explicitly acknowledge the greater uncertainty in newer cohorts.
- The split into a behavior effect and a composition effect, as presented, is a first-order approximation — when changes in both components are large, a residual interaction term exists that the simple decomposition doesn't explicitly attribute to either effect.
- The analysis is observational. Even precisely identifying that a specific cohort worsened, and even with a coincident product change, other concurrent factors in the same period (seasonality, market conditions, competitor moves) can be alternative explanations that cohort decomposition alone doesn't rule out.

## Example

A hypothetical gym chain with a monthly subscription model notices aggregate month-3 retention dropped from 58% last quarter to 52% this quarter. The team decomposes retention by entry

cohort (the month a customer subscribed):

| Cohort                                    | Cohort size | Month-3 retention |
|-------------------------------------------|-------------|-------------------|
| Cohort T-3 (oldest)                       | 1,240       | 59%               |
| Cohort T-2                                | 1,310       | 57%               |
| Cohort T-1                                | 1,180       | 58%               |
| Cohort T (most recent, the drop's period) | 1,850       | 41%               |

Two things stand out: first, cohort T has significantly lower retention (41% against a historical average of roughly 58% across the three prior cohorts) — a behavior effect concentrated in this cohort. Second, cohort T is also considerably larger than the prior ones (1,850 versus an average of about 1,240 in earlier cohorts) — meaning it carries more weight in this quarter's aggregate average, amplifying the impact of its lower retention on the overall metric (an additional composition effect, though secondary to the behavior effect here).

Investigating what specifically changed for cohort T, the team finds it coincides with an aggressive acquisition campaign offering a first-month discount, which brought in an audience historically less likely to continue past the promotional period — a plausible explanation for both the lower retention and the cohort's larger size.

## Code example

```
import pandas as pd
import numpy as np

rng = np.random.default_rng(23)

cohorts = pd.DataFrame({
    "cohort": ["T-3", "T-2", "T-1", "T"],
    "size": [1240, 1310, 1180, 1850],
    "retention_month3": [0.59, 0.57, 0.58, 0.41],
})

# aggregate retention weighted by cohort size
aggregate_retention = np.average(cohorts["retention_month3"], weights=cohorts["size"])
print(f"Aggregate retention this quarter: {aggregate_retention:.1%}")

# counterfactual: what would aggregate retention be if cohort T had the
# average retention of prior cohorts, keeping the same size (isolates
# cohort T's own behavior effect)
historical_avg_retention = cohorts.loc[cohorts.cohort != "T", "retention_month3"].mean()
cohorts_counterfactual = cohorts.copy()
cohorts_counterfactual.loc[cohorts_counterfactual.cohort == "T", "retention_month3"] = historical_avg_retention
counterfactual_retention = np.average(
    cohorts_counterfactual["retention_month3"], weights=cohorts_counterfactual["size"]
)
```

```
cohort_t_behavior_effect = counterfactual_retention - aggregate_retention
print(f"Aggregate retention if cohort T matched history: {counterfactual_retention:.1%}")
print(f"Effect attributable to cohort T's behavior drop: {cohort_t_behavior_effect:.1%} pp")
```

# Growth decomposition: total, comparable, and price-volume-mix

## Concept

Growth decomposition is an accounting-analytical technique (not a statistical model with inferential uncertainty) that rewrites the change in an aggregate metric — typically revenue — as the sum of mutually exclusive, exhaustive components, each answering a well-defined partial causal question: “how much of the growth came from X, holding everything else constant?”

The decomposition happens in two chained levels:

**Level 1 — Total vs. comparable.** Split the units of analysis (stores, regions, product lines, customer accounts) into two populations: those present in both compared periods (“comparable,” *like-for-like*) and those that only exist in the more recent period (new openings, market entry, new SKUs). Total growth is always the sum of comparable-unit growth plus the revenue brought in by new units.

**Level 2 — Price, volume, and mix, within the comparable base.** Within the comparable population, the revenue change is rewritten as average price times quantity sold, aggregated by item or category. Change in that product can come from three sources: the unit price of each item changed (**price effect**); total quantity sold changed while the mix of items stayed the same (**volume effect**); or the composition of items sold changed — for example, higher-ticket items gaining share (**mix effect**).

The underlying logic is the same as any accounting decomposition (structural decomposition analysis, shift-share analysis): hold every factor but one fixed, measure that factor’s marginal effect, repeat for the rest, and ensure the sum reconciles exactly to the observed total — with no unexplained residual, except where the chosen calculation convention deliberately assigns interaction terms to the mix component.

## Mathematical formulation

Let  $R_t = \sum_i p_{i,t} q_{i,t}$  be revenue in period  $t$ , summed over items or categories  $i$ , where  $p_{i,t}$  is the average price of item  $i$  in period  $t$  and  $q_{i,t}$  is the quantity sold.

**Total vs. comparable split.** Let  $C$  be the set of comparable units (present in both periods  $t-1$  and  $t$ ) and  $N$  the set of new units (present only in  $t$ ). Then:

$$\underbrace{R_t - R_{t-1}}_{\text{total growth}} = \underbrace{\left( \sum_{i \in C} p_{i,t} q_{i,t} - \sum_{i \in C} p_{i,t-1} q_{i,t-1} \right)}_{\text{comparable growth } \Delta R_C} + \underbrace{\sum_{i \in N} p_{i,t} q_{i,t}}_{\text{revenue from new units}}$$

There is no separately signed “discontinued units” term in this formulation because, by convention, they are implicitly captured inside  $\Delta R_C$  when  $q_{i,t} = 0$  for items dropped within the comparable set itself; in practice, discontinuations are often treated as an explicit fourth category when relevant.

**Price-volume-mix decomposition, inside  $\Delta R_C$ .** Using the sequential substitution method (one of the most common conventions, equivalent to a modified Laspeyres index):

$$\text{Volume Effect} = \left( \sum_{i \in C} q_{i,t} \right) - \left( \sum_{i \in C} q_{i,t-1} \right) \times \bar{p}_{t-1}$$

$$\text{Price Effect} = (\bar{p}_t - \bar{p}_{t-1}) \times \left( \sum_{i \in C} q_{i,t} \right)$$

$$\text{Mix Effect} = \Delta R_C - \text{Volume Effect} - \text{Price Effect}$$

where  $\bar{p}_t = R_{C,t} / \sum_{i \in C} q_{i,t}$  is the quantity-weighted average price of period  $t$  in the comparable population. The volume effect isolates what would have happened to revenue if only total quantity had changed, at the prior period’s average price. The price effect isolates what would have happened if only average price had changed, applied to the current period’s quantity. The mix effect is defined as a residual: the part of the revenue change explained by neither average price nor total volume — i.e., a shift in the composition of items sold, including higher-ticket items gaining share.

This is not the only possible convention: alternative methods (e.g., symmetric decomposition, or computing each effect at the midpoint between the two periods) distribute the price×volume interaction differently across the three components. The total, however, is invariant to the convention chosen.

## Assumptions

- **Consistent granularity.** Price-volume-mix decomposition requires item/SKU/category-level data — it cannot be computed on aggregate revenue without a matching quantity; without separate unit price and quantity, only total-vs-comparable can be computed.
- **Stable definition of “comparable.”** Classifying a unit as comparable depends on a time-cutoff rule (e.g., a store open for 12+ full months) applied consistently across periods — changing the rule midway through a historical series breaks comparability between quarters.
- **No structural change in the unit of measurement.** If the quantity unit changes (e.g., the business switches to selling larger packs), the computed volume effect mixes real volume with packaging change — the unit needs to be normalized before decomposing.
- **Stable composition within each item.** The mix effect assumes “item  $i$ ” means the same thing in both periods; if a SKU is redefined (new recipe, new size) partway through, the decomposition will attribute the product’s characteristic change to the wrong component.

## Hypotheses

This method is accounting-based, not inferential — there is no null hypothesis, test statistic, or confidence interval built into the decomposition itself. The components (price, volume, mix, new units) are algebraic identities computed from observed data, not estimates with sampling error.

That said, it is common — and recommended — to pair the decomposition with inference when applicable: for example, testing whether the volume effect is significantly different from zero across several units (stores, regions) using a t-test or a random-effects model over the distribution of per-unit volume effects, especially when the goal is to generalize the conclusion beyond the observed set of stores.

## Interpretation

The decomposition lets you state precisely where each dollar of growth came from — not why it happened. A negative volume effect says fewer units were sold at the same comparable stores; it does not say whether that was caused by customer churn, a new competitor, stockouts, or a

deliberate decision to cut back on promotions. The decomposition is diagnostic, not causal — it points the investigation toward the right question (why did volume fall?), but it does not answer it.

It’s also important not to confuse a high percentage growth with a meaningful absolute one: a +2 percentage-point mix effect on a small base can represent fewer dollars than a +0.5 percentage-point volume effect on a large base. Always report absolute values alongside percentages.

## Limitations

- **Sensitivity to the calculation convention.** As shown in the mathematical formulation, the exact value of each component (especially mix) depends on the convention chosen for handling the price×volume interaction. Reports using different conventions are not directly comparable component by component, even when the totals reconcile.
- **Mix as an “error-catching” residual.** Because it is computed as a residual, the mix effect absorbs any measurement error in price or quantity (e.g., discounts not captured correctly, misclassified returns). An unusually large mix effect is often a sign of a data-quality problem, not a genuine change in sales composition.
- **Does not capture causality or a true counterfactual.** “What would have happened if only price had changed” is an accounting question — it assumes quantity sold would have stayed the same regardless of the price change, which ignores price elasticity of demand. Under stricter scrutiny, the price and volume components are interdependent in practice, even though they are computed as if independent.
- **The definition of “comparable” is a choice, not a given fact.** Different comparability windows produce different comparable-growth figures from the same underlying raw data.

## Example

Consider a hypothetical coffee-shop chain with three stores open for over a year (A, B, C) and one new store opened during the period (D), selling two products: coffee (low ticket, high volume) and a coffee+pastry combo (higher ticket, lower volume).

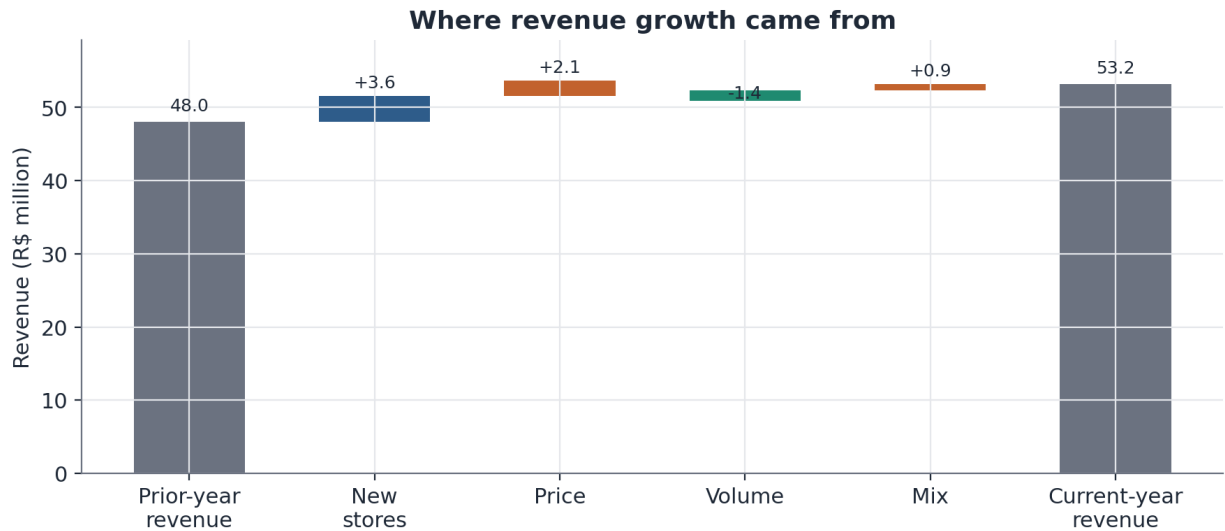
| Store | Comparable | Prior-year revenue | Current-year revenue | Prior-year units | Current-year units |
|-------|------------|--------------------|----------------------|------------------|--------------------|
| A     | Yes        | \$12.0M            | \$12.8M              | 2,400            | 2,280              |
| B     | Yes        | \$15.0M            | \$14.1M              | 3,000            | 2,820              |
| C     | Yes        | \$18.0M            | \$19.7M              | 3,600            | 3,760              |
| D     | No         | —                  | \$3.6M               | —                | 720                |

**Step 1 — total vs. comparable.** Total revenue: \$45.0M → \$50.2M, a \$5.2M increase. Revenue from new stores (D): \$3.6M. Comparable growth: \$5.2M − \$3.6M = **\$1.6M**.

**Step 2 — comparable average price.** Total comparable quantity: 9,000 units (prior year) → 8,860 units (current year). Average price: \$45.0M / 9,000 = \$5,000/unit (prior year) → \$46.6M / 8,860 = \$5,259/unit (current year).

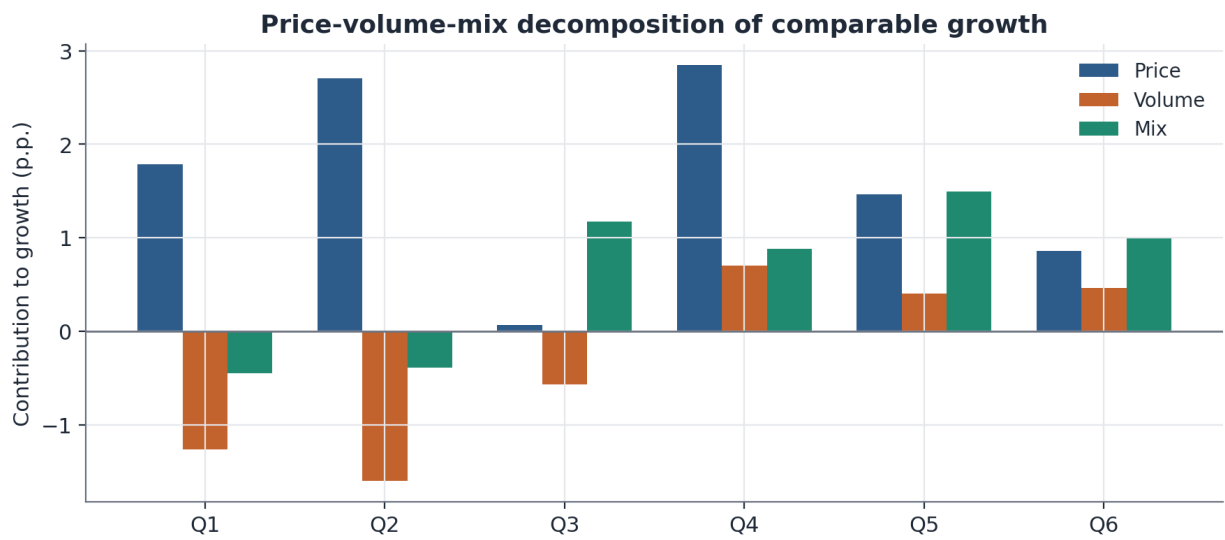
**Step 3 — effects.** - Volume effect:  $(8,860 - 9,000) \times \$5,000 = -\$0.70\text{M}$  - Price effect:  $(\$5,259 - \$5,000) \times 8,860 = +\$2.29\text{M}$  - Mix effect:  $\$1.6\text{M} - (-\$0.70\text{M}) - \$2.29\text{M} = -\$0.01\text{M}$  (rounding; close to zero in this simplified example)

**Reading:** comparable growth of \$1.6M is positive only because of a strong price effect (+\$2.29M), which more than offsets a real volume decline (−\$0.70M) at comparable stores. A near-zero mix effect indicates the split between coffee and combo did not change meaningfully. This pattern — price offsetting volume — is exactly the kind of signal a headline “+11.6% total growth” figure would hide on its own, and it justifies an investigation into price elasticity and possible traffic loss at existing stores.



Illustrative data, not drawn from any real project.

Figure 24: Waterfall decomposition of revenue growth



Illustrative data, not drawn from any real project.

Figure 25: Price-volume-mix decomposition across six quarters

# Poisson regression with fixed effects

## Concept

Poisson regression is a generalized linear model (GLM) built for response variables that are event counts: non-negative integers representing how many times something happened in a fixed interval (a match, an hour, a month). It's the natural choice when the variable of interest violates the two core assumptions of an ordinary linear regression: (1) counts cannot be negative, yet a least-squares fitted line can predict negative values with no restriction; and (2) the distribution of counts is typically right-skewed, with probability mass concentrated at low values (0, 1, 2) and a long tail — very different from the symmetry of a normal distribution.

The central idea is to model the **logarithm** of the expected event rate,  $\log(\lambda)$ , as a linear combination of the explanatory variables — not the rate itself. This logarithmic “link function” guarantees that once the transformation is undone (by exponentiating), the resulting prediction  $\lambda$  is always positive, regardless of the coefficient or variable values.

Fixed effects per unit (individual, team, store, period) add a dummy variable for each unit, letting each one have its own intercept — its own baseline event rate — before the effect of the variable of interest is estimated. This is fundamentally different from simply including the unit as an “average level” control: the fixed effect absorbs every unobserved but time-constant characteristic of that unit (a historically more attacking team, a store in a busier neighborhood), better isolating the effect of the variable of interest (playing at home, it being a Friday) from structural differences between units.

## Mathematical formulation

**Poisson distribution.** A count variable  $Y$  follows a Poisson distribution with parameter  $\lambda > 0$  if

$$P(Y = y) = \frac{e^{-\lambda} \lambda^y}{y!}, \quad y = 0, 1, 2, \dots$$

where  $\lambda = E[Y] = \text{Var}(Y)$  — mean and variance are equal (equidispersion), a property specific to this distribution that becomes a testable assumption of the model.

**Log-linear model.** For observation  $i$  (say, team  $t$  in match  $m$ ), the expected rate is

$$\lambda_i = \exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \alpha_{u(i)})$$

where: -  $x_{1i}, \dots, x_{ki}$  are the explanatory variables (attacking strength, opponent defensive strength, home-field indicator); -  $\beta_1, \dots, \beta_k$  are the coefficients to be estimated, on the log scale; -  $\alpha_{u(i)}$  is the fixed effect of the unit  $u$  that observation  $i$  belongs to (the team, for instance) — it acts as that unit's own intercept; -  $\beta_0$  is the overall intercept (absorbed in practice by the  $\alpha_u$  terms when full per-unit fixed effects are included).

Equivalently, on the log scale:

$$\log(\lambda_i) = \beta_0 + \beta_1 x_{1i} + \dots + \beta_k x_{ki} + \alpha_{u(i)}$$

That’s where the name comes from: “log-linear” — linear on the log-mean scale, not on the original scale.

**Multiplicative interpretation of the coefficient.** Consider two observations identical except for variable  $x_1$  (say, home field:  $x_1 = 1$  vs.  $x_1 = 0$ ). The ratio of expected rates is

$$\frac{\lambda_i | x_1 = 1}{\lambda_i | x_1 = 0} = \frac{\exp(\beta_0 + \beta_1 \cdot 1 + \dots)}{\exp(\beta_0 + \beta_1 \cdot 0 + \dots)} = e^{\beta_1}$$

Every other term cancels because it’s identical across the two observations — which is why  $e^{\beta_1}$  is interpreted as a **multiplicative factor**: how many times the expected rate changes when  $x_1$  increases by one unit, holding everything else constant.

**Estimation.** Coefficients are estimated by maximum likelihood. The model’s log-likelihood, over  $n$  observations, is

$$\ell(\beta, \alpha) = \sum_{i=1}^n \left[ y_i \log(\lambda_i) - \lambda_i - \log(y_i!) \right]$$

There is no closed-form solution for the coefficient vector that maximizes  $\ell$ ; estimation relies on iterative numerical methods (typically Newton-Raphson or variants of iteratively reweighted least squares, IRLS).

## Assumptions

- **Equidispersion** ( $\text{Var}(Y) = E[Y] = \lambda$ ): the assumption most often violated in practice. When observed variance clearly exceeds the mean — **overdispersion** — the maximum-likelihood standard errors from a plain Poisson model are biased downward, producing inflated test statistics and artificially small p-values. Diagnostic: compare the model’s deviance (or Pearson chi-square) statistic to its residual degrees of freedom — a ratio well above 1 suggests overdispersion.
- **Conditional independence of events**: given the vector of explanatory variables and fixed effects, counts must be independent across observations. Time series with strong dependence (a win boosting the odds of scoring in the following match through a confidence effect) violate this.
- **Correct log-linear functional form**: assumes the effect of the variables is multiplicative on the original scale, not additive. If the true relationship takes another shape (e.g., saturation at high values of the explanatory variable), the misspecified model can produce biased predictions at the extremes.
- **Fixed effects require enough within-unit variation**: if a unit (team) never plays both home and away within the sample (a rare but illustrative case), that unit’s fixed effect absorbs all its variation and the coefficient of interest cannot be identified from it.

## Hypotheses

For each coefficient  $\beta_j$ , the usual test is:

- $H_0 : \beta_j = 0$  (variable  $x_j$  has no effect on the expected event rate, controlling for the others)
- $H_1 : \beta_j \neq 0$

The test statistic (Wald test) is  $z = \hat{\beta}_j / \text{SE}(\hat{\beta}_j)$ , compared against a standard normal distribution. A 95% confidence interval for  $\beta_j$  is  $\hat{\beta}_j \pm 1.96 \times \text{SE}(\hat{\beta}_j)$ ; exponentiating the bounds gives the confidence interval for the multiplicative factor  $e^{\beta_j}$ . When the model suffers from overdispersion, it's advisable to recompute standard errors robustly (sandwich estimator) or refit using a negative binomial model before interpreting these tests.

## Interpretation

The exponentiated coefficient tells you how much the expected event rate changes multiplicatively per unit change in the explanatory variable, controlling for the other variables and fixed effects included. That is a statement about conditional association within the specified model — not an automatic causal claim. In an observational setting (like estimating the home-field effect from historical match data, with no controlled experiment), the coefficient reflects the estimated association given the included set of controls; if relevant factors are omitted and correlate with both the variable of interest and the event rate, the estimate can be biased.

Statistical significance (a  $\beta_j$  distinguishable from zero) is not the same as practical relevance: a multiplicative factor of 1.02 (a 2% increase) can be statistically significant in a large sample without being operationally meaningful.

## Limitations

- **Overdispersion is common and is not automatically flagged by the model fit** — it must be checked explicitly. Ignoring it leads to artificially narrow confidence intervals and significance conclusions that don't hold up.
- **Excess zeros** (more zero-count observations than Poisson predicts) is another common form of misspecification, distinct from general overdispersion, which may call for specific models (zero-inflated Poisson).
- **Per-unit fixed effects with many categories and few observations per category** (the incidental parameters problem) can produce unstable estimates, especially combined with low counts.
- **The multiplicative interpretation assumes the effect of  $x_j$  is constant on the log scale across the observed range of values** — extrapolating predictions to variable combinations far outside the training data's observed range is risky.

## Example

Consider a hypothetical example from a generic sports league with five teams (A through E), where the goal is to estimate the effect of playing at home on goals scored, controlling for team fixed effects.

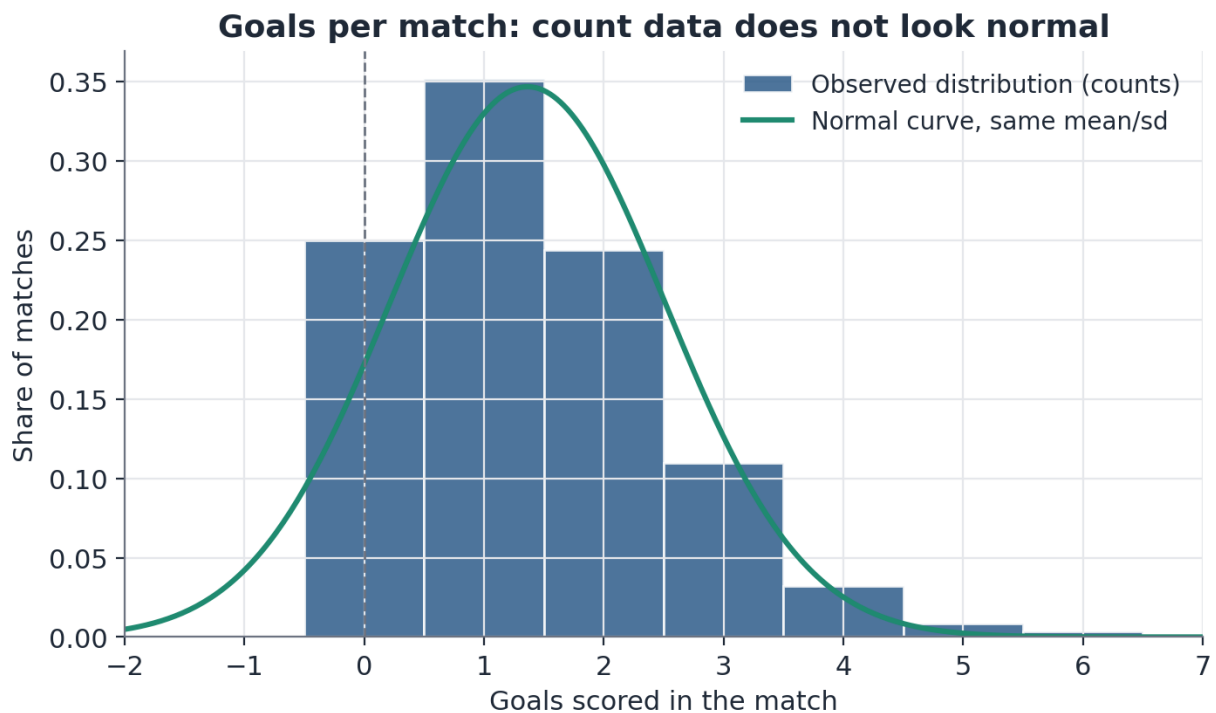
A simplified model, fit on synthetic data from a hypothetical season, estimates:

- Home-field coefficient:  $\hat{\beta}_{\text{home}} = 0.300$ , standard error = 0.085
- Wald statistic:  $z = 0.300 / 0.085 \approx 3.53$ , p-value < 0.001
- 95% confidence interval for  $\beta_{\text{home}}$ : [0.133, 0.467]

Exponentiating the coefficient and the interval bounds: estimated multiplicative factor  $e^{0.300} \approx 1.35$ , with an approximate 95% CI of  $[e^{0.133}, e^{0.467}] \approx [1.14, 1.60]$ .

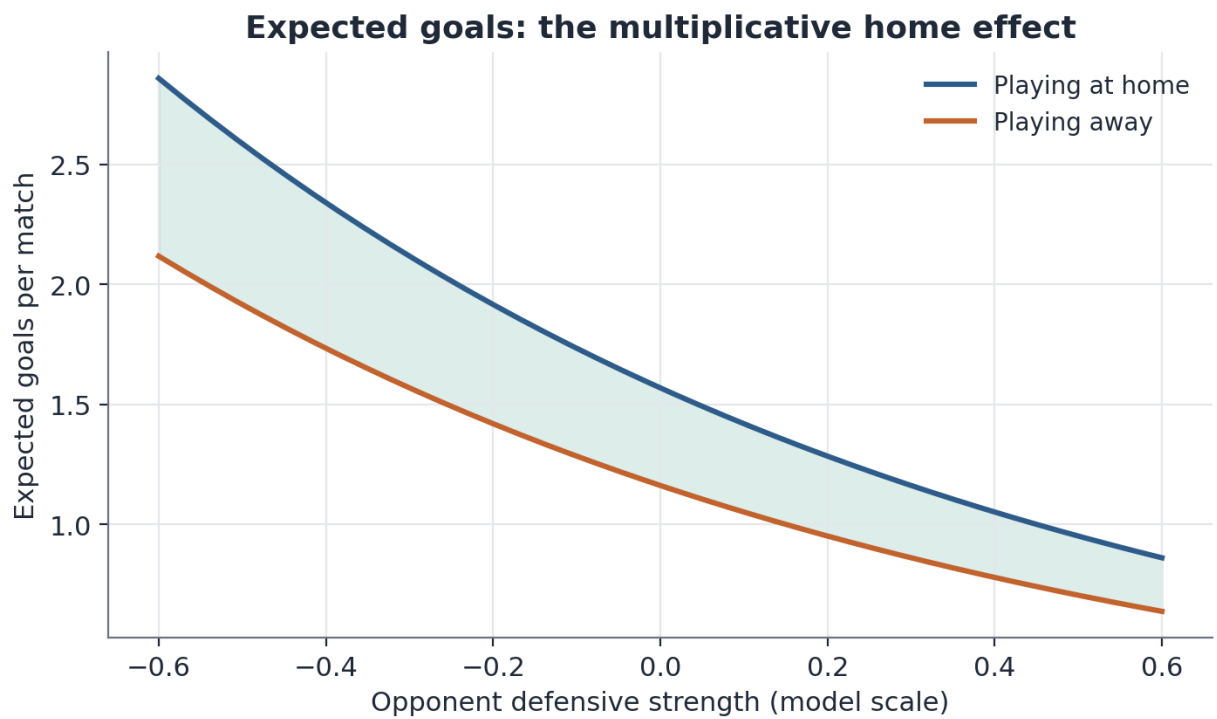
**Reading:** controlling for each team's relative strength (via the per-team fixed effect), playing at home is associated with an estimated 35% increase in expected goals scored (95% CI: an increase

between 14% and 60%), a statistically significant effect ( $p < 0.001$ ). This is a statement about association within this hypothetical season's observed data — generalizing to other seasons or competitions would require refitting the model on data from that context, and checking whether the equidispersion assumption holds before trusting the reported confidence intervals.



Illustrative data, not drawn from any real project.

Figure 26: Goal-count distribution compared to a normal curve



Illustrative data, not drawn from any real project.

Figure 27: Expected goals by opponent defensive strength, home vs. away

# Weighted least squares trend with confidence-interval classification

## Concept

Weighted least squares (WLS) is an extension of ordinary linear regression (OLS) that allows different weights to be assigned to each observation when fitting a line. In the context of trend analysis over time, the most common use is weighting each period by the number of observations (sample size) that back the measured metric in that period — explicitly acknowledging that a point calculated from 400 cases is a more precise estimate (has lower variance) than one calculated from 20 cases, and should therefore carry more influence over the fitted line.

The statistical motivation is known heteroscedasticity: when the measurement error variance of each point is not constant, but is approximately proportional to the inverse of that point's sample size (a direct consequence of the law of large numbers — the variance of a sample mean is  $\sigma^2/n$ ), OLS still produces unbiased estimates of the slope coefficient, but it stops being the minimum-variance estimator (it's no longer BLUE — *Best Linear Unbiased Estimator*, per the generalized Gauss-Markov theorem). WLS, using weights proportional to  $n$ , restores that efficiency property.

The second part of the method — **trend classification** — converts the estimated slope and its confidence interval into a categorical label (rising / falling / stable) using a simple, deterministic rule: the position of the confidence interval relative to zero. Formally, this rule is equivalent to a two-sided hypothesis test on the slope coefficient at a significance level corresponding to the chosen confidence interval.

## Mathematical formulation

**Model.** For a unit (store, player, region) with observations across  $T$  periods, the weighted linear trend model is

$$y_t = \alpha + \beta t + \varepsilon_t, \quad t = 1, \dots, T$$

where  $y_t$  is the observed metric in period  $t$ ,  $\alpha$  is the intercept,  $\beta$  is the slope (the quantity of interest — the trend), and  $\varepsilon_t$  is the error term, with  $\text{Var}(\varepsilon_t) = \sigma^2/w_t$ , where  $w_t$  is period  $t$ 's weight — typically  $w_t = n_t$ , the number of observations underlying  $y_t$ .

**WLS estimator.** The coefficients that minimize the weighted sum of squared residuals,

$$\hat{\alpha}, \hat{\beta} = \arg \min_{\alpha, \beta} \sum_{t=1}^T w_t (y_t - \alpha - \beta t)^2$$

have a closed-form solution analogous to OLS, but with means and products weighted:

$$\hat{\beta} = \frac{\sum_t w_t (t - \bar{t}_w)(y_t - \bar{y}_w)}{\sum_t w_t (t - \bar{t}_w)^2}, \quad \hat{\alpha} = \bar{y}_w - \hat{\beta} \bar{t}_w$$

where  $\bar{t}_w = \frac{\sum_t w_t t}{\sum_t w_t}$  and  $\bar{y}_w = \frac{\sum_t w_t y_t}{\sum_t w_t}$  are the weighted means of  $t$  and  $y$ .

**Standard error of the slope.** Under the assumption that the weights correctly capture the variance structure, the standard error of  $\hat{\beta}$  is

$$\text{SE}(\hat{\beta}) = \sqrt{\frac{\hat{\sigma}^2}{\sum_t w_t (t - \bar{t}_w)^2}}, \quad \hat{\sigma}^2 = \frac{\sum_t w_t (y_t - \hat{\alpha} - \hat{\beta}t)^2}{T - 2}$$

with  $T - 2$  degrees of freedom (two estimated parameters:  $\alpha$  and  $\beta$ ).

**Confidence interval and classification.** The  $(1 - \alpha_{\text{sig}}) \times 100\%$  confidence interval for  $\beta$  is

$$\hat{\beta} \pm t_{T-2, 1-\alpha_{\text{sig}}/2}^* \times \text{SE}(\hat{\beta})$$

where  $t^*$  is the critical value of the Student's t-distribution with  $T - 2$  degrees of freedom. The classification rule is:

$$\text{Trend} = \begin{cases} \text{rising} & \text{if the CI's lower bound} > 0 \\ \text{falling} & \text{if the CI's upper bound} < 0 \\ \text{stable} & \text{if the CI contains zero} \end{cases}$$

This rule is mathematically equivalent to rejecting (or not)  $H_0 : \beta = 0$  in favor of  $H_1 : \beta \neq 0$  at significance level  $\alpha_{\text{sig}}$ , and then using the sign of  $\hat{\beta}$  when  $H_0$  is rejected to decide between “rising” and “falling.”

## Assumptions

- **Weights correctly reflect each point's relative precision.** Using  $n_t$  (observation count) as the weight assumes the variance of  $y_t$  is approximately proportional to  $1/n_t$  — valid when  $y_t$  is itself an average or proportion computed over  $n_t$  cases with individual variance reasonably constant across periods. If within-period individual variability also changes over time, the ideal weights are not simply  $n_t$ .
- **Linearity.** The underlying trend is modeled as a straight line; trends with meaningful curvature (acceleration, deceleration, unremoved seasonality) are poorly captured by this simple model, and the rising/falling/stable classification may not reflect the series' actual behavior.
- **Errors uncorrelated across periods, conditional on the weights.** If residual temporal autocorrelation is present (for example, a good quarter tends to be followed by another good quarter, beyond what the linear trend already captures), standard WLS standard errors underestimate the true uncertainty — in that case, autocorrelation-robust standard errors (Newey-West) are more appropriate.
- **No structural outliers unrelated to the trend.** A single period with an extreme value (even if based on many observations, and therefore carrying high weight) can distort the estimated slope more than is reasonable; it's worth inspecting visually before blindly trusting the automatic classification.

## Hypotheses

- $H_0 : \beta = 0$  — there is no linear trend (the metric does not vary systematically over time, given the weighted precision of the available data).

- $H_1 : \beta \neq 0$  — a linear trend exists (positive or negative).

Typical significance level used: 5% (corresponding to a 95% CI), though the level chosen is an editorial/operational decision — a more permissive level (10%) classifies more series as having a defined trend (fewer “stable”), at the cost of more false positives; a stricter level (1%) does the opposite.

## Interpretation

The slope  $\hat{\beta}$  quantifies the estimated average change in the metric per unit of time (per quarter, for example), under the linearity assumption. The rising/falling/stable classification is a simplified operational translation of that slope and its uncertainty — not a statement about the cause of the trend. A statistically confirmed upward trend doesn’t say why the metric is rising (a mix shift, seasonality, a deliberate action, an external factor); it only says the statistical evidence is sufficient to distinguish the slope from zero, given the available data volume.

It’s important to communicate “stable” correctly: it means insufficient evidence for a defined trend, not proof that the metric truly did not change. Short series (few periods) or ones with little data per period tend to produce more “stable” classifications simply from lack of statistical power, even when a real underlying trend exists.

## Limitations

- **Sensitivity to the chosen confidence level.** Different CI choices (90%, 95%, 99%) produce different classifications for the same data, especially for series whose slope sits near the significance boundary — it’s worth reporting the slope and its full CI, not just the categorical label.
- **Limited statistical power with few periods.** With small  $T$  (three, four points), the confidence interval tends to be wide even with high weights, making it hard to distinguish a real, small trend from noise.
- **A linear model may not fit non-linear patterns.** A metric that rises and then flattens out (an S-curve) might be classified as “rising” overall even though it has already leveled off in the most recent periods — worth supplementing with visual inspection or a local trend model (e.g., local regression, splines) when the pattern looks suspicious.
- **Multiple testing.** Classifying many units simultaneously (hundreds of stores, say) at a 5% significance level each generates, by chance alone, an expected number of spurious “rising” or “falling” classifications — if the goal is to flag genuinely unusual units, it’s worth considering a multiple-comparisons correction.

## Example

Consider two hypothetical stores from a retail chain, each with twelve quarters of average-ticket data, with transaction counts (the weight) varying substantially across quarters — a common situation when some quarters coincide with high-volume promotional dates and others are slow periods.

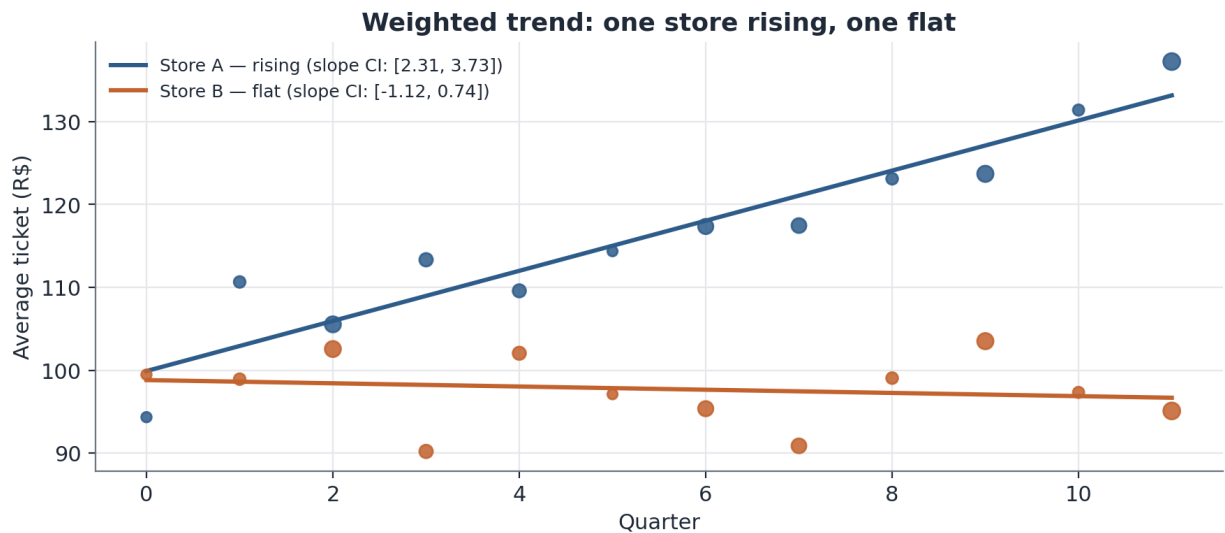
**Store A** — average ticket rising consistently, with a high sample weight in most quarters. WLS fit:  $\hat{\beta} = 2.40$  (dollars per quarter) -  $SE(\hat{\beta}) = 0.58$  -  $t_{10,0.975}^* \approx 2.23$  - 95% CI:  $2.40 \pm 2.23 \times 0.58 = [1.11, 3.69]$

Since the lower bound (1.11) exceeds zero, Store A is classified as **rising**.

**Store B** — average ticket bouncing around with no clear pattern, with several low-weight quarters. WLS fit: -  $\hat{\beta} = 0.05$  -  $SE(\hat{\beta}) = 0.56$  - 95% CI:  $0.05 \pm 2.23 \times 0.56 = [-1.20, 1.30]$

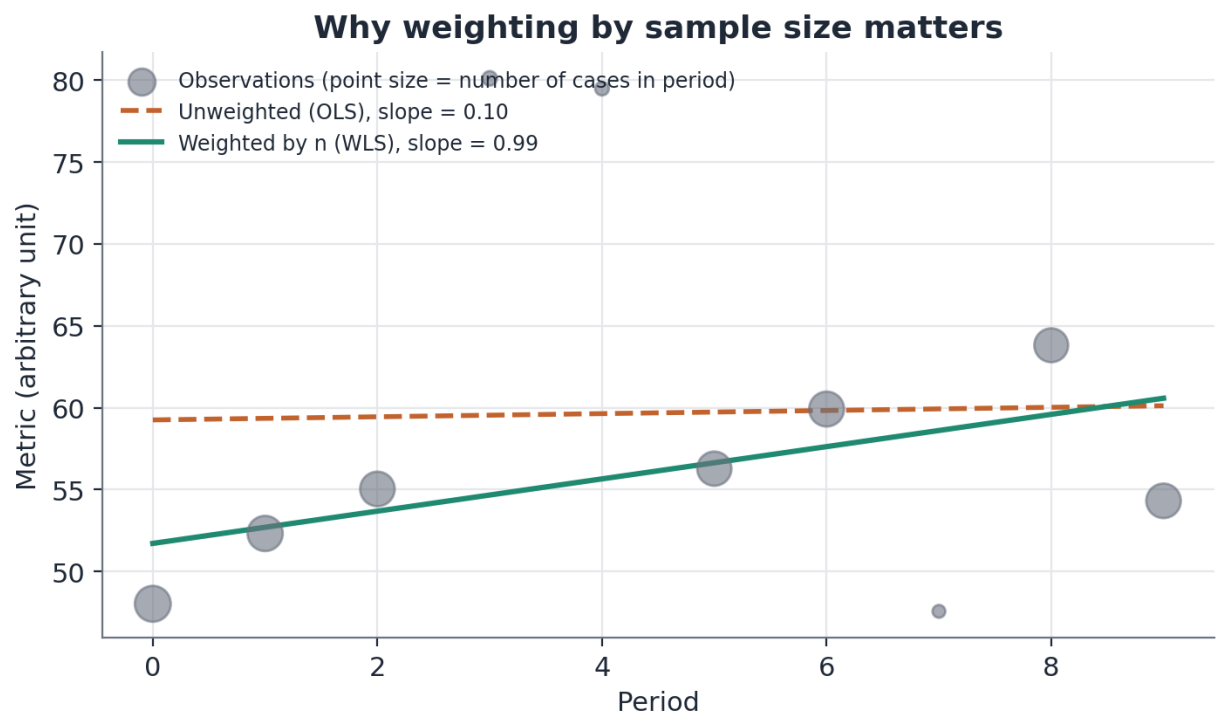
Since the interval contains zero, Store B is classified as **stable** — even though the point estimate is technically positive, there isn't enough statistical evidence, given the observed pattern of sample precision, to distinguish it from zero.

**Contrast with unweighted OLS.** If Store B's same data were fit without weighting (plain OLS), the few low-volume quarters — which happen to show more extreme values — would carry equal weight to the high-volume quarters, potentially producing a quite different estimated slope (and a standard error that doesn't correctly reflect the points' unequal precision). This is exactly the kind of distortion that weighting by sample size exists to correct.



Illustrative data, not drawn from any real project.

Figure 28: Weighted trend classification for two stores



Illustrative data, not drawn from any real project.

Figure 29: OLS vs. WLS when sample size varies sharply across periods

# Logistic regression

## Concept

Logistic regression is a generalized linear model (GLM) designed for binary response variables: an outcome that takes only two values, typically coded 0 (didn't happen) and 1 (happened). It's the natural model when the variable of interest violates the central assumption of an ordinary linear regression applied directly to a 0/1 outcome: a line fit by least squares can predict values outside the  $[0, 1]$  interval, which don't correspond to any valid probability, and the model's errors don't have constant variance (structural heteroskedasticity, a direct consequence of the outcome being binary).

The central idea is to model, not the probability  $p$  directly, but its **logit** — the log odds,  $\log(p/(1-p))$  — as a linear combination of the explanatory variables. The logit can take any real value; the logistic function that converts it back into a probability guarantees the result always stays in the  $(0, 1)$  interval, regardless of the coefficients or the variables.

## Mathematical formulation

**Response variable.**  $Y_i \in \{0, 1\}$  for each observation  $i$ , with  $Y_i \sim \text{Bernoulli}(p_i)$ , where  $p_i = P(Y_i = 1 \mid x_i)$ .

**Logit-linear model.** The logit of the probability is modeled as a linear combination of the explanatory variables:

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki}$$

Solving for  $p_i$  gives the logistic (sigmoid) function:

$$p_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_{1i} + \dots + \beta_k x_{ki})}}$$

**Multiplicative interpretation of the coefficient (odds ratio).** Consider two observations identical except for  $x_1$ , with  $x_1 = a + 1$  in the first and  $x_1 = a$  in the second. The odds ratio between the two is

$$\frac{\text{odds}_i \mid x_1 = a + 1}{\text{odds}_i \mid x_1 = a} = \frac{p/(1-p) \mid x_1 = a + 1}{p/(1-p) \mid x_1 = a} = e^{\beta_1}$$

where  $\text{odds} = p/(1-p)$ . All other terms cancel out because they're identical across the two observations — that's why  $e^{\beta_1}$  is read as the **odds ratio**: how many times the odds of the outcome get multiplied when  $x_1$  increases by one unit, holding everything else constant. Note that an odds ratio is not the same as a probability ratio ( $p_1/p_0$ ) — the two coincide only approximately when  $p$  is small.

**Estimation.** Coefficients are estimated by maximum likelihood. The likelihood function, for  $n$  independent observations, is

$$L(\beta) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}$$

and the log-likelihood,

$$\ell(\beta) = \sum_{i=1}^n \left[ y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right]$$

There's no closed form for the vector  $\hat{\beta}$  that maximizes  $\ell$ ; estimation uses iterative numerical methods (typically Newton-Raphson or iteratively reweighted least squares, IRLS), which converge to the maximum because  $\ell(\beta)$  is concave in the standard logistic case.

## Assumptions

- **Independent observations** (conditional on the explanatory variables). When several observations come from the same unit over time (the same individual, the same team across different rounds of a season), this assumption is violated — maximum-likelihood standard errors come out too small, inflating apparent statistical significance. The usual fix is to cluster standard errors by unit or to validate with splits that respect the group structure (holding out a whole unit, not an isolated observation).
- **Correct logit-linear functional form**: the effect of the variables is assumed to be linear on the logit scale, not the probability scale. The relationship between a continuous variable and the logit may not actually be linear (it could, say, have a U shape); in that case, polynomial terms or splines on the variable help capture the nonlinearity.
- **No severe multicollinearity** among explanatory variables: when two variables are strongly correlated with each other, individual coefficients become unstable (large standard errors) even if the model, as a whole, predicts well.
- **Separation and quasi-separation**: when a combination of explanatory variables predicts the outcome nearly perfectly (say, every observation with  $x$  above some value has  $y = 1$  and every one below has  $y = 0$ ), the likelihood has no well-defined finite maximum — the estimation algorithm doesn't converge stably, and the coefficients (especially their standard errors) come out artificially large. This tends to show up precisely when a variable is highly informative; it isn't a sign of a model error, but it does call for reporting confidence intervals with that caveat.

## Hypotheses

For each coefficient  $\beta_j$ , the usual test is:

- $H_0 : \beta_j = 0$  (variable  $x_j$  has no effect on the logit of the outcome probability, controlling for the other variables)
- $H_1 : \beta_j \neq 0$

The test statistic (Wald test) is  $z = \hat{\beta}_j / \text{SE}(\hat{\beta}_j)$ , compared against a standard normal distribution. A 95% confidence interval for  $\beta_j$  is  $\hat{\beta}_j \pm 1.96 \times \text{SE}(\hat{\beta}_j)$ ; exponentiating the bounds gives the confidence interval for the odds ratio  $e^{\beta_j}$ . Under quasi-separation, the likelihood-ratio test (comparing the full model to one without  $x_j$ , via  $-2[\ell_{\text{restricted}} - \ell_{\text{full}}] \sim \chi_1^2$ ) tends to be more stable than the Wald test.

## Interpretation

The exponentiated coefficient (odds ratio) says how much the odds of the outcome change multiplicatively per unit change in the explanatory variable, controlling for the other variables in the model. As with Poisson regression, this is a statement about conditional association within the specified model, not an automatic causal claim — in observational data, omitted factors correlated with both  $x_j$  and the outcome can bias the estimate.

The model's quality as a **classifier** is a separate question from the significance of its coefficients, and should be assessed independently:

- **AUC (area under the ROC curve):** the probability that the model assigns a higher predicted probability to a randomly chosen positive observation than to a randomly chosen negative one. Equivalent to the normalized Mann-Whitney/Wilcoxon statistic.  $AUC = 0.5$  corresponds to a model with no discriminative power (equivalent to random classification);  $AUC = 1.0$ , perfect separation between the two classes.
- **Brier score:**  $\frac{1}{n} \sum_i (\hat{p}_i - y_i)^2$ , the mean squared error between predicted probability and the observed outcome (0 or 1). It measures discrimination and calibration together — unlike AUC, which only measures the relative ranking of predicted probabilities, not whether they're correctly calibrated in absolute terms.
- **Out-of-sample validation:** as with any predictive model, in-sample performance on the training data tends to overstate performance on new data. With small panels and group structure (e.g., several observations per season), validating by holding out one whole unit at a time (leave-one-group-out) is preferable to a simple random split, which can leak information from the same unit between train and test.

## Limitations

- **Quasi-separation on highly predictive variables:** paradoxically, the better the explanatory variable separates the classes, the higher the risk of instability in estimating the coefficient and its standard error — this should be reported as a caveat, without discarding the direction of the effect.
- **A single regressor can have good AUC and still ignore relevant information:** a one-variable logistic regression measures how much that variable, alone, already separates the classes — it doesn't rule out other variables improving the prediction if included.
- **Odds ratio isn't relative risk:** for common outcomes (base probability far from zero), the odds ratio overstates the corresponding probability ratio — this distinction should be made explicit when communicating results to non-technical audiences, to avoid the wrong reading of “the odds doubled” as “the probability doubled.”
- **Extrapolation outside the observed range:** as with any fitted model, predicting the probability for variable combinations far from the training data is risky, even though the logistic curve always returns a number between 0 and 1.

## Example

Consider a fictional example of predicting subscription cancellation from a satisfaction score (0 to 10), fit on a synthetic customer base.

A single-regressor model, fit on simulated data, estimates:

- Satisfaction coefficient:  $\hat{\beta} = -0.69$ , standard error = 0.10

- Wald statistic:  $z = -0.69/0.10 = -6.90$ ,  $p\text{-value} < 0.001$
- 95% confidence interval for  $\beta$ :  $[-0.89, -0.49]$

Exponentiating the coefficient and the interval bounds: estimated odds ratio  $e^{-0.69} \approx 0.50$ , with an approximate 95% CI of  $[e^{-0.89}, e^{-0.49}] \approx [0.41, 0.61]$ .

**Reading:** each extra point of satisfaction is associated with an estimated 50% reduction in the odds of cancelling (95% CI: reduction between 39% and 59%), a statistically significant effect ( $p < 0.001$ ). The cutoff where predicted probability equals 50% sits at  $x = -\hat{\beta}_0/\hat{\beta}$ ; validated out of sample, the model reaches an AUC of roughly 0.82 in this example — well above 0.50 (random), indicating the satisfaction score alone already discriminates reasonably well between who cancels and who stays.

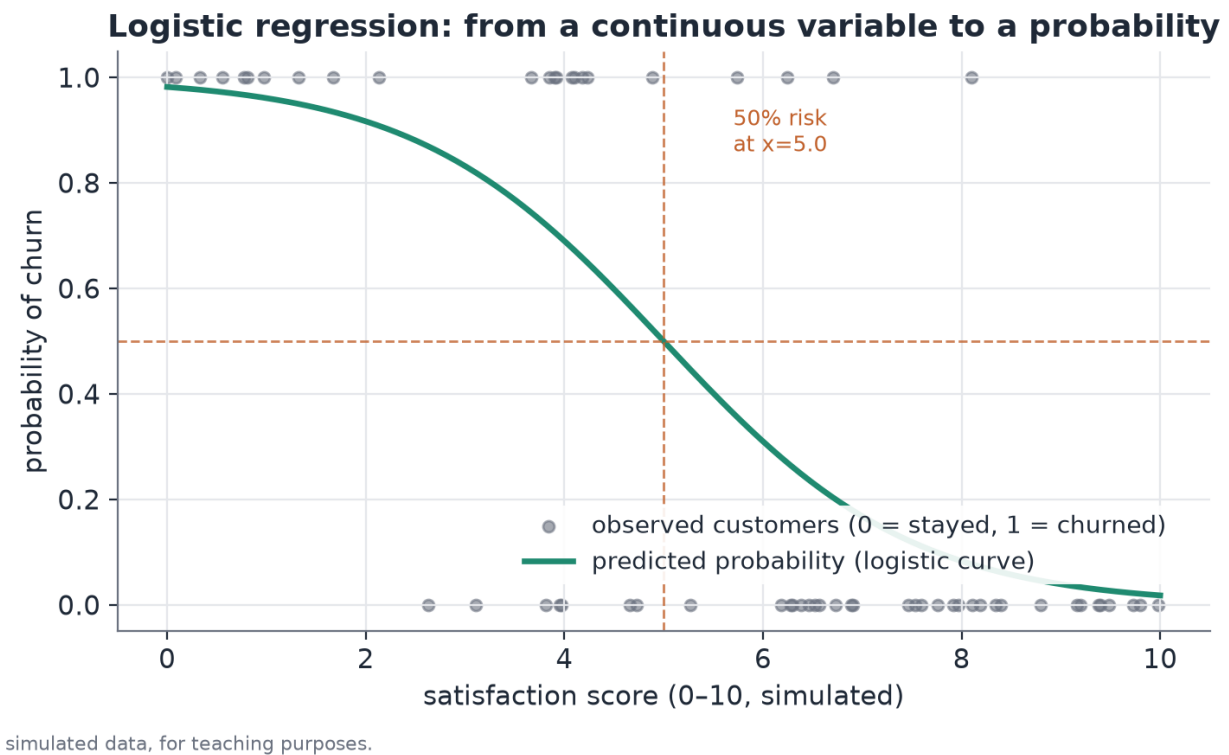
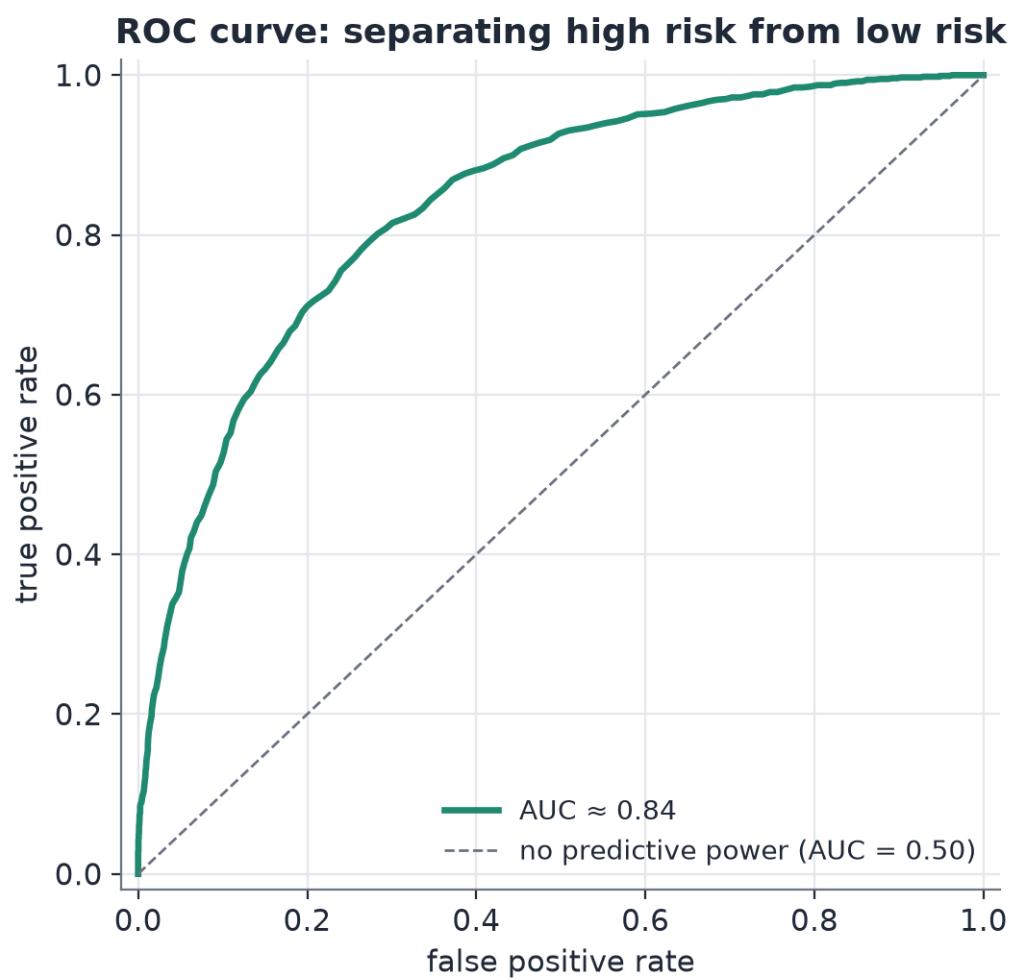


Figure 30: Logistic regression: from a continuous variable to a probability



illustrative classifier, simulated data.

Figure 31: ROC curve for an illustrative classifier